

Probability Theory

Lecture by Chiara Saffirio

Notes taken by David Flückiger at the University of Basel

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Preface

This is a rewriting of the notes initially taken in 2022. There are probably still a lot of typos so the reader is advised to read this text with care.

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1 Introduction

We will put the basic concepts acquired in “Einführung in die Statistik” into a more general mathematical framework using notions from “Real Analysis”.

More on infinite sequences of random variables (stochastic processes in discrete time): asymptotic behaviour, strong law of large numbers, several notions of convergence, random walks and Brownian motion.

1.1 Notation

- We write $\mathbb{N}$ for the natural numbers without zero, i.e.

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

and $\mathbb{N}_0$ for the natural numbers with zero, i.e.

$$\mathbb{N}_0 := \mathbb{N} \cup \{0\}.$$

- We will write λ for the Lebesgue measure (on $\mathbb{R}$).
- $\mathbb{P}$ will mostly denote a probability measure.
- For a set Ω , we write $\Omega^{\mathbb{N}}$ to denote the set of all sequences in Ω i.e.

$$\Omega^{\mathbb{N}} = \bigotimes_{i=1}^{\infty} \Omega = \{\omega = [\omega_n]_{n \in \mathbb{N}} : \omega \text{ is a sequence with values in } \Omega\}.$$

Additionally we will use the notation $\omega(n)$ to indicate the n -th element of a sequence $\omega \in \Omega^{\mathbb{N}}$. We will use the notation $[\omega_i]_{i \in \mathbb{N}}$, $[\omega_i]_{i=1}^{\infty}$ or $\{\omega_i\}_{i \in \mathbb{N}}$ for sequences interchangeably.

- for $\mathbf{x} = [x_1, \dots, x_n] \in \mathbb{R}^n$ we will also write $\mathbf{x} = [x_i]_{i=1}^n$ and likewise for

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & \dots & a_{1,n} \\ \vdots & & \vdots \\ a_{n,1} & \dots & a_{n,n} \end{bmatrix} \in \mathbb{R}^{n \times n}$$

we write $\mathbf{A} = [a_{i,j}]_{i,j=1}^n \in \mathbb{R}^{n \times n}$.

- We define the set $\mathcal{C}^0(X, Y)$ as the collection of all continuous functions $f : X \rightarrow Y$.
- Furthermore we use the symbol $\mathcal{C}^k(X, Y)$ with $k \in \mathbb{N}$ for the collection of functions $f : X \rightarrow Y$ where f is k -times continuously differentiable.

1.2 Examples and motivation

Example 1.1 (Series with random coefficients). We know that

$$x_n := \sum_{i=1}^n \frac{(-1)^i}{i} \xrightarrow{n \rightarrow \infty} -\log 2 \quad \text{and} \quad y_n := \sum_{i=1}^n \frac{1}{i} \xrightarrow{n \rightarrow \infty} \infty.$$

In this course we will look at

$$z_n := \sum_{i=1}^n \frac{w_i}{i} \rightarrow ?$$

where w_i are independent random variables s.t.

$$\mathbb{P}(w_i = 1) = \mathbb{P}(w_i = -1) = \frac{1}{2}.$$

The following questions arise:

- does the random series converge?
- If yes, in which sense?
- if yes, is the limit random or deterministic?

△

Example 1.2 (Random walk). Let $z_i \in \{-1, 1\}$ and define $S : \mathbb{N} \rightarrow \mathbb{R}$ and

$$S_n = \sum_{i=1}^n z_i.$$

By linear interpolation we can define its extensions $S : \mathbb{R}_+ \rightarrow \mathbb{R}$ as a random objects. The following questions arise:

- What happens for $n \rightarrow \infty$? $\frac{1}{\sqrt{n}}S_n \rightarrow \mathcal{N}(0, 1)$ as $n \rightarrow \infty$ where the convergence is in distribution sense

$$\mathbb{P}\left(\frac{1}{\sqrt{n}}S_n \leq a\right) \xrightarrow{n \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^a \exp\left(-\frac{x^2}{2}\right) dx \quad \text{for every } a \in \mathbb{R}.$$

- If we think of this as a function of time: $S_n(t) = \frac{1}{\sqrt{n}}S_{nt} \rightarrow ?$ What is the limit object and which is the right notion of convergence. Spoiler: Here, Brownian motion will pop up!

△

2 Probability spaces and random variables

2.1 Properties of the probability space

Definition 2.1 (Probability measure). A probability measure on a measurable space $(\Omega, \mathcal{A})$ is a measure $\mathbb{P}$ satisfying $\mathbb{P}(\Omega) = 1$. This is also called a **normed measure**. We call a measure space $(\Omega, \mathcal{A}, \mathbb{P})$ where $\mathbb{P}$ is a normed measure a **probability space**.

Interpretation: $\omega \in \Omega$ is a possible *outcome* of the experiment. $A \in \mathcal{A}$ are the *events*.

Lemma 2.2. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. Let $A, B, A_1, A_2, \dots \in \mathcal{A}$. Then

1. $\mathbb{P}(A^C) = 1 - \mathbb{P}(A)$
2. If $A \subset B$ then $\mathbb{P}(A) \leq \mathbb{P}(B)$.
3. $\mathbb{P}(\bigcup_{i \in \mathbb{N}} A_i) \leq \sum_{i \in \mathbb{N}} \mathbb{P}(A_i)$
4. If the A_i 's are pairwise disjoint, then $\mathbb{P}(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n \mathbb{P}(A_i)$
5. If $A_1 \subset A_2 \subset \dots$ then $\mathbb{P}(\bigcup_{i \in \mathbb{N}} A_i) = \lim_{i \rightarrow \infty} \mathbb{P}(A_i)$
6. If $A_1 \supset A_2 \supset \dots$ then $\mathbb{P}(\bigcap_{i \in \mathbb{N}} A_i) = \lim_{i \rightarrow \infty} \mathbb{P}(A_i)$

Proof. We will only proof 5. and 6. The rest is left as an exercise to the reader.

5. Let $\{A_i\}_{i \in \mathbb{N}}$ be a increasing sequence of sets. Now, note that $(A_{i+1} \setminus A_i)_{i \in \mathbb{N}}$ is a sequence of disjoint sets. Therefore we can compute

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i \in \mathbb{N}} A_i\right) &= \mathbb{P}\left(A_1 \cup \bigcup_{i=2}^{\infty} (A_i \setminus A_{i-1})\right) = \mathbb{P}(A_1) + \sum_{i=2}^{\infty} \mathbb{P}(A_i \setminus A_{i-1}) \\ &\stackrel{(!)}{=} \mathbb{P}(A_1) + \sum_{i=2}^{\infty} [\mathbb{P}(A_i) - \mathbb{P}(A_{i-1})] = \lim_{n \rightarrow \infty} \mathbb{P}(A_n). \end{aligned}$$

We now need to prove (!): Let $A \subset B$. Then

$$\mathbb{P}(B) - \mathbb{P}(A) = \mathbb{P}(A \cup B) - \mathbb{P}(A) = \mathbb{P}((B \setminus A) \cup A) - \mathbb{P}(A) = \mathbb{P}(B \setminus A).$$

6. If $\{A_i\}_{i \in \mathbb{N}}$ is an decreasing sequence of sets then $\{A_i^C\}_{i \in \mathbb{N}}$ is increasing. Therefore we can use the above:

$$\mathbb{P} \left(\bigcap_{i \in \mathbb{N}} A_i \right) \stackrel{(1)}{=} 1 - \mathbb{P} \left(\bigcup_{i=1} A_i^C \right) = 1 - \lim_{i \rightarrow \infty} \mathbb{P}(A_i^C) = \lim_{i \rightarrow \infty} \mathbb{P}(A_i).$$

Here we used de Morgan on the first equality.

□

Example 2.3 (Discrete probability measure). Let $\Omega = \{\omega_1, \omega_2, \dots\} \subset \mathbb{R}$, $\mathcal{A} = \mathcal{P}(\Omega)$ and $p_1, p_2, \dots \geq 0$ s.t. $\sum_{i \geq 1} p_i = 1$. Then the function $\mathbb{P} : \mathcal{A} \rightarrow \mathbb{R}$ with $\mathbb{P}(A) = \sum_{i \in \mathbb{N}} \mathbb{1}_A(\omega_i) p_i$ is a probability measure. Indeed we have $\mathbb{P}(\Omega) = \sum_{i \in \mathbb{N}} \mathbb{1}_\Omega(\omega_i) p_i = \sum_{i \in \mathbb{N}} p_i = 1$ and for $\{A_i\}_{i=1}^\infty$ disjoint we have

$$\mathbb{P} \left(\bigcup_{i \geq 1} A_i \right) = \sum_{i \in \mathbb{N}} \mathbb{1}_{\bigcup_{i \geq 1} A_i}(\omega_i) p_i = \sum_{i \in \mathbb{N}} \sum_{j \in \mathbb{N}} \mathbb{1}_{A_j}(\omega_i) p_i = \sum_{j \in \mathbb{N}} \mathbb{P}(A_j).$$

△

Example 2.4 (Absolutely continuous probability measure). Let $(\Omega, \mathcal{A}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a measure space and let $f : \mathbb{R} \rightarrow [0, \infty)$ be a Borel measurable function s.t. $\int_{\mathbb{R}} f dx = 1$. Then

$$\mathbb{P} : \mathcal{B}(\mathbb{R}) \rightarrow \mathbb{R}, \quad A \mapsto \int_A f dx = \int_{\Omega} \mathbb{1}_A f dx$$

is a probability measure. Indeed we have $\mathbb{P}(\mathbb{R}) = 1$ by assumption. Furthermore let $\{A_i\}_{i \geq 1}$ be a sequence of disjoint sets. Then we have

$$\mathbb{P} \left(\bigcup_{i \in \mathbb{N}} A_i \right) = \int_{\Omega} \mathbb{1}_{\bigcup_{i \in \mathbb{N}} A_i} f dx = \int_{\Omega} \sum_{i=1}^{\infty} \mathbb{1}_{A_i} f dx = \sum_{i=1}^{\infty} \int_{\Omega} \mathbb{1}_{A_i} f dx = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

where we used the Lebesgue dominated convergence theorem to swap integral and sum. In this case f is called the *density* of the probability measure $\mathbb{P}$.

△

Definition 2.5. On a measurable space $(\Omega, \mathcal{A})$, s.t. for each $\omega \in \Omega$ the set $\{\omega\} \in \mathcal{A}$, a probability measure $\mathbb{P}$ is **continuous** if $\mathbb{P}(\{\omega\}) = 0$ for every $\omega \in \Omega$ and it is **discrete** if there is $D \subseteq \Omega$ countable such that $\mathbb{P}(D^C) = 0$.

Example 2.6 (multivariate continuous probability measure). Let $\Omega = \mathbb{R}$, $\mathcal{A} = \mathcal{B}(\mathbb{R}^n)$. Let $f : \mathbb{R}^n \rightarrow [0, \infty)$ measurable and s.t. $\int_{\Omega} f dx = 1$. Then the probability measure

$$\mathbb{P} : \mathcal{A} \rightarrow \mathbb{R}, \quad A \mapsto \int_A f dx$$

is continuous.

△

Definition 2.7 (Product space). Let $(\Omega_1, \mathcal{A}_1, \mathbb{P}_1)$ and $(\Omega_2, \mathcal{A}_2, \mathbb{P}_2)$ be two probability spaces. Then we can define the **product space** $(\Omega, \mathcal{A}, \mathbb{P})$ with $\Omega = \Omega_1 \times \Omega_2$, $\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2$ (the product σ -algebra) and $\mathbb{P} = \mathbb{P}_1 \otimes \mathbb{P}_2$ the product measure which satisfies $\mathbb{P}(A_1 \times A_2) = \mathbb{P}_1(A_1)\mathbb{P}_2(A_2)$ for every $A_1 \in \mathcal{A}_1$ and $A_2 \in \mathcal{A}_2$.

Remark 2.8. We proved in “Real Analysis” that the product measure with $\mathbb{P}(A_1 \times A_2) = \mathbb{P}_1(A_1)\mathbb{P}_2(A_2)$ for every $A_1 \in \mathcal{A}_1$ and $A_2 \in \mathcal{A}_2$ exists. We will prove the uniqueness later in this lecture. $\triangle$

Example 2.9. Important examples of absolutely continuous (with regards to the Lebesgue measure) probability measures are *normal distributions (Gaussians)*. In one dimension:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

is the *density* of a normal distribution with mean m and variance σ^2 . We will use the notation $\mathcal{N}(m, \sigma^2)$. For higher dimensions ($\mathbb{R}^n$ with $n \geq 2$) one can define the probability measure $\mathbb{P}$ on $(\Omega, \mathcal{A}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with

$$\mathbb{P} : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1], \quad A \mapsto \int_A \frac{1}{(2\pi)^{n/2}} \exp\left(-\frac{x_1^2 + x_2^2 + \cdots + x_n^2}{2}\right) d(x_1, \dots, x_n). \quad \triangle$$

2.2 Random variables and their distribution

Definition 2.10. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and $(S, \mathcal{S})$ a measurable space. We call a function

$$X : \Omega \rightarrow S$$

an **S -valued random variable** if it is a measurable function i.e. $X^{-1}(B) \in \mathcal{A}$ for every $B \in \mathcal{S}$. If $(S, \mathcal{S}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ then we call X a **real valued random variable**.

Definition 2.11. A subset of $\mathcal{E}$ of S is a **generator** if $\mathcal{S} = \sigma(\mathcal{E})$ i.e. $\mathcal{S}$ is the smallest σ -algebra containing $\mathcal{E}$.

Example 2.12. Generators of $\mathcal{B}(\mathbb{R})$ are:

- the collection of all open subsets of $\mathbb{R}$,
- the collection of all closed subsets of $\mathbb{R}$,
- the collection of all open intervals,
- the collection of all closed intervals,
- the collection of all half infinite intervals like $(-\infty, a]$ or $[b, \infty)$ for $a, b \in \mathbb{R}$. $\triangle$

The generator is very useful for checking if a function is measurable. Precisely: To see if $X : \Omega \rightarrow \mathbb{R}$ is a random variable it suffices to check that

$$X^{-1}((-\infty, a]) = \{X \leq a\} \in \mathcal{A} \quad \forall a \in \mathbb{R}.$$

The following proposition justifies that:

Proposition 2.13. If $\mathcal{E}$ is a generator of $\mathcal{S}$ and X is a random variable as in described in 3.1, then the following are equivalent:

1. $X^{-1}(B) \in \mathcal{A}$ for every $B \in \mathcal{S}$
2. $X^{-1}(B) \in \mathcal{A}$ for every $B \in \mathcal{E}$.

Proof. We will prove that $\{B \subset S : X^{-1}(B) \in \mathcal{A}\}$ is a σ -algebra. Clearly $X^{-1}(S) = \Omega \in \mathcal{A}$ and

$$X^{-1}(B^C) = \{\omega \in \Omega : X(\omega) \in B^C\} = \{\omega \in \Omega : X(\omega) \in B\}^C = (X^{-1}(B))^C \in \mathcal{A}$$

because $B^C \in \mathcal{S}$.

Let $\{B_i\}_{i \in \mathbb{N}}$ be a sequence of disjoint sets in S . Then

$$X^{-1}\left(\bigcup_{i \in \mathbb{N}} B_i\right) = \left\{\omega \in \Omega : X(\omega) \in \bigcup_{i \in \mathbb{N}} B_i\right\} = \bigcup_{i \in \mathbb{N}} \{\omega \in \Omega : X(\omega) \in B_i\} = \bigcup_{i \in \mathbb{N}} X^{-1}(B_i) \in \mathcal{A}$$

because $\bigcup_{i \in \mathbb{N}} B_i \in \mathcal{S}$. This proves that $\{B \subset S : X^{-1}(B) \in \mathcal{A}\}$ is a σ -algebra. This σ -algebra contains all elements of $\mathcal{E}$ hence $\sigma(\mathcal{E}) \subseteq \mathcal{S}$. $\square$

Definition 2.14. Given a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and a measurable space $(S, \mathcal{S})$, we consider the random variable $X : \Omega \rightarrow S$. The **distribution of X** is the probability measure μ_X on $(S, \mathcal{S})$ defined by

$$\mu_X(B) := \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in B\})$$

for every $B \in \mathcal{S}$. Alternative notations are $\mu_X = X_{\#}\mathbb{P} = \mathbb{P} \circ X^{-1} = \mathbb{P}^X = \mathbb{P}_X$.

Example 2.15 (Marginal distribution). Let $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), \mathbb{P})$ be a probability space with

$$\mathbb{P}(A) = \int_A f \, d\mathbf{x}$$

for some measurable function $f : \mathbb{R}^n \rightarrow [0, \infty)$. For $i = 1, \dots, n$ let $X_i : \mathbb{R}^n \rightarrow \mathbb{R}$, s.t $X_i(\boldsymbol{\omega}) = \omega_i$ with $\boldsymbol{\omega} = [\omega_i]_{i=1}^n \in \mathbb{R}^n$, the projection on the i -th coordinate. Take $B \in \mathcal{B}(\mathbb{R})$. Then

$$X_i^{-1}(B) = \mathbb{R}^{i-1} \times B \times \mathbb{R}^{n-i} \in \mathcal{B}(\mathbb{R}^n)$$

which implies that X_i is a random-variable. Furthermore

$$\mu_{X_i}(B) = \mathbb{P}(X_i^{-1}(B)) = \int_{\mathbb{R}^{i-1} \times B \times \mathbb{R}^{n-i}} f \, d\mathbf{x}$$

is called the i -th marginal of the probability measure P . In this case f_{X_i} is called the i -th

marginal density of P . Indeed, by Fubini's theorem

$$\begin{aligned}
 \mu_{X_i}(B) &= \int_{\mathbb{R}^{i-1}} \int_B \int_{\mathbb{R}^{n-i}} f \, d\mathbf{x} \\
 &= \int_B \left(\underbrace{\int_{\mathbb{R}^{n-1}} f(x_1 \dots x_n) \, dx_1 \dots dx_{i-1}, dx_{i+1} \dots dx_n}_{=f_{X_i}(x_i)} \right) dx_i \\
 &= \int_B f_{X_i} \, dx. \quad \triangle
 \end{aligned}
 \tag{2.1}$$

Example 2.16. Let $f : \mathbf{x} = [x_i]_{i=1}^n \in \mathbb{R}^n \mapsto (1/(2\pi)^{n/2}) \cdot \exp\left(-\frac{x_1^2 + \dots + x_n^2}{2}\right) \in \mathbb{R}$. Then

$$\mu_{X_j}(B) = \frac{1}{\sqrt{2\pi}} \int_B \exp\left(-\frac{x_j^2}{2}\right) dx_j$$

hence μ_{X_j} is the standard normal distribution $\mathcal{N}(0, 1)$. $\triangle$

Remark 2.17. This definition indeed makes sense because μ_X as defined in definition 2.14 satisfies the properties of a probability measure:

- $\mu_X(S) = \mathbb{P}(X^{-1}(S)) = \mathbb{P}(\Omega) = 1$.
- Let $\{B_i\}_{i \in \mathbb{N}}$ be a sequence of disjoint sets in $\mathcal{S}$. Then we have that $\{X^{-1}(B_i)\}_{i \in \mathbb{N}}$ is a disjoint sequence of sets in $\mathcal{A}$. Then

$$\begin{aligned}
 \mu_X \left(\bigcup_{i \in \mathbb{N}} B_i \right) &= \mathbb{P} \left(X^{-1} \left(\bigcup_{i \in \mathbb{N}} B_i \right) \right) = \mathbb{P} \left(\bigcup_{i \in \mathbb{N}} X^{-1}(B_i) \right) \\
 &= \sum_{i \in \mathbb{N}} \mathbb{P}(X^{-1}(B_i)) = \sum_{i \in \mathbb{N}} \mu_X(B_i). \quad \triangle
 \end{aligned}$$

Definition 2.18 (Distribution function). let X be a real-valued random variable. The **distribution function** of X is defined by

$$F_X : \mathbb{R} \rightarrow [0, 1], \quad x \mapsto \mathbb{P}(X \leq x) = \mu_X((-\infty, x])$$

Remark 2.19. You should already know this from “Einführung in die Statistik”. We have seen it in more generality in real Analysis. $\triangle$

Proposition 2.20. Let F_X be the distribution function of a random variable X . Then

- (1) F_X is non-decreasing,
- (2) $\lim_{t \rightarrow \infty} F_X(t) = 1$, $\lim_{t \rightarrow -\infty} F_X(t) = 0$ and
- (3) F_X is right continuous.

Proof. (1) Note that $F_X(a) = \mathbb{P}(X \leq a) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq a\})$ is a non-decreasing function because if $a < b$ then $\{\omega \in \Omega : X(\omega) \leq a\} \subseteq \{\omega \in \Omega : X(\omega) \leq b\}$ hence $\mathbb{P}(X \leq a) \leq \mathbb{P}(X \leq b)$ because $\mathbb{P}$ is a measure.

(2) We take a monotonously increasing sequence $\{a_n\}_{n \in \mathbb{N}} \subset \mathbb{R}$ with $\lim_{n \rightarrow \infty} a_n = \infty$. Then, using lemma 2.2

$$\lim_{n \rightarrow \infty} F_X(a_n) = \lim_{n \rightarrow \infty} \mu_X((-\infty, a_n]) = \mu_X\left(\bigcup_{n \in \mathbb{N}} (-\infty, a_n)\right) = \mu_X(\mathbb{R}) = \mathbb{P}(X \in \mathbb{R}) = 1.$$

For the other limit let $\{b_n\}_{n \in \mathbb{N}} \subset \mathbb{R}$ a monotonously decreasing sequence with $\lim_{n \rightarrow \infty} b_n = -\infty$ then

$$\lim_{n \rightarrow \infty} F_X(b_n) = \lim_{n \rightarrow \infty} \mu_X((-\infty, b_n]) = \mu_X\left(\bigcap_{n \in \mathbb{N}} (-\infty, b_n]\right) = \mu(\emptyset) = 0$$

again using lemma 2.2. Since we chose the sequence arbitrarily it holds that

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow \infty} F(x) = 1,$$

the desired property.

(3) Let $\{t_n\}_{n \in \mathbb{N}} \subset \mathbb{R}$ be a non-increasing sequence s.t. $\lim_{n \rightarrow \infty} t_n = t$ for a $t \in \mathbb{R}$. Then we have

$$(-\infty, t] = \bigcap_{n \in \mathbb{N}} (-\infty, t_n].$$

This is a decreasing sequence of sets, so Lemma 2.2 implies $F_X(t) = \lim_{n \rightarrow \infty} F_X(t_n)$. $\square$

Remark 2.21. Through F_X we can compute the following probabilities

- $\mathbb{P}(X \in (a, b]) = F_X(b) - F_X(a)$
- $\mathbb{P}(X \in [a, b]) = F_X(b) - F_X(a^-)$
- $\mathbb{P}(X \in [a, b)) = F_X(b^-) - F_X(a^-)$
- $\mathbb{P}(X \in (a, b)) = F_X(b^-) - F_X(a)$
- $\mathbb{P}(X = a) = F_X(a) - F_X(a^-)$.

Here we are using the notation $F_X(a^-) := \lim_{t \downarrow a} F_X(t)$. Observe that by proposition 2.20 (1). F_X is non-decreasing, hence its left limit exists i.e. $F_X(a^-)$ is well defined.

Moreover observe that F_X is continuous everywhere if and only if $\mathbb{P}(X = a) = 0$ for every $a \in \mathbb{R}$. $\triangle$

So far we know that given a distribution function of a random variable the properties mentioned in prop. 2.20 hold. We now show the inverse, that is: if those properties hold for a function F then there exists a unique probability measure μ defined through F .

Theorem 2.22 (Lebesgue-Stieltjes). If $F : \mathbb{R} \rightarrow [0, 1]$ is s.t.

- F is non-decreasing,
- $\lim_{t \rightarrow \infty} F(t) = 1$, $\lim_{t \rightarrow -\infty} F(t) = 0$ and
- F is right continuous.

then there is a unique probability measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with

$$F(x) = \mu((-\infty, x]) \quad \forall x \in \mathbb{R}.$$

Proof. We will prove the existence: Let $\Omega = (0, 1)$, $\mathcal{A} = \mathcal{B}((0, 1))$ and $\mathbb{P} = \lambda$ the Lebesgue-measure on $(0, 1)$. Let

$$X : (0, 1) \rightarrow \mathbb{R}, \quad \omega \mapsto \sup\{y \in \mathbb{R} : F(y) < \omega\}.$$

We claim

$$\{\omega : X(\omega) \leq x\} = \{\omega : \omega \leq F(x)\}, \quad (2.2)$$

for every $x \in \mathbb{R}$, which would imply that X is a random variable because $\{\omega : \omega \leq F(x)\} \in \mathcal{B}((0, 1))$. Furthermore if this is true we have

$$\begin{aligned} \mathbb{P}(X \leq x) &= \mathbb{P}(\{\omega : X(\omega) \leq x\}) \stackrel{(2.2)}{=} \mathbb{P}(\{\omega : \omega \leq F(x)\}) \\ &= \mathbb{P}(\{\omega : \omega \in [0, F(x)]\}) = F(x) \end{aligned}$$

hence F is the distribution function of X :

$$\mu((-\infty, x]) = \mathbb{P}(X \leq x) = F(x).$$

We now prove (2.2):

- We first prove $\{\omega : X(\omega) \leq x\} \supset \{\omega : \omega \leq F(x)\}$:

Let $\omega \in (0, 1)$ s.t. $\omega \leq F(x)$ for a given $x \in \mathbb{R}$. From that we see that

$$x \notin \{y \in \mathbb{R} : F(y) < \omega\}$$

so we have

$$x \geq X(\omega) = \sup\{y \in \mathbb{R} : F(y) < \omega\}$$

which implies

$$\{\omega : X(\omega) \leq x\} \supset \{\omega : \omega \leq F(x)\}.$$

- We now prove $\{\omega : X(\omega) \leq x\} \subset \{\omega : \omega \leq F(x)\}$:

To do that we prove

$$\{\omega : X(\omega) \leq x\}^C \supset \{\omega : \omega \leq F(x)\}^C. \quad (2.3)$$

We start by picking $\omega \in (0, 1)$ s.t. $\omega > F(x)$ for a given $x \in \mathbb{R}$. By the right-continuity of F there is an $\varepsilon > 0$ s.t. $F(x + \varepsilon) < \omega$ hence $X(\omega) \geq x + \varepsilon > x$. This implies (2.3) which leads to

$$\{\omega : X(\omega) \leq x\} \subset \{\omega : \omega \leq F(x)\}.$$

We still would need to show the uniqueness but this is easier using Dynkin's lemma. We do this proof in the corresponding chapter. $\square$

We have already pointed out that random variables are nothing else than measurable functions. In particular the properties holding for measurable functions hold true for random variables as well.

Lemma 2.23. Consider the measurable spaces $(\Omega, \mathcal{A})$ and $(S, \mathcal{S})$.

1. Let $X : \Omega \rightarrow S$ be an S valued random variable and $f : S \rightarrow \mathbb{R}$ be a measurable function. Then $f \circ X : \Omega \rightarrow \mathbb{R}$ is a random variable
2. Let $X_1, \dots, X_n$ be random variables on $(\Omega, \mathcal{A}, \mathbb{P})$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ a measurable function. Then $f \circ \mathbf{X}$ with $\mathbf{X} = [X_1, \dots, X_n]$ is a random variable.

Proof. 1. Note that f is measurable if and only if $f^{-1}(B) \in \mathcal{S}$ for every $B \in \mathcal{B}(\mathbb{R})$. We have $(f \circ X)^{-1}(B) = X^{-1}(f^{-1}(B)) \in \mathcal{A}$ because $f^{-1}(B) \in \mathcal{S}$ because f is measurable and X is a random variable. So we get that $f \circ X$ is a random variable.

2. We have $\mathbf{X} : \omega \in \Omega \mapsto [X_1(\omega), \dots, X_n(\omega)] \in \mathbb{R}^n$. Using that $\mathcal{B}(\mathbb{R}^n) = \mathcal{B}(\mathbb{R})^{\otimes n}$ (which we know from real Analysis), we have that for every $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R})$

$$\{\omega \in \Omega : \mathbf{X}(\omega) \in A_1 \times \dots \times A_n\} = \bigcup_{i=1}^n \underbrace{\{\omega \in \Omega : X_i(\omega) \in A_i\}}_{\in \mathcal{A}} \in \mathcal{A}$$

because $\{\omega \in \Omega : X_i(\omega) \in A_i\} \in \mathcal{A}$ as X_i are random variables. Therefore we get that $\mathbf{X}$ is measurable thus the first point applies. Here we used that the family $\{A_1 \times \dots \times A_n : A_i \in \mathcal{B}(\mathbb{R}), i = 1, \dots, n\}$ is a generator of $\mathcal{B}(\mathbb{R}^n)$. $\square$

Lemma 2.24. Let $\{X_i\}_{i=1}^\infty$ be a sequence of $[-\infty, +\infty]$ -real valued random variables on $(\Omega, \mathcal{A}, \mathbb{P})$. Then $\sup_{n \in \mathbb{N}} X_n$, $\inf_{n \in \mathbb{N}} X_n$, $\liminf_{n \rightarrow \infty} X_n$ and $\limsup_{n \rightarrow \infty} X_n$ are random variables.

Proof. See “Real Analysis”. $\square$

3 Expectation of a random variable

3.1 Definition and basic properties

Recall from “Einführung in die Statistik” that the expectation value was defined by

$$\mathbb{E}[X] = \sum_{x \in X(\Omega)} x P(X = x)$$

if the random variable X is discrete and

$$\mathbb{E}[X] = \int_{\mathbb{R}} x f_X(x) dx$$

if X is a continuous random variable with density f_X . We can unify these definitions using notions from “Real Analysis”:

Definition 3.1. Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable on $(\Omega, \mathcal{A}, \mathbb{P})$. We define the **expectation value of X** as

$$\mathbb{E}[X] := \int_{\Omega} X d\mathbb{P}.$$

We will also use L^p -spaces from “Real Analysis” but recontextualize it a bit.

Definition 3.2. For $p \in [1, \infty)$ we denote by

$$L^p := L^p(\Omega, \mathcal{A}, \mathbb{P}) = \{X : \Omega \rightarrow \mathbb{R} \mid X \text{ is a r.v. (random variable) and } \mathbb{E}(|X|^p) < \infty\}$$

and

$$L^\infty := L^\infty(\Omega, \mathcal{A}, \mathbb{P}) = \{X : \Omega \rightarrow \mathbb{R} \text{ r.v.} \mid \exists M < \infty, \text{ s.t. } \mathbb{P}(|X| < M) = 1\}.$$

We endow these spaces with the norms

$$\|X\|_{L^p} = \mathbb{E}[|X|^p]^{1/p}, \quad X \in L^p$$

and

$$\|X\|_{L^\infty} = \inf\{M : \mathbb{P}(|X| > M) = 0\} = \text{ess sup } |X|, \quad X \in L^\infty.$$

One can see that these definition are basically the same as the L^p norm in Analysis but since we are on a probability space we can make write the norm with the expectation value. The next definition introduces the so-called moments:

Definition 3.3. Let X and Y be random variables. We call

- $\mathbb{E}[X^p]$ the **p -th moment** of X ,
- $\mathbb{E}[|X|^p]$ the **p -th absolute moment** of X ,
- $\mathbb{E}[(X - \mathbb{E}(X))^p]$ the **p -th central moment** of X and
- $\text{Var}(X) := \mathbb{E}[(X - \mathbb{E}(X))^2]$ the **variance** of X for $X \in L^1$ and
- $\text{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$ the **covariance** of X and Y ($X, Y \in L^2$).

Proposition 3.4. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and $X, Y : \Omega \rightarrow \mathbb{R}$ random variables. Then

1. $\mathbb{P}(A) = \mathbb{E}[\mathbf{1}_A]$ for every $A \in \mathcal{A}$.
2. For every $\alpha \in \mathbb{R}$ we have

$$\mathbb{E}[\alpha X + Y] = \alpha \mathbb{E}[X] + \mathbb{E}[Y].$$

3. If $\mathbb{E}[X], \mathbb{E}[Y] < \infty$ and $X \leq Y$ on Ω then $\mathbb{E}[X] \leq \mathbb{E}[Y]$.

Proof. Those properties are an immediate consequence of the definition and properties of the Lebesgue integral. $\square$

3.2 Analysis reminders and their consequences in probability theory

We will now see some Real Analysis reminders but formulated differently to suit the context of probability theory better:

Proposition 3.5 (Hölder's inequality). Let $X \in L^p$ and $Y \in L^q$ where $\frac{1}{p} + \frac{1}{q} = 1$ (with the convention $\frac{1}{0} = \infty$ and $\frac{1}{\infty} = 0$) then

$$\mathbb{E}(|XY|) \leq \mathbb{E}[|X|^p]^{1/p} \mathbb{E}[|Y|^q]^{1/q}.$$

A special case of this is if $p = q = 2$. Then

$$\mathbb{E}[|XY|] \leq \sqrt{\mathbb{E}[|X|^2] \mathbb{E}[|Y|^2]}$$

and we speak of the **Cauchy-Schwarz Inequality**.

Proof. First we treat the case where $p = 1$, so $q = \infty$. So we have

$$\|XY\|_{L^1} = \mathbb{E}[|XY|] = \int_{\Omega} |XY| d\mathbb{P} \leq \|Y\|_{\infty} \underbrace{\int_{\Omega} |X| d\mathbb{P}}_{=\|X\|_1} = \|Y\|_{\infty} \|X\|_1.$$

Now let $p > 1$ which leads to a finite q . Without loss of generality we assume $\mathbb{E}[|X|^p] = \mathbb{E}[|Y|^q] = 1$.

$$\mathbb{E}[|XY|] = \int_{\Omega} |XY| d\mathbb{P}$$

Now we will use Young's inequality which gets us

$$\begin{aligned} \int_{\Omega} |XY| d\mathbb{P} &\leq \int_{\Omega} \frac{|X|^p}{p} + \frac{|Y|^q}{q} d\mathbb{P} = \int_{\Omega} \frac{|X|^p}{p} d\mathbb{P} + \int_{\Omega} \frac{|Y|^q}{q} d\mathbb{P} \\ &= \frac{\mathbb{E}[|X|^p]^{1/p}}{p} + \frac{\mathbb{E}[|Y|^q]^{1/q}}{q} = \frac{1}{p} + \frac{1}{q} = 1. \end{aligned}$$

To see the general case we look at the expectation value of $X/\mathbb{E}[|X|^p]^{1/p}$ and $Y/\mathbb{E}[|Y|^q]^{1/q}$ which gives us

$$\mathbb{E} \left[\left| \frac{X}{\mathbb{E}[|X|^p]^{1/p}} \cdot \frac{Y}{\mathbb{E}[|Y|^q]^{1/q}} \right| \right] \leq \frac{\mathbb{E}[|X|^p]^{1/p}}{\mathbb{E}[|X|^p]^{1/p}} \cdot \frac{\mathbb{E}[|Y|^q]^{1/q}}{\mathbb{E}[|Y|^q]^{1/q}} = 1.$$

Multiplying both sides by $\mathbb{E}[|X|^p]^{1/p} \cdot \mathbb{E}[|Y|^q]^{1/q}$ yields the result, thanks to the linearity of $\mathbb{E}$. $\square$

Proposition 3.6 (Minkowski). Let $X, Y \in L^p$. Then

$$\|X + Y\|_{L^p} = \mathbb{E}[|X + Y|^p]^{1/p} \leq \mathbb{E}[|X|^p]^{1/p} + \mathbb{E}[|Y|^p]^{1/p} = \|X\|_{L^p} + \|Y\|_{L^p}.$$

Proof. See real Analysis. $\square$

Lemma 3.7 (Fatou). let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of real-valued nonnegative random variables (valued in $[0, \infty]$). Then

$$\mathbb{E}[\liminf_{n \rightarrow \infty} X_n] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n].$$

Proof. See real Analysis. $\square$

Theorem 3.8 (Monotone convergence). Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of random variables valued in $[0, \infty]$ satisfying $X_n(\omega) \leq X_{n+1}(\omega)$ and $\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)$ for $\mathbb{P}$ -almost every $\omega \in \Omega$, then

$$\mathbb{E}[X_n] \xrightarrow{n \rightarrow \infty} \mathbb{E}[X].$$

Theorem 3.9 (Lebesgue dominated convergence thm.). Let X, Y be two random variables and $\{X_n\}_{n \in \mathbb{N}}$ a sequence of random variables satisfying

- $X_n \rightarrow X$ $\mathbb{P}$ -a.e
- $|X_n| \leq Y$ for $\mathbb{P}$ -a.e $\omega \in \Omega$
- $Y \in L^1(\Omega, \mathcal{A}, \mathbb{P})$

then

$$\mathbb{E}[X_n] \xrightarrow{n \rightarrow \infty} \mathbb{E}[X].$$

Theorem 3.10 (Jensen's inequality). Let $X \in L^1(\Omega, \mathcal{A}, \mathbb{P})$ a random variable and $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ convex s.t. $\mathbb{E}(\varphi(X)) < +\infty$. Then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi \circ X].$$

Remark 3.11. This is not true for general measures because it requires $\mathbb{P}(\Omega) = 1$ i.e. $\mathbb{P}$ has to be a probability measure. $\triangle$

Proof. Since φ is convex we have

$$\varphi(\alpha x + (1 - \alpha)y) \leq \alpha\varphi(x) + (1 - \alpha)\varphi(y)$$

for every $x, y \in \mathbb{R}$ and $\alpha \in [0, 1]$. If φ is convex then for every $x \in \mathbb{R}$ there exists $m_x \in \mathbb{R}$ s.t

$$\varphi(y) \geq \varphi(x) + m_x(y - x)$$

for every $y \in \mathbb{R}$. In particular we can choose $x = \mathbb{E}[X]$, $y = X$ and $m_x = m_{\mathbb{E}[X]}$. This implies $\varphi \circ X \geq \varphi(\mathbb{E}[X]) + m_{\mathbb{E}[X]}(X - \mathbb{E}[X])$. Taking the expectation on both sides preserves the inequality because $\mathbb{E}$ is monotone:

$$\mathbb{E}[\varphi \circ X] \geq \mathbb{E}[\varphi(\mathbb{E}[X]) + m_{\mathbb{E}[X]}(X - \mathbb{E}[X])] = \varphi(\mathbb{E}[X]) + m_{\mathbb{E}[X]} \underbrace{(\mathbb{E}[X] - \mathbb{E}[X])}_{=0}.$$

□

Remark 3.12. for $1 \leq p \leq q$ and $X \in L^p(\Omega, \mathcal{A}, \mathbb{P})$. Then $\|X\|_{L^q} \leq \|X\|_{L^p}$ hence $L^p \subseteq L^q$. $\triangle$

Proposition 3.13 (Generalized Chebistev's inequality). Let $\varphi : S \rightarrow [0, +\infty]$ measurable. Let $A \in \mathcal{S}$ where $\mathcal{S}$ is a σ -algebra on S , and let X be an S -valued random variable. Then

$$\inf_{y \in A} [\varphi(y)] \mathbb{P}(X \in A) \leq \int_{\Omega} (\varphi \circ X)(\omega) \mathbf{1}_{\{X \in A\}} d\mathbb{P}(\omega) =: \mathbb{E}[(\varphi \circ X) \cdot \mathbf{1}_{\{X \in A\}}] \leq \mathbb{E}[\varphi \circ X].$$

Proof. We have

$$\inf_{z \in A} [\varphi(z)] \mathbb{1}_{\{X \in A\}}(y) \leq \varphi(y) \mathbb{1}_{\{X \in A\}}(y) \leq \varphi(y).$$

Then we take the expectation on both sides ($\mathbb{E}$ is monotone)

$$\begin{aligned} \inf_{z \in A} [\varphi(z)] \cdot \mathbb{E}[\mathbb{1}_{\{X \in A\}}(y)] &\leq \mathbb{E}[\varphi(y) \mathbb{1}_{\{X \in A\}}(y)] \\ &= \int_{\Omega} (\varphi \circ X)(\omega) \mathbb{1}_{\{X \in A\}}(\omega) d\mathbb{P}(\omega) \\ &\leq \mathbb{E}[\varphi \circ X] \end{aligned}$$

where we used the monotonicity of $\mathbb{E}$ in the last step. $\square$

Corollary 3.14. 1. Markov's inequality: Let $X \in L^p(\Omega, \mathcal{A}, \mathbb{P})$ $p \geq 1$, $a > 0$ then

$$\mathbb{P}(|X| \geq a) \leq \frac{1}{a^p} \mathbb{E}[|X|^p].$$

2. Chebistev's inequality: Let $X \in L^2(\Omega, \mathcal{A}, \mathbb{P})$ and $a > 0$ then

$$\mathbb{P}(|X - \mathbb{E}(X)| \geq a) \leq \frac{1}{a^2} \underbrace{\mathbb{E}[|X - \mathbb{E}(X)|^2]}_{=\text{Var}(X)}.$$

Proof. 1. We reduce the case to the generalized Chebistev's Inequality. Let $\varphi(x) = |x|^p$. Using prop. 3.13 we have

$$\mathbb{P}(|X| \geq a) \leq \frac{\mathbb{E}[|X|^p]}{\inf\{|y|^p : y \in [a, \infty)\}} = \frac{\mathbb{E}[|X|^p]}{a^p}.$$

2. Let $\varphi(x) = (x - \mathbb{E}[X])^2$ in prop. 3.13:

$$\mathbb{P}(|X - \mathbb{E}(X)| \geq a) \leq \frac{\mathbb{E}[|X - \mathbb{E}(X)|^2]}{a^2} = \frac{\text{Var}(X)}{a^2},$$

which concludes the proof. $\square$

Another useful theorem from "Real Analysis" would be:

Theorem 3.15 (Change of variables). Let $(\Omega, \mathcal{A}, \mathbb{P})$ a probability space and $(S, \mathcal{S})$ a measurable space. Let $X : \Omega \rightarrow S$, $g : S \rightarrow \mathbb{R}$ two random variables. Then $g \circ X$ is a random variable. Let $\mu_X := \mathbb{P} \circ X^{-1}$ the distribution of X on $(S, \mathcal{S})$. Then

$$\int_{\Omega} |g| d\mu_X < \infty \Leftrightarrow \int_{\Omega} |g \circ X| d\mathbb{P} < \infty \quad (3.1)$$

and

$$\mu_X(g) = \mathbb{E}_{\mu_X}[g] := \int_S g d\mu_X = \int_{\Omega} (g \circ X) d\mathbb{P} =: \mathbb{E}_{\mathbb{P}}[g \circ X]$$

Remark 3.16. If $g(x) = x$ then

$$\mathbb{E}_{\mathbb{P}}[X] = \int_{\Omega} x \, d\mu_X(x).$$

△

Proof of theorem 3.15. Clearly $g \circ X$ is a random variable as it is the composition of two measurable functions.

1. Let $g = \mathbb{1}_B$ for $B \in \mathcal{S}$. For this function (3.1) is obvious. Furthermore

$$\mathbb{E}_{\mu_X}[\mathbb{1}_B] = \int_S \mathbb{1}_B \, d\mu_X = \mu_X(B) = \mathbb{P} \circ X^{-1}(B) = \mathbb{E}_{\mathbb{P}}[\mathbb{1}_B \cdot X]$$

so the proposition holds for indicator functions.

2. Let $g = \sum_{i=1}^n c_i \mathbb{1}_{B_i}$ with $B_i \in \mathcal{S}$ so we get the result for linear-combinations of indicator functions (which we call *simple functions*) by using the linearity of the integral and the expectation:

$$\mathbb{E}_{\mu_X} \left[\sum_{i=1}^n c_i \mathbb{1}_{B_i} \right] = \sum_{i=1}^n c_i \mathbb{E}_{\mu_X}[\mathbb{1}_{B_i}] = \sum_{i=1}^n c_i \mathbb{E}_{\mathbb{P}}[\mathbb{1}_{B_i} \cdot X] = \mathbb{E}_{\mathbb{P}} \left[\sum_{i=1}^n c_i (\mathbb{1}_{B_i} \cdot X) \right]$$

where we used point 1. Hence $\mathbb{E}_{\mu_X}[g] = \mathbb{E}_{\mathbb{P}}[g \cdot X]$.

3. Let g be a non-negative measurable function ($[0, \infty]$ -valued). Let

$$g_n = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \mathbb{1}_{\{\frac{k}{2^n} \leq g < \frac{k+1}{2^n}\}} + n \mathbb{1}_{\{g \geq n\}}.$$

By construction $g_n \rightarrow g$ as $n \rightarrow \infty$ pointwise and g_n is also pointwise increasing. On the other hand: $g_n \circ X \rightarrow g \circ X$ as $n \rightarrow \infty$ also pointwise and is also increasing. So we can apply the monotone convergence theorem and get $\mu_X(g_n) \rightarrow \mu_X(g)$ as $n \rightarrow \infty$. The same holds for the composition: $\mathbb{E}_{\mathbb{P}}[g_n \circ X] \rightarrow \mathbb{E}_{\mathbb{P}}[g \circ X]$ as $n \rightarrow \infty$ again using the monotone convergence theorem (theorem 3.8). Now we have

$$\mu_X(g_n) = \mathbb{E}_{\mu_X}[g_n] = \mathbb{E}_{\mathbb{P}}[g_n \circ X]$$

thanks to the steps before. Taking the limit on both sides yields

$$\mu_X(g) = \lim_{n \rightarrow \infty} \mu_X(g_n) = \lim_{n \rightarrow \infty} \mathbb{E}_{\mathbb{P}}[g_n \circ X] = \mathbb{E}_{\mathbb{P}}[g \circ X]$$

so we have the result for non-negative functions.

4. Let g be real valued and measurable ($\hat{\mathbb{R}} = [-\infty, \infty]$ -valued). We can write $g = g^+ - g^-$ where $g^+ = \max\{g, 0\}$ and $g^- = \max\{-g, 0\}$. These are both non-negative measurable functions so we can apply step 3 and use the linearity of the expectation yielding

$$\mathbb{E}_{\mathbb{P}}[g \circ X] = \mathbb{E}_{\mathbb{P}}[g^+ \circ X] - \mathbb{E}_{\mathbb{P}}[g^- \circ X] = \mathbb{E}_{\mu_X}[g^+] - \mathbb{E}_{\mu_X}[g^-].$$

□

4 Dynkin's lemma and independence

A σ -algebra is quite a restrictive concept which can be hard to use in proofs. A π -system or a Dynkin-system are more loosely defined but very essential in some circumstances.

4.1 Definitions, Dynkin's lemma and Independence of events and σ -algebras

First two definitions:

Definition 4.1. Fix a set Ω . Then a **Dynkin system** on Ω is a family $\mathcal{D} \subset \mathcal{P}(\Omega)$ of subsets of Ω satisfying

1. $\Omega \in \mathcal{D}$,
2. if $A \in \mathcal{D}$ then $A^C \in \mathcal{D}$ and
3. for any disjoint sequence of sets $\{A_n\}_{n \in \mathbb{N}} \subset \mathcal{D}$ we have $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{D}$.

Remark 4.2. This is very similar to a σ -algebra but the σ -algebra doesn't care about the sets being disjoint or not. Clearly every σ -algebra is a Dynkin-system but the converse doesn't hold. $\triangle$

Definition 4.3. Fix Ω a set. A family $\mathcal{C} \subset \mathcal{P}(\Omega)$ is a **π -system** if $\mathcal{C}$ is closed under intersection i.e. for every $A, B \in \mathcal{C}$ holds $A \cap B \in \mathcal{C}$.

We now can state Dynkin's lemma:

Lemma 4.4 (Dynkin's lemma). Let $\mathcal{D}$ be a Dynkin-system and $\mathcal{C}$ a π -system, both on a set Ω with $\mathcal{C} \subset \mathcal{D}$. Then $\sigma(\mathcal{C}) \subset \mathcal{D}$.

Proof. The proof is long and quite technical. The interested reader can find it in the literature. However we will see similar proofs. $\square$

Lemma 4.5. Let $(\Omega, \mathcal{A})$ be a measurable space and $\mathbb{P}, \mathbb{P}'$ two probability measures on said space. Then $\mathcal{D} = \{A \in \mathcal{A} : \mathbb{P}(A) = \mathbb{P}'(A)\}$ is a Dynkin system.

Proof. We will check the three points from the definition of the Dynkin-system:

1. $\mathbb{P}(\Omega) = 1 = \mathbb{P}'(\Omega)$ hence $\Omega \in \mathcal{D}$.
2. Let $A \in \mathcal{D}$. Then $\mathbb{P}(A) = \mathbb{P}'(A)$ and $\mathbb{P}(A^C) = 1 - \mathbb{P}(A) = 1 - \mathbb{P}'(A) = \mathbb{P}'(A^C)$.
3. Let $\{A_i\}_{i \in \mathbb{N}}$ be a sequence of disjoint sets s.t. $\mathbb{P}(A_i) = \mathbb{P}'(A_i)$ for every $i \in \mathbb{N}$. Using the σ -additivity of the probability measure we get

$$\mathbb{P}\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sum_{i \in \mathbb{N}} \mathbb{P}(A_i) = \sum_{i \in \mathbb{N}} \mathbb{P}'(A_i) = \mathbb{P}'\left(\bigcup_{i \in \mathbb{N}} A_i\right)$$

hence $\bigcup_{i \in \mathbb{N}} A_i \in \mathcal{D}$.

□

Lemma 4.6. let $\mathbb{P}$ and $\mathbb{P}'$ be two probability measures on $(\Omega, \mathcal{A})$, a measurable space. Let $\mathcal{C}$ be a π -system s.t. $\mathbb{P}(C) = \mathbb{P}'(C)$ for every $C \in \mathcal{C}$. Then $\mathbb{P}(A) = \mathbb{P}'(A)$ for every $A \in \sigma(\mathcal{D})$.

Proof. Using lemma 4.5 we get that $\mathcal{C}$ is a Dynkin-system. Using Dynkin's lemma we get the result. □

Example 4.7. In lemma 4.6 we really need $\mathcal{C}$ to be a π -system. If we take for example a generator of the σ -algebra, then the lemma fails. For example take $\Omega = \{1, 2, 3, 4\}$ and $\mathcal{E} = \{\{1, 2\}, \{2, 3\}\}$ which would be a generator of a σ -algebra over Ω . If we take $\mathbb{P} = \frac{1}{2}(\delta_1 + \delta_3)$ and $\mathbb{P}' = \frac{1}{2}(\delta_2 + \delta_4)$ where δ is the *Dirac measure* i.e

$$\delta_n : \mathcal{B}(\mathbb{R}) \rightarrow \{0, 1\}, \quad A \mapsto \mathbb{1}_A(n)$$

then $\sigma(\mathcal{E}) = \mathcal{P}(\Omega)$ the power set of Ω and $\mathbb{P}|_{\mathcal{E} \cup \{\Omega\}} = \mathbb{P}'|_{\mathcal{E} \cup \{\Omega\}}$. But $\mathbb{P} \neq \mathbb{P}'$ on $\mathcal{P}(\Omega)$. △

We can now prove that the Lebesgue-Stieltjes measure is unique:

Proof (continuation of the proof of theorem 2.22). Let μ_1 and μ_2 be two probability measures both satisfying the conditions stated in theorem 2.22. We already proved that such a probability measure exists so we are going to prove uniqueness. Thanks to lemma 4.5 we know that

$$\mathcal{D} := \{A \in \mathcal{B}(\mathbb{R}) : \mu_1(A) = \mu_2(A)\}$$

is a Dynkin system. The assumptions state that

$$\mu_1((-\infty, x]) = \mu_2((-\infty, x]) \quad \text{for every } x \in \mathbb{R}.$$

Note that

$$\mathcal{C} := \{(-\infty, a] : a \in \mathbb{R}\}$$

is a π -systems (the properties are easily checked) and a generator of $\mathcal{B}(\mathbb{R})$. Since

$$\mu_1|_{\mathcal{C}} = \mu_2|_{\mathcal{C}},$$

lemma 4.4 (Dynkin's lemma) implies that $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C}) \subseteq \mathcal{D}$ i.e. $\mu_1(A) = \mu_2(A)$ for every $A \in \mathcal{B}(\mathbb{R})$ hence the uniqueness of the described measure. □

In “Einführung in die Statistik” we have seen independence for sets. Now we will define independence for measures and σ -algebras. Recall the definition from “Einführung in die Statistik”:

Definition 4.8. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. $A, B \in \mathcal{A}$ are **independent** if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Remark 4.9. If $\mathbb{P}(B) > 0$ then $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$. So if A and B are independent then $\mathbb{P}(A|B) = \mathbb{P}(A)$. Note that the concept of independence always depends on the probability measure and is not a global concept i.e. two sets can be independent for one probability measure but not for another one. $\triangle$

Definition 4.10. Let $(\Omega, \mathcal{A})$ and $(S, \mathcal{S})$ be two measurable spaces. Let $X : \Omega \rightarrow S$ a random variable. Then the σ -algebra generated by the random variable X (denoted by $\sigma(X)$) is the smallest σ -algebra s.t. X is measurable. In particular:

$$\sigma(X) = \{X^{-1}(S) : S \in \mathcal{S}\}. \quad (4.1)$$

and for several random variables $X_1, \dots, X_n$,

$$\sigma(X_1, \dots, X_n) = \sigma\left(\bigcup_{i=1}^n \sigma(X_i)\right)$$

Remark 4.11. Since X is a random variable we know that X is a $\mathcal{A}$ -measurable which implies $\sigma(X) \subset \mathcal{A}$. $\triangle$

We will now extend the definition of independence to σ -algebras:

Definition 4.12. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. Let $\mathcal{G}_1, \dots, \mathcal{G}_n \subset \mathcal{A}$ all σ -algebras. We say that $\mathcal{G}_1, \dots, \mathcal{G}_n$ are **independent** if for every $A_1 \in \mathcal{G}_1, \dots, A_n \in \mathcal{G}_n$ we have $\mathbb{P}(A_1 \cap \dots \cap A_n) = \prod_{i=1}^n \mathbb{P}(A_i)$.

Remark 4.13. $A_1, \dots, A_n \in \mathcal{A}$ are independent if the σ -algebras $\{\{\emptyset, A_i, A_i^C, \Omega\}\}_{i=1}^n$ are independent. $\triangle$

Definition 4.14. The random variables $X_1, \dots, X_n$ on Ω are independent if the σ -algebras generated by X_i for $i = 1, \dots, n$ are independent.

This allows us to state a first theorem about the connection between independence of elements of a π -system and the σ -algebras generated by it:

Theorem 4.15. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. Let $\mathcal{C}_1, \dots, \mathcal{C}_n \subset \mathcal{A}$ a π -system with $\Omega \in \mathcal{C}_i$ for every $i = 1, \dots, n$. If $\{\mathcal{C}_i\}_{i=1}^n$ are independent, then $\{\sigma(\mathcal{C}_i)\}_{i=1}^n$ are independent.

Proof. Fix $C_2 \in \mathcal{C}_2, \dots, C_n \in \mathcal{C}_n$ and define

$$\mathcal{D}_1 := \left\{ A \in \mathcal{A} : \mathbb{P}(A \cap C_2 \cap \dots \cap C_n) = \mathbb{P}(A) \prod_{i=2}^n \mathbb{P}(C_i) \right\}.$$

By assumption we know that $C_1 \subset \mathcal{A}$ and therefore $\mathbb{P}(C_1 \cap \dots \cap C_n) = \prod_{i=1}^n \mathbb{P}(C_i)$ for every $C_1 \in \mathcal{C}_1$ which implies $\mathcal{C}_1 \subset \mathcal{D}_1$. $\mathcal{D}_1$ is a Dynkin system:

Indeed, the first two points ($\Omega \in \mathcal{D}_1$ and $A \in \mathcal{D}_1 \Rightarrow A^c \in \mathcal{D}_1$) clearly hold. For the third point ($\mathcal{D}_1$ is closed under disjoint union) let $\{A_n\}_{n \in \mathbb{N}} \subset \mathcal{D}_1$ be disjoint and define $A := \bigcup_{i \in \mathbb{N}} A_i$. Then

$$\begin{aligned} \mathbb{P} \left[A \cap \left(\bigcap_{j=2}^n C_j \right) \right] &= \mathbb{P} \left[\left(\bigcup_{i \in \mathbb{N}} A_i \right) \cap \left(\bigcap_{j=2}^n C_j \right) \right] = \mathbb{P} \left[\bigcup_{i \in \mathbb{N}} \left(A_i \cap \left(\bigcap_{j=2}^n C_j \right) \right) \right] \\ &= \sum_{i \in \mathbb{N}} \mathbb{P} \left[A_i \cap \left(\bigcap_{j=2}^n C_j \right) \right] = \sum_{i \in \mathbb{N}} \mathbb{P}(A_i) \prod_{j=2}^n \mathbb{P}(C_j) \\ &= \mathbb{P} \left(\bigcup_{i \in \mathbb{N}} A_i \right) \prod_{j=2}^n \mathbb{P}(C_j) = \mathbb{P}(A) \prod_{j=2}^n \mathbb{P}(C_j) \end{aligned}$$

implies $A \in \mathcal{D}_1$.

Thanks to the argumentation before we can apply lemma 4.4 (Dynkin's lemma) which yields $\sigma(\mathcal{C}_1) \subset \mathcal{D}_1$.

Then we iterate this argument:

$$\mathcal{D}_2 := \left\{ A \in \mathcal{A} : \mathbb{P}(D_1 \cap A \cap C_3 \cap \dots \cap C_n) = \mathbb{P}(D_1) \mathbb{P}(A) \prod_{i=3}^n \mathbb{P}(C_i) \forall D_1 \in \sigma(\mathcal{C}_1) \right\}$$

and show

1. $\mathcal{C}_2 \subset \mathcal{D}_2$
2. $\mathcal{D}_2$ is a Dynkin-system
3. Apply Lemma 4.4 implying $\sigma(\mathcal{C}_2) \subset \mathcal{D}_2$

and so on, n times up to

$$\mathcal{D}_n = \left\{ A \in \mathcal{A} : \mathbb{P}(D_1 \cap \dots \cap D_{n-1} \cap A) = \left(\prod_{i=1}^n \mathbb{P}(D_i) \right) \cdot \mathbb{P}(A), \forall D_i \in \sigma(\mathcal{C}_i) \right\}.$$

This gives us $\sigma(\mathcal{C}_1) \subset \mathcal{D}_1, \dots, \sigma(\mathcal{C}_n) \subset \mathcal{D}_n$ independent. □

4.2 Independence of random variables and product spaces

We now exploit the relation between independence and product spaces. Some of the properties are again Analysis reminders with focus on probability theory.

Theorem 4.16. Let X and Y be *independent* $\mathbb{R}$ -valued random variables on $(\Omega, \mathcal{A}, \mathbb{P})$ a probability space. Let μ_X and μ_Y be the distribution associated with X and Y respec-

tively. Then for $\varphi = [X, Y]$ we have $\mathbb{P} \circ \varphi^{-1} = \mu_X \otimes \mu_Y$. Furthermore for $h : \mathbb{R}^2 \rightarrow \mathbb{R}$, a measurable function,

$$\mathbb{E}[|h(X, Y)|] < \infty \Leftrightarrow h \text{ is } \mu_X \otimes \mu_Y \text{ measurable}$$

and

$$\mathbb{E}[h(X, Y)] = \int_{\mathbb{R}^2} h(x, y) d\mu_X(x) d\mu_Y(y).$$

Proof. We can prove this using Dynkin-systems by proving $\mathbb{P} \circ \varphi^{-1} = \mu_X \otimes \mu_Y$: First we define

$$\mathcal{C} = \{A_1 \times A_2 \in \mathcal{B}(\mathbb{R}^2) : A_1, A_2 \in \mathcal{B}(\mathbb{R})\}$$

a collection of rectangles in $\mathbb{R}^2$. Then $\mathcal{C}$ is clearly a π -system. Indeed, for every $A_1, A_2, B_1, B_2 \in \mathcal{B}(\mathbb{R})$,

$$(A_1 \times A_2) \cap (B_1 \times B_2) = (A_1 \cap B_1) \times (A_2 \cap B_2) \in \mathcal{C}$$

so $\mathcal{C}$ is closed under intersection. Let's take a set $A = A_1 \times A_2 \in \mathcal{C}$. Then

$$(\mu_X \otimes \mu_Y)(A) = \mu_X(A_1)\mu_Y(A_2)$$

by the properties of a product measure and

$$(\mathbb{P} \circ \varphi^{-1})(A) = \mathbb{P}((X, Y) \in A) = \mathbb{P}(X \in A_1, Y \in A_2) = \mathbb{P}(X \in A_1)\mathbb{P}(Y \in A_2)$$

because X and Y are independent. But by definition of the distribution we have $\mu_X(A_1) = \mathbb{P}(X \in A_1)$ and the same for Y . So we have for every $A = A_1 \times A_2 \in \mathcal{C}$ that $(\mathbb{P} \circ \varphi^{-1})(A) = (\mu_X \otimes \mu_Y)(A)$. By lemma 4.5 $(\mathbb{P} \circ \varphi^{-1})(C) = (\mu_X \otimes \mu_Y)(C)$ for every $C \in \sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R}^2)$. The fact that $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R}^2)$ follows from the construction of $\mathcal{C}$. $\square$

Remark 4.17. If the assumptions of Theorem 4.16 are satisfied then we can define

$$h : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad (x, y) \mapsto x \cdot y$$

so we have by Fubini,

$$\begin{aligned} \mathbb{E}[XY] &= \mathbb{E}[h(X, Y)] = \int_{\mathbb{R}^2} h(x, y) d\mu_X(x) d\mu_Y(y) \\ &= \int_{\mathbb{R}^2} xy d\mu_X(x) d\mu_Y(y) \stackrel{\text{Fubini}}{=} \int_{\mathbb{R}} \int_{\mathbb{R}} xy d\mu_X(x) d\mu_Y(y) \\ &= \left(\int_{\mathbb{R}} x d\mu_X(x) \right) \left(\int_{\mathbb{R}} y d\mu_Y(y) \right) = \mathbb{E}[X]\mathbb{E}[Y] \end{aligned}$$

if the two expectations are finite and similarly

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

$\triangle$

Remark 4.18. By Fubini's Theorem, if $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ is measurable then

$$\int_{\mathbb{R}^2} h d(\mu_X \otimes \mu_Y) = \int_{\mathbb{R}} \int_{\mathbb{R}} h(x, y) d\mu_X(x) d\mu_Y(y) = \int_{\mathbb{R}} \int_{\mathbb{R}} h(x, y) d\mu_Y(y) d\mu_X(x).$$

$\triangle$

Proposition 4.19. Let X and Y two independent random variables (on the probability space $(\Omega, \mathcal{A}, \mathbb{P})$) with distribution functions F_X and F_Y respectively. Let $S := X + Y$. Then

1. we have

$$F_S(z) = \mathbb{P}(S \leq z) = \mathbb{P}(X + Y \leq z) = \int_{\Omega} F_X(z - y) \underbrace{d\mu_Y(y)}_{:= dF_Y(y)}.$$

2. If X is a random variable with density f_X then S has density

$$f_S(z) = \int_{\Omega} f_X(z - y) d\mu_X(y).$$

Proof. 1. We can choose $h(x, y) = \mathbb{1}_{\{(x, y) \in \mathbb{R}^2 : x + y \leq z\}}(x, y)$ as in Theorem 4.16 and get

$$\begin{aligned} \int_{\mathbb{R}} h(x, y) d\mu_X(x) &= \int_{\mathbb{R}} \mathbb{1}_{\{(x, y) \in \mathbb{R}^2 : x + y \leq z\}} d\mu_X(x) = \int_{\mathbb{R}} \mathbb{1}_{\{(x, y) \in \mathbb{R}^2 : x \leq z - y\}} d\mu_X(x) \\ &= \mu_X((-\infty, z - y]) = F_X(z - y) \end{aligned}$$

Then using Theorem 4.16, Fubini and change of variables we have

$$\begin{aligned} F_S(z) &= \mathbb{P}(X + Y \leq z) = \int_{\Omega} h(X(\omega), Y(\omega)) d\mathbb{P}(\omega) \\ &= \int_{\Omega} h(x, y) d(\mathbb{P} \circ h^{-1})(x, y) = \int_{\mathbb{R}^2} h d(\mu_X \otimes \mu_Y) \\ &= \int_{\mathbb{R}} \underbrace{\int_{\mathbb{R}} h(x, y) d\mu_X(x)}_{= F_X(z - y)} d\mu_Y(y) \end{aligned}$$

which proves the first point.

2. Recall the definition of the density: if f_X satisfies $\mu_X(A) = \int_A f_X d\mu_X$ then f_X is called the *density* of X . Using the formula of point 1. we have

$$F_S(x) = \int_{\Omega} F_X(x - y) d\mu_Y(y)$$

and using the definition of the density this leads to

$$\int_{\Omega} F_X(x - y) d\mu_Y(y) = \int_{\Omega} \int_{-\infty}^{x-y} f_X(z) dz d\mu_Y(y).$$

Performing a change of variables yields

$$\begin{aligned} \int_{\Omega} \int_{-\infty}^{x-y} f_X(z) dz d\mu_Y(y) &= \int_{\Omega} \int_{-\infty}^x f_X(z - y) dz d\mu_Y(y) \\ &= \int_{-\infty}^x \underbrace{\int_{\Omega} f_X(z - y) d\mu_Y(y)}_{= f_S(z)} dz \end{aligned}$$

so

$$F_S(x) = \int_{-\infty}^x \underbrace{\int_{\mathbb{R}} f_X(z-y) d\mu_Y(y)}_{=f_S(z)} dz,$$

hence

$$f_S(x) = \int_{\mathbb{R}} f(x-y) d\mu_Y(y)$$

is the density function of S which concludes the proof. □

Remark 4.20. If Y in Prop. 4.19.2 has density f_Y ($d\mu_Y = f_Y(y) dy$) then

$$f_S(x) = \int_{\Omega} f_X(x-y) f_Y(y) dy = (f_X \star f_Y)(x)$$

so the density is just the *convolution* of f_X and f_Y . △

4.3 Infinite product spaces

What if we take the cartesian product of infinitely many sets? And can we define a measure on this? The answer is yes!

Definition 4.21. Fix I a set of infinite (not necessarily countable) indices (we often call this an **infite index set**). Let

- $\Omega = \times_{i \in I} \Omega_i$ the cartesian product of the sets $\{\Omega_i\}_{i \in I}$ and
- $\bigotimes_{i \in I} \mathcal{A}_i$ the product σ -algebra of the σ -algebras $\mathcal{A}_i$'s
- $\mathbb{P} = \bigotimes_{i \in I} \mathbb{P}_i$ the probability measure on $(\Omega, \mathcal{A})$.

$\mathbb{P}$ is the **unique** probability measure on $(\Omega, \mathcal{A})$ s.t.

$$\mathbb{P} \left(\left(\times_{i \in J} A_i \right) \times \left(\times_{i \in I \setminus J} \Omega_i \right) \right) = \prod_{i \in J} \mathbb{P}_i(A_i)$$

for $A_i \in \mathcal{A}_i$. Furthermore $(\Omega, \mathcal{A}, \mathbb{P})$ form a probability space.

Remark 4.22. We started with $(\Omega_i, \mathcal{A}_i, \mathbb{P}_i)$, $i \in I$. Then we constructed $(\Omega, \mathcal{A}, \mathbb{P})$. This space is called the probability space of products of $(\Omega_i, \mathcal{A}_i, \mathbb{P}_i)$.

$$\mathcal{A} = \sigma \left(\left\{ \left(\times_{i \in J} A_i \right) \times \left(\times_{i \in I \setminus J} \Omega_i \right) : J \subset I \text{ finite } \forall A_i \in \mathcal{A}_i, \forall i \in I \right\} \right)$$

is the smallest σ -algebra on Ω containing all cylinders of finite size. However we still need to prove that the measure $\mathbb{P}$ in definition. To show that one can again apply lemma 4.5 (so this is a further application of the Dynkin lemma). △

Remark 4.23. How to construct a sequence of random variables? Take $(S, \mathcal{S}, \mu)$ a probability space with $\Omega := S^{\mathbb{N}}$, $\mathcal{A} := \mathcal{S}^{\otimes \mathbb{N}} = \mathcal{S} \otimes \mathcal{S} \otimes \cdots$ and $\mathbb{P} := \mu^{\otimes \mathbb{N}}$. Then $(\Omega, \mathcal{A}, \mathbb{P})$ is a probability space. Consider the projection on the i -th component of an element $\omega = \{\omega_i\}_{i=1}^{\infty} \in \Omega$ with $X_i : \omega = \{\omega_i\}_{i \in \mathbb{N}} \in \Omega \mapsto \omega_i \in S$ for every $i \in \mathbb{N}$. This function is clearly measurable hence X_i is a random variable. By definition 4.21 and remark 4.22

$$\mathbb{P}(X_i \in A_i, i \in J) = \prod_{i \in J} \mathbb{P}(A_i)$$

for $J \subset \mathbb{N}$ finite. In particular $\mathbb{P}(X_i \in A_i) = \mathbb{P}(A_i) = \mu(A_i)$ for every $i \in J$. It follows that X_i are indentially distributed i.e. they all have the distribution μ and X_i 's are independent random variables. So we constructed a sequence of random variables $\{X_i\}_{i \in \mathbb{N}}$ independent identically distributed (i.i.d.). $\triangle$

5 Convergence of random variables

We now enter the heart of the course where we look at the asymptotic behaviour of sequences of random variables.

5.1 Basics of convergence

Fix X and $\{X_n\}_{n \in \mathbb{N}}$ a sequence of random variables on $(\Omega, \mathcal{A}, \mathbb{P})$.

Definition 5.1 (Convergence in probability). A sequence of random variables $\{X_n\}_{n \in \mathbb{N}}$ converges in probability (written $X_n \xrightarrow{\mathbb{P}} X$ as $n \rightarrow \infty$) if

$$\lim_{n \rightarrow \infty} \mathbb{P}(\{|X_n - X| > \varepsilon\}) = 0$$

for every $\varepsilon > 0$.

Remark 5.2. This is the same as convergence in measure from “Real Analysis” $\triangle$

Definition 5.3 (Almost sure convergence). Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of random variables. We say that X_n **converges almost surely** to X (written $X_n \xrightarrow{\text{a.s.}} X$ as $n \rightarrow \infty$ or $X_n \xrightarrow{\mathbb{P}\text{-a.s.}} X$) if

$$\mathbb{P}(\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}) = 1.$$

Remark 5.4. This would be the same as convergence almost everywhere from “Real Analysis”. $\triangle$

Definition 5.5 (L^p convergence). For $p \in [1, \infty]$, $\{X_n\}_{n \in \mathbb{N}}$ a sequence of random variable converges to X in L^p (written $X_n \xrightarrow{L^p} X$) if

$$\lim_{n \rightarrow \infty} \|X_n - X\|_{L^p} = \lim_{n \rightarrow \infty} \mathbb{E}_{\mathbb{P}}[|X_n - X|^p]^{1/p} = 0$$

The following theorem will compare some of these definitions.

Theorem 5.6. Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of random variables on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. Let $1 \leq p \leq q \leq \infty$. Then

1. if $X_n \xrightarrow{\text{a.s.}} X$ then $X_n \xrightarrow{\mathbb{P}} X$ for $n \rightarrow \infty$.

2. If $X_n \xrightarrow{L^p} X$ then $X_n \xrightarrow{\mathbb{P}} X$ and $X_n \xrightarrow{L^q} X$ for $n \rightarrow \infty$.

Proof. 1. Assume $\mathbb{P}(\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\}) = 1$. This implies that

$$\mathbb{1}_{\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\}} \xrightarrow{\text{a.s.}} 0, \quad n \rightarrow \infty.$$

But we know that

$$\mathbb{P}(\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\}) = \mathbb{E}[\mathbb{1}_{\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\}}].$$

Using the Lebesgue dominated convergence theorem (because the function $f(x) \equiv 1$ dominates the indicator function a.e) we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{P}(\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\}) &= \lim_{n \rightarrow \infty} \mathbb{E}[\mathbb{1}_{\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\}}] \\ &= \mathbb{E}[\lim_{n \rightarrow \infty} \mathbb{1}_{\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\}}] \\ &= \mathbb{E}[0] = 0 \end{aligned}$$

so this implies $X_n \xrightarrow{\mathbb{P}} X$.

2. For this we are going to use Markov's inequality (Corollary 3.14). From Markov we get that

$$\mathbb{P}(\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\}) \leq \frac{\|X_n - X\|_{L^p}^p}{\varepsilon} \rightarrow 0$$

so the right hand side gets dominated by something that goes to zero hence

$$\mathbb{P}(\{|X_n - X| > \varepsilon\}) \xrightarrow{n \rightarrow \infty} 0$$

for every $\varepsilon > 0$.

For the second part we can use Hölder's inequality:

$$\begin{aligned} \|X_n - X\|_{L^q}^q &= \int_{\Omega} 1 \cdot |X_n - X|^q d\mathbb{P} \leq \underbrace{\left(\int_{\Omega} d\mathbb{P} \right)^{\frac{p-q}{p}}}_{=1} \left(\int_{\Omega} |X_n - X|^p d\mathbb{P} \right)^{\frac{q}{p}} \\ &\leq \|X_n - X\|_{L^p}^q \xrightarrow{n \rightarrow \infty} 0, \end{aligned}$$

which proves the result. □

Theorem 5.7 (Weak law of large numbers). Let $\{X_n\}_{n \in \mathbb{N}}$ a sequence of identically distributed random variables. Assume $\mathbb{E}[X_n^2] < \infty$ and $\text{Cov}(X_i, X_j) \leq C_{|i-j|}$ for every $i, j \in \mathbb{N}$ with $C_n \downarrow 0$ as $n \rightarrow \infty$. Let $S_n := \sum_{i=1}^n X_i$. Then

$$\frac{S_n}{n} \xrightarrow{L^2} \mathbb{E}[X_i] \quad \text{as } n \rightarrow \infty.$$

Corollary 5.8. Because of Theorem 5.6 Theorem 5.7 implies that

$$\frac{S_n}{n} \xrightarrow{\mathbb{P}} \mathbb{E}[X_i].$$

Proof of theorem 5.7. We will use the so-called 2nd moment method. We want to show that $\left\| \frac{S_n}{n} - \mathbb{E}[X_i] \right\|_{L^2} \rightarrow 0$ as $n \rightarrow \infty$. We have

$$\begin{aligned} \left\| \frac{S_n}{n} - \mathbb{E}[X_i] \right\|_{L^2}^2 &= \mathbb{E} \left[\left(\frac{S_n}{n} - \mathbb{E}[X_i] \right)^2 \right] = \mathbb{E} \left[\frac{1}{n^2} \left(\sum_{i=1}^n (X_i - \mathbb{E}[X_i]) \right)^2 \right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \mathbb{E}[(X_i - \mathbb{E}[X_i]) \cdot (X_j - \mathbb{E}[X_j])] \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j) \leq \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n C_{[i-j]} \leq \frac{n}{n^2} \sum_{k=0}^n C_k \xrightarrow{n \rightarrow \infty} 0. \end{aligned}$$

Comment: Prove that $\frac{1}{n} \sum_{k=0}^n C_k \xrightarrow{n \rightarrow \infty} 0$ if $C_n \xrightarrow{n \rightarrow \infty} 0$: Truncate the sum at $\varepsilon(n)$ s.t. $\frac{\varepsilon(n)}{n} \rightarrow 0$ as $n \rightarrow \infty$ (e.g. choose $\varepsilon(n) = \log(n)$ or $\varepsilon(n) = \sqrt{n}$). Then

$$\begin{aligned} \frac{1}{n} \sum_{k=0}^n C_k &= \frac{1}{n} \sum_{k=0}^{\lfloor \varepsilon(n) \rfloor} C_k + \frac{1}{n} \sum_{k=\lfloor \varepsilon(n) \rfloor + 1}^n C_k \\ &\leq \underbrace{\frac{\varepsilon(n)}{n} \max_{k=0, \dots, \lfloor \varepsilon(n) \rfloor} C_k}_{\rightarrow 0} + \underbrace{\frac{1}{n} \cdot \max_{k=\lfloor \varepsilon(n) \rfloor + 1, \dots, n} C_k}_{\rightarrow 0}. \end{aligned}$$

This concludes the proof that $\frac{1}{n} S_n \rightarrow \mathbb{E}[X_i]$ in L^2 as $n \rightarrow \infty$. The convergence in probability follows from that thanks to theorem 5.6. $\square$

5.2 Almost sure convergence

Recall the definition of lim sup and lim inf for sets:

Definition 5.9. Let $\{A_n\}_{n \in \mathbb{N}}$ a sequence on $(\Omega, \mathcal{A})$. We define

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{n \in \mathbb{N}} \bigcup_{k \geq n} A_k = \{\omega \in \Omega : \omega \text{ is contained in infinitely many } A_k\}$$

and

$$\liminf_{n \rightarrow \infty} A_n = \bigcup_{n \in \mathbb{N}} \bigcap_{k \geq n} A_k = \{\omega \in \Omega : \omega \text{ is not contained finitely many } A_k\}$$

Remark 5.10. Definition 5.9 implies $\mathbb{1}_{\limsup_{n \rightarrow \infty} A_n} = \limsup_{n \rightarrow \infty} \mathbb{1}_{A_n}$ and $\mathbb{1}_{\liminf_{n \rightarrow \infty} A_n} = \liminf_{n \rightarrow \infty} \mathbb{1}_{A_n}$. $\triangle$

Lemma 5.11 (Borel-Cantelli 1). Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence of sets in $\mathcal{A}$ s.t. $\sum_{n \in \mathbb{N}} \mathbb{P}(A_n) < \infty$ then

$$\mathbb{P}(\limsup_{n \rightarrow \infty} A_n) = 0.$$

Proof. The proof relies a lot on the definition of $\liminf$ and $\limsup$. We start by looking at

$$\mathbb{E} \left[\sum_{n \in \mathbb{N}} \mathbf{1}_{A_n} \right] = \sum_{n \in \mathbb{N}} \mathbb{E}[\mathbf{1}_{A_n}],$$

where we used Beppo-Levi to exchange sum and expectation. Now we use the fact that $\mathbb{E}[\mathbf{1}_{A_n}] = \mathbb{P}(A_n)$ and get

$$\mathbb{E} \left[\sum_{n \in \mathbb{N}} \mathbf{1}_{A_n} \right] = \sum_{n \in \mathbb{N}} \mathbb{P}(A_n) < \infty$$

by assumption. This implies that $\sum_{n \in \mathbb{N}} \mathbf{1}_{A_n} < \infty$ almost surely and

$$\mathbb{P}(\limsup_{n \rightarrow \infty} A_n) = 0$$

because otherwise the sum $\sum_{n \in \mathbb{N}} \mathbb{P}(A_n)$ would be infinite (A_n cannot occur infinitely often). $\square$

Remark 5.12. We can also prove this by the following computation: For $M \in \mathbb{N}$,

$$\mathbb{P}(\limsup_{n \rightarrow \infty} A_n) = \mathbb{P} \left(\bigcap_{n \in \mathbb{N}} \bigcup_{k \geq n} A_k \right) \leq \mathbb{P} \left(\bigcup_{n=M}^{\infty} A_n \right) \leq \sum_{n=M}^{\infty} \mathbb{P}(A_n) \xrightarrow{M \rightarrow \infty} 0$$

because $\sum_n \mathbb{P}(A_n)$ converges. $\triangle$

Remark 5.13. Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of random variables with $\sum_{n \in \mathbb{N}} \mathbb{P}(|X_n| \geq \varepsilon) < \infty$ for every $\varepsilon > 0$ then the Borel-Cantelli-Lemma (Lemma 5.11) implies

$$\mathbb{P}(\limsup_{n \rightarrow \infty} \{|X_n| \geq \varepsilon\}) = 0 \quad \text{for every } \varepsilon > 0,$$

hence

$$\begin{aligned} \mathbb{P}(\limsup_{n \rightarrow \infty} |X_n| > 0) &= \mathbb{P}(\limsup_{n \rightarrow \infty} \{|X_n| > 0\}) = \mathbb{P} \left(\bigcup_{k \in \mathbb{N}} \limsup_{n \rightarrow \infty} \left\{ |X_n| \geq \frac{1}{k} \right\} \right) \\ &\leq \sum_{k \in \mathbb{N}} \underbrace{\mathbb{P} \left(\limsup_{n \rightarrow \infty} \left\{ |X_n| \geq \frac{1}{k} \right\} \right)}_{=0} = 0, \end{aligned}$$

so

$$\lim_{n \rightarrow \infty} X_n = 0 \quad \mathbb{P}\text{-a.s.}$$

$\triangle$

Theorem 5.14 (Strong law of large numbers). Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of identically distributed and independent (i.i.d) random variables on $(\Omega, \mathcal{A}, \mathbb{P})$, a probability space with $\mathbb{E}[X_n] = m$ for every $n \in \mathbb{N}$. Furthermore $\mathbb{E}[X_n^4] < \infty$ for every $n \in \mathbb{N}$. If $S_n = \sum_{i=1}^n X_i$ then

$$\frac{1}{n} S_n \xrightarrow{\text{a.s.}} m, \quad \text{as } n \rightarrow \infty.$$

Proof. Define $\bar{X}_n := X_n - \mathbb{E}[X_n] = X_n - m$. Considering this new random variable

$$\mathbb{E}[\bar{X}_n] = \mathbb{E}[X_n] - m = m - m = 0$$

we can without loss of generality assume that $m = 0$.

$$\begin{aligned} \mathbb{E}[S_n^4] &= \mathbb{E} \left[\sum_{i=1}^n X_i \sum_{j=1}^n X_j \sum_{k=1}^n X_k \sum_{l=1}^n X_l \right] \\ &= \sum_{i,j,k,l=1}^n \mathbb{E}[X_i X_j X_k X_l] + \binom{4}{2} \sum_{i,j,k=1}^n \mathbb{E}[X_i X_j X_k^2] \\ &\quad + \binom{4}{1} \sum_{i,j=1}^n \mathbb{E}[X_i X_j^3] + \binom{4}{2} \sum_{i,j=1}^n \mathbb{E}[X_i^2 X_j^2] + \sum_{i=1}^n \mathbb{E}[X_i^4]. \end{aligned}$$

Using the independence of those random variables and the fact that we have chosen $\mathbb{E}[X_i] = 0$ we obtain

$$\begin{aligned} \sum_{i,j,k,l=1}^n \mathbb{E}[X_i X_j X_k X_l] &= \sum_{i,j,k,l=1}^n \mathbb{E}[X_i] \mathbb{E}[X_j] \mathbb{E}[X_k] \mathbb{E}[X_l] = 0, \\ \sum_{i,j,k=1}^n \mathbb{E}[X_i X_j X_k^2] &= \sum_{i,j,k=1}^n \mathbb{E}[X_i] \mathbb{E}[X_j] \mathbb{E}[X_k^2] = 0 \end{aligned}$$

and

$$\sum_{i,j=1}^n \mathbb{E}[X_i X_j^3] = \sum_{i,j=1}^n \mathbb{E}[X_i] \mathbb{E}[X_j^3] = 0.$$

We are left with

$$\sum_{i,j=1}^n \mathbb{E}[X_i^2 X_j^2] = \sum_{i,j=1}^n \mathbb{E}[X_i^2] \mathbb{E}[X_j^2]$$

and

$$\sum_{i=1}^n \mathbb{E}[X_i^4] < \infty \quad \text{and} \quad \mathbb{E}[X_i^2] < \infty,$$

thus

$$\mathbb{E}[S_n^4] \leq 6n^2 \mathbb{E}[X_1^2]^2 + n \mathbb{E}[X_1^4] = \mathcal{O}(n^2)$$

for every $i \in \{1, \dots, n\}$. Markov's Inequality and remark 5.13 imply

$$\mathbb{P} \left(\left| \frac{S_n}{n} \right| \geq \varepsilon \right) = \mathbb{P}(S_n^4 \geq \varepsilon^4 n^4) \leq \frac{\mathbb{E}[S_n^4]}{\varepsilon^4 n^4} \leq \frac{C}{\varepsilon^4 n^2}$$

for every $\varepsilon > 0$. Taking the sum

$$\sum_{n \in \mathbb{N}} \frac{C}{\varepsilon^4 n^2} < \infty,$$

we apply the argument used in remark 5.13 to conclude that $\frac{1}{n} S_n \xrightarrow{\text{a.s.}} 0$ as $n \rightarrow \infty$ i.e.

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} \frac{1}{n} S_n = 0 \right) = 1.$$

□

Notation 5.15. We use the notation

$$\begin{aligned} \sum_{i,j,k,l}^n \mathbb{E}[X_i X_j X_k X_l] &= \sum_{i,j,k,l=1}^n (\mathbb{E}[X_i X_j X_k X_l] \cdot \delta_{i,j}^C \cdot \delta_{i,k}^C \cdot \delta_{i,l}^C \cdot \delta_{j,k}^C \cdot \delta_{j,l}^C \cdot \delta_{k,l}^C) \\ &= \sum_{i=1}^n \sum_{j=1, j \neq i}^n \sum_{k=1, k \notin \{i,j\}}^n \sum_{l=1, l \notin \{i,j,k\}}^n \mathbb{E}[X_i X_j X_k X_l], \end{aligned}$$

with the convention $\delta_{i,j}^C = (1 - \delta_{i,j})$ with δ the Kronecker Delta.

This is basically a sum that doesn't repeat indices.

Lemma 5.16 (Borel-Cantelli 2). Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence of independent events in $\mathcal{A}$. Then if $\sum_{n \in \mathbb{N}} \mathbb{P}(A_n) = \infty$ then

$$\mathbb{P}(\limsup_{n \rightarrow \infty} A_n) = 1. \quad (5.1)$$

Proof. We will show

$$\mathbb{P}((\limsup_{n \rightarrow \infty} A_n)^C) = 0 \quad (5.2)$$

which is equivalent to (5.1). First note that

$$\mathbb{P}((\limsup_{n \rightarrow \infty} A_n)^C) = \mathbb{P} \left(\left(\bigcap_{n \in \mathbb{N}} \bigcup_{k \geq n} A_k \right)^C \right) = \mathbb{P} \left(\bigcup_{n \in \mathbb{N}} \bigcap_{k \geq n} A_k^C \right) = \mathbb{P}(\liminf_{n \rightarrow \infty} A_k^C).$$

Then with $1 - x \leq e^{-x}$ for every $x \in \mathbb{R}$ follows that

$$\begin{aligned} \mathbb{P} \left(\bigcap_{n=m}^{\infty} A_n^C \right) &\stackrel{(!)}{\leq} \mathbb{P} \left(\bigcap_{n=m}^M A_n^C \right) \stackrel{\text{independence}}{=} \prod_{n=m}^M (1 - \mathbb{P}(A_n)) \\ &\leq \prod_{n=m}^M e^{-\mathbb{P}(A_n)} = \exp \left(- \sum_{n=m}^M \mathbb{P}(A_n) \right) \xrightarrow{M \rightarrow \infty} 0 \quad \forall m \in \mathbb{N} \end{aligned}$$

by assumption (at (!) we used the monotonicity of $\mathbb{P}$). So $\mathbb{P}(\bigcap_{n \geq m} A_n^C) = 0$ for every $m \in \mathbb{N}$ hence

$$\mathbb{P}(\liminf_{n \rightarrow \infty} A_n^C) \leq \underbrace{\sum_{m=1}^{\infty} \mathbb{P} \left(\bigcap_{n \geq m} A_n^C \right)}_{=0} = 0$$

which proves (5.2). □

Remember: We have shown that for $\mathcal{C}_1, \dots, \mathcal{C}_n$ π -systems containing Ω and independent, then $\sigma(\mathcal{C}_1), \dots, \sigma(\mathcal{C}_n)$ are independent.

We will now see an application of the second Borel-Cantelli lemma (the proposition itself is not that important but the proof is a good exercise on the Borel-Cantelli Lemma):

Proposition 5.17. Let $\{X_n\}_{n \in \mathbb{N}}$ a sequence of i.i.d random variables s.t. $X_n \sim \mathcal{N}(0, 1)$ (normally distributed) for every $n \in \mathbb{N}$. Then

$$\limsup_{n \rightarrow \infty} \frac{X_n}{\sqrt{2 \log(n)}} = 1 \quad \mathbb{P}\text{-a.s.} \quad (5.3)$$

Proof. Claim: If $X \sim \mathcal{N}(0, 1)$ then for every $x > 0$,

$$\frac{1}{\sqrt{2\pi}} \left(x + \frac{1}{x}\right)^{-1} e^{-x^2/2} \leq \mathbb{P}(X \geq x) = \int_x^\infty \frac{e^{-y^2/2}}{\sqrt{2\pi}} dy \leq \frac{e^{-x^2/2}}{x\sqrt{2\pi}}.$$

To prove this claim we first look at the lower bound

$$\frac{1}{\sqrt{2\pi}} \left(x + \frac{1}{x}\right)^{-1} e^{-x^2/2} = \frac{1}{\sqrt{2\pi} \cdot x} \left(1 + \frac{1}{x^2}\right)^{-1} e^{-x^2/2}.$$

Note that

$$\frac{d}{dy} \left(\frac{1}{y} e^{-y^2/2} \right) = - \left(1 + \frac{1}{y^2} \right) e^{-y^2/2} \quad (5.4)$$

so

$$\int_x^\infty \frac{d}{dy} \left(\frac{1}{y} e^{-y^2/2} \right) dy = \left(\frac{1}{x} e^{-x^2/2} \right).$$

Integrating the RHS of (5.4) yields

$$\left(\frac{1}{x} e^{-x^2/2} \right) = \int_x^\infty \frac{d}{dy} \left(\frac{1}{y} e^{-y^2/2} \right) dy = - \int_x^\infty \left(1 + \frac{1}{y^2} \right) e^{-y^2/2} dy$$

which implies

$$\frac{1}{x} e^{-x^2/2} = \int_x^\infty \left(1 + \frac{1}{y^2} \right) e^{-y^2/2} dy \leq \left(1 + \frac{1}{x^2} \right) \int_x^\infty e^{-y^2/2} dy$$

implying the lower bound. For the upper bound consider

$$\int_x^\infty e^{-y^2/2} dy = \int_x^\infty \frac{y}{y} e^{-y^2/2} dy \leq \int_x^\infty \frac{y}{x} e^{-y^2/2} dy = \frac{1}{x} e^{-x^2/2}$$

so we have the upper bound by multiplying with $1/\sqrt{2\pi}$. Now we can prove the initial statement. To prove (5.3) we first look for an upper bound. To do this we will apply lemma 5.11 (Borel-Cantelli-1). Then we fix $\varepsilon > 0$ and define

$$A_n := \left\{ X_n \geq (1 + \varepsilon) \sqrt{2 \log(n)} \right\}, \quad n \in \mathbb{N}.$$

Now we compute $\mathbb{P}(A_n)$ by first using the upper bound

$$\mathbb{P}(A_n) \leq \frac{e^{-(1+\varepsilon)^2 \log(n)}}{\sqrt{2\pi}(1+\varepsilon)\sqrt{2\log(n)}} = \frac{1}{\sqrt{2\pi}(1+\varepsilon)\sqrt{2\log(n)}n^{(1+\varepsilon)^2}},$$

so

$$\sum_{n \in \mathbb{N}} \mathbb{P}(A_n) \leq \frac{1}{\sqrt{2\pi}(1+\varepsilon)} \sum_{n \in \mathbb{N}} \frac{1}{\sqrt{2\log(n)}n^{(1+\varepsilon)^2}} < \infty.$$

This allows us to use lemma 5.11 obtaining

$$\mathbb{P}(\limsup_{n \rightarrow \infty} A_n) = 0,$$

hence for every $n \in \mathbb{N}$, $X_n \leq (1+\varepsilon)\sqrt{2\log(n)}$ $\mathbb{P}$ -a.s. This implies

$$\limsup_{n \rightarrow \infty} \frac{X_n}{\sqrt{2\log(n)}} \leq (1+\varepsilon) \quad \mathbb{P}\text{-a.s.}$$

for every $\varepsilon > 0$. For the lower bound let $\varepsilon \in (0, 1)$ and define

$$B_n := \left\{ X_n \geq (1-\varepsilon)\sqrt{2\log(n)} \right\}.$$

Using the lower bound on $\mathbb{P}(X_n \geq x)$ we can apply the earlier calculations with $x = (1-\varepsilon)\sqrt{2\log(n)}$

$$\mathbb{P}(B_n) \geq \frac{1}{\sqrt{2\pi}} \left((1-\varepsilon)\sqrt{2\log(n)} + \frac{1}{(1-\varepsilon)\sqrt{2\log(n)}} \right)^{-1} e^{-(1-\varepsilon)^2 \log(n)} \geq \frac{1}{n^a}$$

for $a \in \{(1-\varepsilon)^2, 1\}$ and $n \geq \bar{n}(a, \varepsilon)$. Summing over all $n \in \mathbb{N}$ yields

$$\sum_{n \in \mathbb{N}} \mathbb{P}(B_n) \geq \sum_{n \in \mathbb{N}} \frac{1}{n^a} = \infty.$$

So we can use lemma 5.16 (the B_n are independent) to see that

$$\mathbb{P}(\limsup_{n \rightarrow \infty} B_n) = 1.$$

In particular this implies that

$$\limsup_{n \rightarrow \infty} X_n \geq (1-\varepsilon)\sqrt{2\log(n)} \quad \mathbb{P}\text{-a.s.}$$

for every $\varepsilon > 0$ hence

$$\limsup_{n \rightarrow \infty} X_n \geq \sqrt{2\log(n)}, \quad \mathbb{P}\text{-a.s.}$$

Together with the estimations for the upper bound we get

$$\limsup_{n \rightarrow \infty} \frac{X_n}{\sqrt{2\log(n)}} = 1 \quad \mathbb{P}\text{-a.s.}$$

□

5.3 Kolomogorov's law 0-1

A further application of the second Borel-Cantelli lemma would be Kolmogorov's law 0-1. This is another method (other than the 2nd moment method) to prove a.s. convergence of random variables. However we need some new structures like the tail σ -algebra

Definition 5.18 (Tail σ -algebra). Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of random variables on $(\Omega, \mathcal{A}, \mathbb{P})$. For $n \in \mathbb{N}$ we define the σ -algebra $\mathcal{T}_n$ s.t.

$$\mathcal{T}_n = \sigma(X_n, X_{n+1}, \dots).$$

By definition this is the smallest σ -algebra containing the σ -algebras generated by X_k for $k \geq n$. Notice that for $p > n$ we have $\mathcal{T}_p \subset \mathcal{T}_n$ and furthermore $\sigma(X_n, \dots, X_p) \subset \mathcal{T}_n$. We call a **tail σ -algebra** the intersection over all $\mathcal{T}_n$ i.e.

$$\mathcal{T} := \bigcap_{n \in \mathbb{N}} \mathcal{T}_n.$$

Remark 5.19. $\mathcal{T}$ defined as before contains the so-called convergence set of $\sum_{n \in \mathbb{N}} X_n = S$. Indeed if we look at the convergent set of the series S

$$\Omega_1 := \left\{ \omega \in \Omega : \limsup_{p \rightarrow \infty} \sum_{n=1}^p X_n(\omega) = \liminf_{p \rightarrow \infty} \sum_{n=1}^p X_n(\omega) \right\}$$

(note that this expression also can take on the value of ∞). Then for every $k \geq 1$

$$\Omega_1 = \left\{ \omega \in \Omega : \limsup_{p \rightarrow \infty} \sum_{n=k}^p X_n(\omega) = \lim_{p \rightarrow \infty} \sum_{n=k}^p X_n(\omega) \right\}.$$

This implies in particular that $\Omega_1 \subset \mathcal{T}_n$ for all $n \in \mathbb{N}$. So $\Omega_1 \in \bigcap_{n \in \mathbb{N}} \mathcal{T}_n = \mathcal{T}$. $\triangle$

Remark 5.20. In remark 5.19. Ω_1 contains all the elements s.t.

$$\lim_{p \rightarrow \infty} \sum_{n=1}^p X_n(\omega) = \pm \infty.$$

But with the exact same reasoning of the previous remark we can show that

$$\left\{ \omega \in \Omega : \lim_{p \rightarrow \infty} \sum_{n=1}^p X_n = \infty \right\} \in \mathcal{T}$$

or

$$\Omega_2 = \left\{ \omega \in \Omega : -\infty < \liminf_{p \rightarrow \infty} \sum_{n=1}^p X_n(\omega) = \limsup_{p \rightarrow \infty} \sum_{n=1}^p X_n(\omega) < \infty \right\} \in \mathcal{T}.$$

$\triangle$

We now can state Kolmogorov's law 0-1 using the definitions from earlier.

Theorem 5.21 (Kolmogorov's law 0-1). Let $\{X_n\}_{n \in \mathbb{N}}$ a sequence of independent random variables on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. Then $\mathbb{P}(A) \in \{0, 1\}$ for every $A \in \mathcal{T}$.

The following corollary is an immediate consequence of Kolmogorov's law 0-1:

Corollary 5.22. If $\{X_n\}_{n \in \mathbb{N}}$ is a sequence of independent random variables on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. Then either

$$\sum_{n \in \mathbb{N}} X_n < \infty \quad \mathbb{P}\text{-a.s.}$$

or

$$\sum_{n \in \mathbb{N}} X_n = \infty \quad \mathbb{P}\text{-a.s.}$$

Proof of theorem 5.21. The proof is divided into several steps.

1. Claim: for every $n \in \mathbb{N}$, $\mathcal{T}_n$ and $\mathcal{A}_{n-1} := \sigma(X_1, \dots, X_{n-1})$ are independent (because the random variables $\{X_n\}_{n \in \mathbb{N}}$ are all independent i.e. $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ for every $A \in \mathcal{A}_{n-1}$ and every $B \in \mathcal{T}$). Indeed,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \quad \forall A \in \mathcal{A}_{n-1}, B \in \sigma(X_n, \dots, X_p)$$

for $p \geq n$. Moreover $\mathcal{A}_{n-1}$ and $\bigcup_{p \geq n} \sigma(X_n, \dots, X_p)$ are π -systems and

$$\sigma\left(\bigcup_{p \geq n} \sigma(X_n, \dots, X_p)\right) = \mathcal{T}_n.$$

Then theorem 4.15 implies $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ for every $A \in \mathcal{A}_{n-1}$ and $B \in \mathcal{T}_n$ hence we have proven that $\mathcal{T}_n$ and $\mathcal{A}_{n-1}$ are independent.

2. Claim: $\mathcal{T}$ and $\sigma(X_1, X_2, \dots)$ are independent i.e. $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ for every $A \in \sigma(X_1, X_2, \dots)$ and $B \in \mathcal{T}$. Indeed, by step 1 and by the fact that $\mathcal{T} \subset \mathcal{T}_n$ we see that $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ for every $A \in \bigcup_{n \geq 1} \sigma(X_1, \dots, X_n)$ and $B \in \mathcal{T}$. Observe that $\bigcup_{n \geq 1} \sigma(X_1, \dots, X_n)$ is a π -system and also $\mathcal{T}$ is a π -system. Then by theorem 4.15

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \quad \forall A \in \sigma(X_1, X_2, \dots), B \in \mathcal{T}$$

which concludes step 2.

3. Let $A \in \mathcal{T}$ which implies $A \in \sigma(X_1, X_2, \dots)$ since $\mathcal{T} \subset \sigma(X_1, X_2, \dots)$ so

$$\mathbb{P}(A) = \mathbb{P}(A \cap A) = \mathbb{P}(A)^2$$

by step 2 hence $\mathbb{P}(A) \in \{0, 1\}$.

This proves the theorem. □

6 Strong law of large numbers

We have already seen a version of the statement and the proof of the law of large numbers. However it is also possible to drop some of the assumptions in the earlier version making the statement more general.

6.1 Strong law of large numbers theorem and its proof

Theorem 6.1 (Strong law of large numbers). Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of indentially distributed and *pairwise independent* random variables on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ s.t. $\mathbb{E}[|X_n|] < \infty$ (for every $n \in \mathbb{N}$). Let $S_n = \sum_{i=1}^n X_i$ then

$$\frac{S_n}{n} \xrightarrow{n \rightarrow \infty} \mathbb{E}[X_1], \quad \mathbb{P}\text{-a.s.}$$

Remark 6.2 (comparison with earlier version of the law of large numbers). We had that the random variables had to be all independent and $\mathbb{E}[|X_n|^2] < \infty$. For the previous proof we used the 2nd moment method making the proof shorter. $\triangle$

Proof of theorem 6.1. We will divide the prove into several steps:

1. Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of general random variables. In this step we reduce the proof to the case of non-negative random variables. Recast the probability in order to consider only $X_n \geq 0$. So we define

$$X_n^+ = \max\{X_n, 0\}, \quad X_n^- = \max\{-X_n, 0\}.$$

By assumption $\{X_n\}_{n \in \mathbb{N}}$ are pairwise independent hence their postive and negatives parts (X_n^+ and X_n^-) are pairwise independent and identically distributed. Furthermore $\mathbb{E}[|X_n|] < \infty$ implies that $\mathbb{E}[X_n^+] < \infty$ and $\mathbb{E}[X_n^-] < \infty$. Without loss of generality we can consider $X_n \geq 0$ due to the fact that $X_n = X_n^+ - X_n^-$.

2. This step deals with the truncation: We assume $X_n \geq 0$ and define $Y_n := X_n \mathbb{1}_{\{X_n \leq n\}}$ for every $n \in \mathbb{N}$. Let $A := \liminf_{n \rightarrow \infty} \{X_n = Y_n\}$. Then $\mathbb{P}(A) = 1 - \mathbb{P}(\limsup_{n \rightarrow \infty} \{X_n \neq Y_n\})$. Now we have

$$\sum_{n \in \mathbb{N}} \mathbb{P}(X_n \neq Y_n) = \sum_{n \in \mathbb{N}} \mathbb{P}(X_n > n) = \sum_{n \in \mathbb{N}} \mathbb{E}[\mathbb{1}_{\{X_n > n\}}].$$

Using the fact that the X_n 's are indentially distributed we obtain

$$\sum_{n \in \mathbb{N}} \mathbb{E}[\mathbb{1}_{\{X_n > n\}}] = \sum_{n \in \mathbb{N}} \mathbb{E}[\mathbb{1}_{\{X_1 > n\}}]$$

Since this is a series of positive values we can use Beppo-Levi to see

$$\sum_{n \in \mathbb{N}} \mathbb{E}[\mathbb{1}_{\{X_1 > n\}}] = \mathbb{E} \left[\sum_{n \in \mathbb{N}} \mathbb{1}_{\{X_1 > n\}} \right] \leq \mathbb{E}[X_1] < \infty$$

hence

$$\sum_{n \in \mathbb{N}} \mathbb{P}(X_n \neq Y_n) < \infty.$$

This allows us to apply Borel-Cantelli yielding

$$\mathbb{P}(A) = 1 - \underbrace{\mathbb{P}(\limsup_{n \rightarrow \infty} \{X_n \neq Y_n\})}_{=0} = 1.$$

Moreover in A we have

$$\frac{\sum_{k=1}^n X_k}{n} \xrightarrow{n \rightarrow \infty} \mathbb{E}[X_1] \quad \mathbb{P}\text{-a.s.} \quad \Leftrightarrow \quad \frac{\sum_{k=1}^n Y_k}{n} \xrightarrow{n \rightarrow \infty} \mathbb{E}[X_1] \quad \mathbb{P}\text{-a.s.}$$

so we observe that the conclusion of theorem 6.1 follows if

$$\frac{T_n}{n} \xrightarrow{n \rightarrow \infty} \mathbb{E}[X_1] \quad \mathbb{P}\text{-a.s.}$$

with

$$T_n = \begin{cases} \sum_{k=1}^n Y_k, & n \geq 1 \\ 0, & n = 0 \end{cases},$$

so we will prove that in the third step.

3. Fix $\alpha > 1$. We will prove that if $\frac{T_{\lfloor \alpha^n \rfloor}}{\lfloor \alpha^n \rfloor} \xrightarrow{n \rightarrow \infty} \mathbb{E}[X_1] \quad \mathbb{P}\text{-a.s.}$ then $\frac{T_n}{n} \xrightarrow{n \rightarrow \infty} \mathbb{E}[X_1] \quad \mathbb{P}\text{-a.s.}$
 Indeed, let $\alpha_M = 1 + \frac{1}{M}$ for $M > 1$ and $\tilde{\Omega} = \bigcap_{M \geq 1} \Omega_M$ with

$$\Omega_M := \left\{ \omega \in \Omega : \frac{T_{\lfloor \alpha_M^n \rfloor}}{\lfloor \alpha_M^n \rfloor} \xrightarrow{n \rightarrow \infty} \mathbb{E}[X_1] \right\}.$$

In particular $\mathbb{P}(\Omega_M) = 1$ for every $M \geq 1$ and by construction also $\mathbb{P}(\tilde{\Omega}) = 1$. Fix an $M \geq 1$ and let $k_n := \lfloor \alpha_M^n \rfloor$. For every $\omega \in \tilde{\Omega}$ and $m \in [k_n, k_{n+1}]$ look at

$$\frac{T_{k_n}}{k_{n+1}} \leq \frac{T_m}{m} \leq \frac{T_{k_{n+1}}}{k_n}.$$

Observe that by definition of k_n we have $\lim_{n \rightarrow \infty} \frac{k_{n+1}}{k_n} = \alpha_M$. We will consider

$$\frac{1}{\alpha_M} \mathbb{E}[X_1] \leq \liminf_{n \rightarrow \infty} \frac{T_n}{n} \leq \limsup_{n \rightarrow \infty} \frac{T_n}{n} \leq \mathbb{E}[X_1] \alpha_M.$$

Taking $M \rightarrow \infty$ implies

$$\frac{T_n}{n} \xrightarrow{n \rightarrow \infty} \mathbb{E}[X_1] \quad \mathbb{P}\text{-a.s.}$$

because $\alpha_M = 1 + 1/M \rightarrow 1$ as $M \rightarrow \infty$.

4. Now we show that the assumption of step 3. i.e. $\frac{T_{\lfloor \alpha^n \rfloor}}{\lfloor \alpha^n \rfloor} \xrightarrow{n \rightarrow \infty} \mathbb{E}[X_1]$ holds true. For that we fix $\alpha > 1$, $k_n = \lfloor \alpha^n \rfloor$ and $\varepsilon > 0$. By Chebistevs inequality

$$\sum_{n=1}^{\infty} \mathbb{P}(|T_{k_n} - \mathbb{E}[T_{k_n}]| > \varepsilon k_n) \leq \sum_{n=1}^{\infty} \frac{\text{Var}(T_{k_n})}{\varepsilon^2 k_n^2} = \frac{1}{\varepsilon^2} \sum_{n=1}^{\infty} \frac{1}{k_n^2} \text{Var} \left(\sum_{m=1}^{k_n} Y_m \right)$$

holds. Using that the Y_m are pairwise independent we obtain

$$\frac{1}{\varepsilon^2} \sum_{n=1}^{\infty} \frac{1}{k_n^2} \text{Var} \left(\sum_{m=1}^{k_n} Y_m \right) = \frac{1}{\varepsilon^2} \sum_{n=1}^{\infty} \frac{1}{k_n^2} \sum_{m=1}^{k_n} \text{Var}(Y_m).$$

Using a similar argument as Tonelli or Fubini we can swap sums which leads to

$$\frac{1}{\varepsilon^2} \sum_{n=1}^{\infty} \frac{1}{k_n^2} \sum_{m=1}^{k_n} \text{Var}(Y_m) = \frac{1}{\varepsilon^2} \sum_{m=1}^{\infty} \sum_{n \in \{n \in \mathbb{N} : k_n \geq m\}} \frac{\text{Var}(Y_m)}{k_n}.$$

Then for every $n \in \mathbb{N}$ we have $k_n \geq \frac{\alpha^n}{2}$ so

$$\sum_{n \in \{n \in \mathbb{N} : k_n \geq m\}} \frac{1}{k_n^2} \leq 4 \sum_{n \in \{n \in \mathbb{N} : \alpha^n \geq m\}} \alpha^{-2n} = 4 \frac{\alpha^{-2n_0(m)}}{1 - \alpha^{-2}}$$

with $n_0(m) := \inf\{n \in \mathbb{N} : \alpha^n \geq m\}$. So

$$\sum_{n=1}^{\infty} \mathbb{P}(|T_{k_n} - \mathbb{E}[T_{k_n}]| \geq \varepsilon k_n) \leq \frac{4}{\varepsilon^2} \cdot \frac{1}{1 - \alpha^{-2}} \sum_{m=1}^{\infty} \frac{\mathbb{E}[Y_m^2]}{m^2}.$$

It is left to show that

- $\sum_{j=1}^m \frac{\mathbb{E}[Y_j^2]}{j^2}$ is bounded
- $\frac{\mathbb{E}[T_n]}{n} \rightarrow \mathbb{E}[X_1]$.

Indeed, if these two would hold then by Borel Cantelli:

$$\mathbb{P} \left(\limsup_{n \rightarrow \infty} \left| \frac{T_{k_n}}{k_n} - \frac{\mathbb{E}[T_{k_n}]}{k_n} \right| \leq \varepsilon \right) = 1,$$

which is equivalent to

$$\mathbb{P} \left(\limsup_{n \rightarrow \infty} \left| \frac{T_{k_n}}{k_n} - \mathbb{E}[X_1] \right| > \varepsilon \right) = 0.$$

5. Now we prove $\frac{\mathbb{E}[T_n]}{n} \rightarrow \mathbb{E}[X_1]$. Recall that by definition of Y_n , then

$$0 \leq \mathbb{E}[X_n - Y_n] = \mathbb{E}[X_n] - \mathbb{E}[Y_n] = \mathbb{E}[X_n \mathbf{1}_{\{X_n > n\}}] = \mathbb{E}[X_1 \mathbf{1}_{\{X_1 > n\}}].$$

because $\{X_n\}_{n \in \mathbb{N}}$ are i.i.d. Taking the limit yields

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_1 \mathbf{1}_{\{X_1 > n\}}] = 0$$

which one can easily see by applying the Lebesgue dominated convergence theorem. So $\mathbb{E}[X_1 - Y_n] \rightarrow 0$ as $n \rightarrow \infty$ which is equivalent to $\mathbb{E}[Y_n] \rightarrow \mathbb{E}[X_1]$ as $n \rightarrow \infty$. Moreover

$$\frac{\mathbb{E}[T_n]}{n} = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \mathbb{E}[X_1] \quad \Rightarrow \quad \lim_{n \rightarrow \infty} \frac{\mathbb{E}[T_n]}{n} = \mathbb{E}[X_1]$$

6. Next we prove that $\sum_{j=1}^{\infty} \frac{\mathbb{E}[Y_j^2]}{j^2} < \infty$ holds true. We have

$$\mathbb{E}[Y_n^2] = \mathbb{E} \left[\int_0^{\infty} 2y \cdot \mathbf{1}_{\{Y_n \geq y\}} dy \right]$$

Using Fubini this transforms into

$$\mathbb{E} \left[\int_0^{\infty} 2y \cdot \mathbf{1}_{\{Y_n \geq y\}} dy \right] = 2 \int_0^{\infty} y \mathbb{P}(Y_n \geq y) dy.$$

Since Y_n is such that $Y_n \leq n$ and $X_n \geq Y_n$ it holds that

$$\mathbb{E}[Y_n^2] \leq 2 \int_0^{\infty} y \mathbb{P}(Y_n \geq y) dy \leq 2 \int_0^{\infty} y \mathbb{P}(X_n \geq y) dy.$$

Consider

$$\sum_{n \in \mathbb{N}} \frac{\mathbb{E}[Y_n^2]}{n^2} \leq \sum_{n \in \mathbb{N}} \frac{1}{n^2} \int_0^n y \mathbb{P}(X_n \geq y) dy = 2 \int_0^{\infty} \sum_{n \in \mathbb{N}} \frac{1}{n^2} y \mathbb{P}(X_n \geq y) \mathbf{1}_{\{y \leq n\}} dy$$

where we used Beppo-Levi at the equality. Using the fact that all the random variables are equally distributed we obtain

$$\int_0^{\infty} \sum_{n \in \mathbb{N}} \frac{1}{n^2} y \mathbb{P}(X_n \geq y) \mathbf{1}_{\{y \leq n\}} dy = \int_0^{\infty} \sum_{n \in \mathbb{N}} \frac{y}{n^2} \mathbb{P}(X_1 \geq y) \mathbf{1}_{\{y \leq n\}} dy,$$

hence

$$\sum_{n \in \mathbb{N}} \frac{\mathbb{E}[Y_n^2]}{n^2} \leq 2 \int_0^{\infty} \sum_{n \in \mathbb{N}} \frac{y}{n^2} \mathbb{P}(X_1 \geq y) \mathbf{1}_{\{y \leq n\}} dy. \quad (6.1)$$

For $y \geq 2$

$$\sum_{n \in \mathbb{N}, n \geq y} \frac{1}{n^2} \leq \sum_{n \in \mathbb{N}, n \geq y} \int_{n-1}^n \frac{dx}{x^2} = \int_{y-1}^{\infty} \frac{dx}{x^2} = \frac{1}{y-1}, \quad (6.2)$$

which implies

$$\sum_{n \in \mathbb{N}} \frac{1}{n^2} = 1 + \sum_{n=2}^{\infty} \frac{1}{n^2} \leq 1 + C$$

for some $C > 0$ by (6.2). Using said bound in (6.1):

$$\begin{aligned} \sum_{n \in \mathbb{N}} \frac{\mathbb{E}[Y_n^2]}{n^2} &\leq \int_0^2 \underbrace{2y}_{\leq 4} \mathbb{P}(X_1 \geq y) dy + 2 \int_2^{\infty} \underbrace{\frac{y}{y-1}}_{\leq 2} \mathbb{P}(X_1 \geq y) dy \\ &\leq 4 \int_0^{\infty} \mathbb{P}(X_1 \geq y) dy + 4 \int_0^{\infty} \mathbb{P}(X_1 \geq y) dy \\ &\leq 8 \cdot \mathbb{E} \left[\int_0^{X_1} dy \right] = \mathbb{E}[X_1] < \infty, \end{aligned}$$

hence $\sum_{n \in \mathbb{N}} \frac{\mathbb{E}[Y_n^2]}{n^2} < \infty$.

Using Borel-Cantelli gives us the result. □

6.2 Examples and applications

Example 6.3 (Monte Carlo's method). Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function. Monte Carlo's method is a numerical scheme to compute integrals like

$$\int_0^1 f(x) \, dx.$$

Let $X_1, Z_1, X_2, Z_2, \dots$ be a sequence of independent random variables uniformly distributed on $[0, 1]$. Let

$$\rho_n = \begin{cases} 1, & f \circ X_n \geq Z_n \\ 0, & f \circ X_n < Z_n \end{cases}.$$

Then

$$\mathbb{E}[\rho_n] = \mathbb{P}(\{\omega \in [0, 1] : f(X_n(\omega)) \geq Z_n(\omega)\}) = \int_0^1 f(x) \, dx.$$

Since f is continuous on a compact interval $\mathbb{E}[\rho_n] < \infty$. We are within the assumptions of the strong law of large numbers (thm 6.1) hence

$$\frac{1}{n} \sum_{i=1}^n \rho_i \xrightarrow{n \rightarrow \infty} \int_0^1 f(x) \, dx \quad \mathbb{P}\text{-a.s.}$$

△

Example 6.4. Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of independent identically distributed random variables s.t. $0 < X_n < \infty$ for every $n \in \mathbb{N}$. Let $T_n = \sum_{i=1}^n T_i$ and $N_t := \sup\{n \in \mathbb{N} : T_n \leq t\}$. This value N_t can be interpreted as the number of events before a point in time t . We shall prove that if $\mathbb{E}[X_n] = \mu < \infty$ for every $n \in \mathbb{N}$ then

$$\lim_{n \rightarrow \infty} \frac{N_t}{t} = \frac{1}{\mu}.$$

To see this we will again use the strong law of large numbers. Said law gives us that

$$\frac{T_n}{n} \xrightarrow{n \rightarrow \infty} \mu \quad \mathbb{P}\text{-a.s.}$$

Observe that by the definition of N_t that $T_{N_t} \leq t \leq T_{(N_t+1)}$ which implies

$$\underbrace{\frac{T_{N_t}}{N_t}}_{\xrightarrow{t \rightarrow \infty} \mu} \leq \frac{t}{N_t} \leq \underbrace{\frac{T_{(N_t+1)}}{N_t+1}}_{\xrightarrow{t \rightarrow \infty} \mu} \cdot \frac{N_t+1}{N_t}.$$

Regarding $(N_t+1)/N_t$: As $t \rightarrow \infty$, clearly $N_t \rightarrow \infty$ $\mathbb{P}$ -a.s. so

$$\lim_{t \rightarrow \infty} \frac{N_t+1}{N_t} = 1$$

therefore

$$\frac{T_{(N_t+1)}}{N_t+1} \cdot \frac{N_t+1}{N_t} \xrightarrow{t \rightarrow \infty} \mu$$

hence

$$\lim_{t \rightarrow \infty} \frac{t}{N_t} = \mu \quad \mathbb{P}\text{-a.s.} \quad \Rightarrow \quad \lim_{t \rightarrow \infty} \frac{N_t}{t} = \frac{1}{\mu} \quad \mathbb{P}\text{-a.s.}$$

△

7 Kolmogorov's three series theorem

We have seen Kolmogorov's law 0-1 that told us when the events in the tail σ -algebra are either zero or one (see theorem 5.21.). The problem with this theorem is it is difficult to see whether $\mathbb{P}(A) = 1$ or $\mathbb{P}(A) = 0$. We aim at studying the behaviour and properties of the stochastic series $\sum_{n \in \mathbb{N}} X_n$. Kolmogorov's law 0-1 implies that the convergence set of said series,

$$\bar{\Omega} = \left\{ \omega \in \Omega : -\infty < \liminf_{n \rightarrow \infty} \sum_{k=1}^n X_k = \limsup_{n \rightarrow \infty} \sum_{k=1}^n X_k < \infty \right\}$$

satisfies $\mathbb{P}(\bar{\Omega}) \in \{0, 1\}$. But how can we determine whether $\mathbb{P}(\bar{\Omega}) = 1$ or $\mathbb{P}(\bar{\Omega}) = 0$? The following lemma helps to find conditions to clarify the two cases:

Lemma 7.1 (Kolmogorov's inequality). Let $S_0 = 0$ and $S_k = \sum_{j=1}^k X_j$ for $k \geq 1$ where $\{X_i\}_{i \in \mathbb{N}}$ is a sequence of independent random variables s.t. $\mathbb{E}[X_n] = 0$ and $\mathbb{E}[X_n^2] < \infty$ for every $n \in \mathbb{N}$. Then

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| \geq \alpha\right) \leq \frac{\text{Var}(S_n)}{\alpha^2} \quad \text{for } \alpha > 0.$$

Remark 7.2. Note that while Chebystev's inequality gives a control on the tail of one partial sum (for k fixed), Kolmogorov's inequality is a “maximal” inequality. The maximum of the series (partial sums) is controlled by the variance of only the last term. $\triangle$

Proof of lemma 7.1. Define $A_k := \{\omega \in \Omega : |S_k(\omega)| \geq \alpha, |S_j(\omega)| < \alpha, \forall j < k\}$. By construction $\{A_k\}_{k \in \mathbb{N}}$ is a disjoint sequence of sets, thus

$$1 \equiv \mathbb{1}_{(\cup_{k=1}^n A_k)^c} + \mathbb{1}_{\cup_{k=1}^n A_k} = \mathbb{1}_{(\cup_{k=1}^n A_k)^c} + \sum_{k=1}^n \mathbb{1}_{A_k}.$$

By definition of S_n we have $\text{Var}(S_n) = \mathbb{E}[S_n^2]$ so this term is bounded from below by

$$\begin{aligned} \mathbb{E}[S_n^2] &\geq \mathbb{E}[S_n^2 \mathbb{1}_{\cup_{k=1}^n A_k}] = \sum_{k=1}^n \mathbb{E}[S_n^2 \mathbb{1}_{A_k}] \\ &\geq \sum_{k=1}^n \mathbb{E}[(S_k^2 + 2S_k(S_n - S_k) + (S_n - S_k)^2) \mathbb{1}_{A_k}] \\ &\geq \sum_{k=1}^n \mathbb{E}[S_k^2 \mathbb{1}_{A_k}] + 2 \sum_{k=1}^n \mathbb{E}[S_k(S_n - S_k) \mathbb{1}_{A_k}]. \end{aligned}$$

Notice that S_k and $(S_n - S_k)$ are independent thanks to the independence of $\{X_n\}_{n \in \mathbb{N}}$. Furthermore observe that $\mathbb{E}[S_n - S_k] = 0$ because $S_n - S_k = X_{k+1} + \dots + X_n$ and $\mathbb{E}[X_j] = 0$ for every $j \in \mathbb{N}$. Then using independence:

$$\mathbb{E}[S_k \mathbf{1}_{A_k} (S_n - S_k)] = \mathbb{E}[S_k \mathbf{1}_{A_k}] \underbrace{\mathbb{E}[S_n - S_k]}_{=0} = 0,$$

so

$$\mathbb{E}[S_n^2] \geq \sum_{k=1}^n \mathbb{E}[S_k^2 \mathbf{1}_{A_k}] + 2 \sum_{k=1}^n \mathbb{E}[S_k (S_n - S_k) \mathbf{1}_{A_k}] = \sum_{k=1}^n \mathbb{E}[S_k^2 \mathbf{1}_{A_k}].$$

Focusing on the LHS we obtain

$$\sum_{k=1}^n \mathbb{E}[S_k^2 \mathbf{1}_{A_k}] \geq \sum_{k=1}^n \alpha^2 \mathbb{P}(A_k) = \alpha^2 \mathbb{P}\left(\bigcup_{k=1}^n A_k\right) \geq \alpha^2 \mathbb{P}\left(\max_{1 \leq k \leq n} |S_n| \geq \alpha\right)$$

hence the result. $\square$

This allows us to get the first criterion to determine if the stochastic series converges:

Theorem 7.3. Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of independent random variables with $\mathbb{E}[X_i] = 0$ for every $i \in \mathbb{N}$ and $\sum_{n=1}^{\infty} \text{Var}(X_n) < \infty$. Then $\mathbb{P}(\bar{\Omega}) = 1$ where

$$\bar{\Omega} = \left\{ \omega \in \Omega : -\infty < \liminf_{n \rightarrow \infty} \sum_{k=1}^n X_k = \limsup_{n \rightarrow \infty} \sum_{k=1}^n X_k < \infty \right\}$$

the convergence set of the random variable $S_n := \sum_{k=1}^n X_k$.

Before proving this theorem we look at an example:

Example 7.4. Let $\{Z_n\}_{n \in \mathbb{N}}$ be a sequence of independent random variable with $\mathbb{P}(Z_k = \pm 1) = \frac{1}{2}$. Then

$$\sum_{k=1}^{\infty} \frac{Z_k}{k} < \infty$$

converges. Indeed we have $\mathbb{E}[Z_k] = 0$ and $\text{Var}(Z_k) = 1 < \infty$. Moreover

$$\sum_{k=1}^{\infty} \text{Var}\left(\frac{Z_k}{k}\right) = \sum_{k=1}^{\infty} \frac{\text{Var}(Z_k)}{k^2} = \sum_{k=1}^{\infty} \frac{1}{k^2} < \infty$$

so the assumption of theorem 7.3 are satisfied implying $\mathbb{P}(\bar{\Omega}) = 1$. $\triangle$

Proof of theorem 7.3. Let $W_M = \sup_{n, m \geq M} |S_n - S_m|$ with $S_n = \sum_{k=1}^n X_k$. By construction, $W_M \downarrow W_{\infty}$ as $M \rightarrow \infty$. We shall prove that

$$\mathbb{P}(W_{\infty} = 0) = 1. \tag{7.1}$$

Indeed, if we can prove (7.1) it means that $\{S_n\}_{n \in \mathbb{N}}$ is a Cauchy-sequence $\mathbb{P}$ -a.s. and therefore converges $\mathbb{P}$ -a.s. To show (7.1) let $\varepsilon > 0$ and $M \geq 1$. We observe that

$$\sup_{n \geq M} |S_n - S_M| \leq \varepsilon \quad \Rightarrow \quad \sup_{n, m \geq M} |S_n - S_m| \leq 2\varepsilon \quad (7.2)$$

by the triangle inequality. Furthermore using (7.2)

$$\begin{aligned} \mathbb{P}(W_M > 2\varepsilon) &= \mathbb{P}\left(\sup_{n, m \geq M} |S_n - S_m| > 2\varepsilon\right) \leq \mathbb{P}\left(\sup_{m \geq M} |S_m - S_M| > \varepsilon\right) \\ &= \lim_{N \rightarrow \infty} \mathbb{P}\left(\sup_{M \leq n \leq N} |S_n - S_M| > \varepsilon\right). \end{aligned}$$

Applying the monotone convergence theorem we have

$$\mathbb{P}\left(\lim_{N \rightarrow \infty} \sup_{M \leq n \leq N} |S_n - S_M| > \varepsilon\right) = \lim_{N \rightarrow \infty} \mathbb{P}\left(\sup_{M \leq n \leq N} |S_n - S_M| > \varepsilon\right),$$

so we're at a point where we can utilize Kolmogorov's inequality yielding

$$\mathbb{P}(W_M > 2\varepsilon) \leq \lim_{N \rightarrow \infty} \mathbb{P}\left(\sup_{M \leq n \leq N} |S_n - S_M| > \varepsilon\right) \leq \lim_{N \rightarrow \infty} \frac{1}{\varepsilon^2} \sum_{k=M+1}^N \text{Var}(X_k).$$

Using that W_M converges in a decreasing way to W_∞ , we have

$$\mathbb{P}(W_\infty \geq 2\varepsilon) \leq \mathbb{P}(W_M \geq 2\varepsilon) \leq \frac{1}{\varepsilon^2} \sum_{k>M}^\infty \text{Var}(X_k)$$

and

$$\mathbb{P}(W_\infty \geq 2\varepsilon) = \lim_{M \rightarrow \infty} \mathbb{P}(W_M \geq 2\varepsilon) \leq \lim_{M \rightarrow \infty} \frac{1}{\varepsilon^2} \sum_{k>M}^\infty \text{Var}(X_k) = 0,$$

hence $\mathbb{P}(W_\infty \geq 2\varepsilon) = 0$ due to the convergence of the series. This implies $\mathbb{P}(W_\infty < 2\varepsilon) = 1$ for every $\varepsilon > 0$. This concludes the proof. $\square$

Theorem 7.5 (Kolmogorov's three series theorem). Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of independent random variables. Let $\alpha > 0$ and $\{Y_n\}_{n \in \mathbb{N}}$ s.t. $Y_n = X_n \mathbf{1}_{\{|X_n| \leq \alpha\}}$. Then $\sum_{n \in \mathbb{N}} X_n$ converges $\mathbb{P}$ -a.s. if and only if for some $\alpha > 0$ the following holds

1. $\sum_{n \in \mathbb{N}} \mathbb{P}(|X_n| > \alpha) < \infty$
2. $\sum_{n \in \mathbb{N}} \mathbb{E}[Y_n] < \infty$
3. $\sum_{n \in \mathbb{N}} \text{Var}(Y_n) < \infty$.

Remark 7.6. This result can also be generalized for weakly independent random variables i.e. $|\mathbb{E}[X_k X_j] - \mathbb{E}[X_k] \mathbb{E}[X_j]| < \varepsilon$ for some $\varepsilon > 0$. $\triangle$

Proof of theorem 7.5. We have to prove both directions:

- "⇐": Let $\bar{Y}_n := Y_n - \mathbb{E}[Y_n]$ then $\mathbb{E}[\bar{Y}_n] = 0$ and $\text{Var}(\bar{Y}_n) = \text{Var}(Y_n)$. By assumption 3. we have $\sum_{n \in \mathbb{N}} \text{Var}(\bar{Y}_n) < \infty$ so we can apply theorem 7.3 (using also the independence of the X_k 's) hence $\sum_{n \in \mathbb{N}} \bar{Y}_n$ converges $\mathbb{P}$ -a.s. By the definition of $\bar{Y}_n$ we can write $Y_n = \bar{Y}_n + \mathbb{E}[Y_n]$ so

$$\sum_{n \in \mathbb{N}} Y_n = \sum_{n \in \mathbb{N}} \bar{Y}_n + \sum_{n \in \mathbb{N}} \mathbb{E}[Y_n] \quad (7.3)$$

where the RHS consists of two convergent sums thanks to our assumptions and the reasoning above. Now assumption 1. and Borel-Cantelli give

$$\mathbb{P}(\limsup_{n \rightarrow \infty} |X_n| > \alpha) = 0$$

this implies that

$$\sum_{n \in \mathbb{N}} X_n \text{ converges} \quad \Leftrightarrow \quad \sum_{n \in \mathbb{N}} Y_n \text{ converges.}$$

But (7.3) shows the convergence of $\sum_{n \in \mathbb{N}} Y_n$ so $\sum_{n \in \mathbb{N}} X_n$ converges.

- "⇒": To prove "⇒" we need to show that all those three series converge. If $\sum_{n \in \mathbb{N}} X_n$ converges $\mathbb{P}$ -a.s. then assumption 1. holds for every $\alpha > 0$. Indeed if 1. was not true then there is $\bar{\alpha} > 0$ s.t. $\{|X_n| > \bar{\alpha}\}$ happens infinitely often $\mathbb{P}$ -a.s. by Borel-Cantelli. But this would contradict the hypothesis ($\sum_{n \in \mathbb{N}} X_n < \infty$ $\mathbb{P}$ -a.s.) hence 1. holds true.

If $\sum_{n \in \mathbb{N}} X_n$ converges then also $\sum_{n \in \mathbb{N}} Y_n$ converges. Now observe that if 3. holds then 2. follows: indeed 3. together with theorem 7.3 give the convergence of

$$\sum_{n \in \mathbb{N}} (Y_n - \mathbb{E}[Y_n]) < \infty$$

so

$$\sum_{n \in \mathbb{N}} \mathbb{E}[Y_n] < \infty$$

because of the convergence of $\sum_{n \in \mathbb{N}} Y_n$ and 2. So it is left to prove that $\sum_{n \in \mathbb{N}} \text{Var}(Y_n) < \infty$. Without loss of generality we can consider Y_n s.t. $\mathbb{E}[Y_n] = 0$. Indeed let $\{\bar{Y}_n\}$ an independent copy of $\{Y_n\}$ and define $Z_n = Y_n - \bar{Y}_n$. Note that $\{Z_n\}$ are independent random variables and $\mathbb{E}[Z_n] = 0$. Furthermore Z_n is bounded by 2 (because 1. holds with $\alpha = 1$ and $Y_n := X_n \mathbb{1}_{\{|X_n| \leq 1\}}$). Then $\text{Var}(Z_n) = 2 \text{Var}(Y_n)$ and

$$\sum_{n \in \mathbb{N}} Z_n = \sum_{n \in \mathbb{N}} Y_n - \sum_{n \in \mathbb{N}} \bar{Y}_n < \infty \quad \mathbb{P}\text{-a.s.} \quad (7.4)$$

using that $\sum_{n \in \mathbb{N}} Y_n \leq \sum_{n \in \mathbb{N}} X_n < \infty$ $\mathbb{P}$ -a.s. by hypothesis. So it is enough to prove 3. for this new sequence of random variables $\{Z_n\}_{n \in \mathbb{N}}$. Let S_n denote the sequence of partial sums

$$S_n = \sum_{k=1}^n Z_k \quad \text{and} \quad \sigma_k^2 = \text{Var}(Z_k)$$

and let $\tau_L := \min\{n \geq 0 : |S_n| \geq L\}$. We know by (7.4) that S_n converges $\mathbb{P}$ -a.s. as $n \rightarrow \infty$, thus

$$\lim_{L \rightarrow \infty} \mathbb{P}(\tau_L = \infty) = 1. \quad (7.5)$$

Furthermore

$$|S_{\tau_L}| \leq L + 2 \quad \text{if } \tau_L < \infty.$$

Take the random variable $S_{\min\{n, \tau_L\}}^2$ and compute its expectation:

$$\begin{aligned} \mathbb{E}[S_{\min\{n, \tau_L\}}^2] &= \mathbb{E} \left[\sum_{k=1}^n Z_k \mathbf{1}_{\{k \leq \tau_L\}} + 2 \sum_{1 \leq i < j \leq n} Z_i Z_j \mathbf{1}_{\{j \leq \tau_L\}} \right] \\ &= \sum_{k=1}^n \mathbb{E}[Z_k \mathbf{1}_{\{k \leq \tau_L\}}] + 2 \sum_{1 \leq i < j \leq n} \mathbb{E}[Z_i Z_j \mathbf{1}_{\{j \leq \tau_L\}}]. \end{aligned}$$

Remark that $\{j \leq \tau_L\} = \{j > \tau_k + 1\}^C$ are completely determined by Z_i and Z_{j-1} therefore Z_i and Z_j in the sum are independent. Using this yields

$$\begin{aligned} (L + 2)^2 &\geq S_{\min\{n, \tau_L\}}^2 = \sum_{j=1}^n \sigma_j^2 \mathbb{P}(j \leq \tau_L) + \sum_{1 \leq i < j \leq n} \underbrace{\mathbb{E}[Z_i] \mathbb{E}[Z_j \mathbf{1}_{\{j \leq \tau_L\}}]}_{=0} \\ &\geq \sum_{j=1}^n \sigma_j^2 \mathbb{P}(\tau_L = \infty). \end{aligned}$$

For L large enough such that $\mathbb{P}(\tau_L = \infty) > 0$ which is possible due to (7.5) we let $n \rightarrow \infty$ so

$$\sum_{j \in \mathbb{N}} \sigma_j^2 \leq (L + 2)^2 < \infty$$

hence we have shown property 3. This concludes the proof. □

Corollary 7.7. Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of independent random variables with $\mathbb{E}[X_n] = 0$ for every $n \in \mathbb{N}$. Then if

$$\sum_{n \in \mathbb{N}} \mathbb{E} \left[\frac{X_n^2}{1 + |X_n|} \right] < \infty$$

then $\sum_{n \in \mathbb{N}} X_n$ converges $\mathbb{P}$ -a.s.

Remark 7.8. This corollary isn't particularly useful but its proof shows good applications of the discussed theorem. △

Proof. The condition in the statement can be rewritten in an equivalent way: Note that

$$\sum_{n \in \mathbb{N}} \mathbb{E} \left[\frac{X_n^2}{1 + |X_n|} \right] < \infty \quad \Leftrightarrow \quad \sum_{n \in \mathbb{N}} \mathbb{E}[X_n^2 \mathbf{1}_{\{|X_n| \leq 1\}} + |X_n| \mathbf{1}_{\{|X_n| > 1\}}] < \infty.$$

Let $Y_n := X_n \mathbf{1}_{\{|X_n| \leq 1\}}$. Then we have $\sum_{n \in \mathbb{N}} \mathbb{E}[Y_n^2] < \infty$ and since $\mathbb{E}[X] = 0$ we get

$$\sum_{n \in \mathbb{N}} \mathbb{E}[Y_n] = \sum_{n \in \mathbb{N}} |\mathbb{E}[X_n \mathbf{1}_{\{|X_n| \leq 1\}}]| = \sum_{n \in \mathbb{N}} |\mathbb{E}[X_n \mathbf{1}_{\{|X_n| < 1\}}]| \leq \sum_{n \in \mathbb{N}} \mathbb{E}[|X_n| \mathbf{1}_{\{|X_n| > 1\}}] < \infty,$$

so the mentioned sums converge. These equalities yield

- $\sum_{n \in \mathbb{N}} \mathbb{E}[Y_n]$ converges and
- $\sum_{n \in \mathbb{N}} \text{Var}(Y_n)$ converges.

Using Markov's inequality

$$\sum_{n \in \mathbb{N}} \mathbb{P}(|X_n| > 1) = \sum_{n \in \mathbb{N}} \mathbb{P}(|X_n| \mathbf{1}_{\{|X_n| > 1\}} > 1) \leq \sum_{n \in \mathbb{N}} \mathbb{E}[|X_n| \mathbf{1}_{\{|X_n| > 1\}}] < \infty.$$

So $\sum_{n \in \mathbb{N}} X_n$ converges $\mathbb{P}$ -a.s. thanks to the three series theorem (theorem 7.5). $\square$

Recall the strong law of large numbers (theorem 6.1). We had that $\mathbb{E}[X_n] < \infty$ must hold true in order to apply the theorem. This condition is necessary to get a finite limit as the following proposition will show

Proposition 7.9. If $\{X_n\}_{n \in \mathbb{N}}$ are independent identically distributed random variables s.t. there is $C \in \mathbb{R}$ with

$$\frac{S_n}{n} \xrightarrow{n \rightarrow \infty} C \quad \mathbb{P}\text{-a.s.}$$

Then $\mathbb{E}[X_1] < \infty$ and $C = \mathbb{E}[X_1]$.

Proof. If $\frac{S_n}{n} \xrightarrow{n \rightarrow \infty} C$ $\mathbb{P}$ -a.s. then

$$\frac{X_n}{n} = \frac{S_n - S_{n-1}}{n} = \underbrace{\frac{S_n}{n}}_{\rightarrow C} - \underbrace{\frac{S_{n-1}}{n-1}}_{\rightarrow C} \cdot \underbrace{\frac{n-1}{n}}_{\rightarrow 1} \xrightarrow{n \rightarrow \infty} 0 \quad \mathbb{P}\text{-a.s.}$$

hence $\mathbb{P}(|X_n| > n \text{ infinitely often}) = 0$. By Borel-Cantelli we have

$$\sum_{n \in \mathbb{N}} \mathbb{P}(|X_1| > n) < \infty. \quad (7.6)$$

Observe that for a non-negative random variable X it holds that

$$\sum_{n \in \mathbb{N}} \mathbb{P}(X \geq n) \leq \mathbb{E}[X] \leq 1 + \sum_{n \in \mathbb{N}} \mathbb{P}(X \geq n). \quad (7.7)$$

Indeed,

$$\begin{aligned} \sum_{n \in \mathbb{N}} \mathbb{P}(X \geq n) &= \sum_{n \in \mathbb{N}} \sum_{k \geq n} \mathbb{P}(k \leq X \leq k+1) = \sum_{k \in \mathbb{N}} k \mathbb{P}(k \leq X < k+1) \\ &= \sum_{k \in \mathbb{N}} \mathbb{E}[k \cdot \mathbf{1}_{\{k \leq X < k+1\}}] \leq \sum_{k \in \mathbb{N}} \mathbb{E}[X \cdot \mathbf{1}_{\{k \leq X < k+1\}}] \\ &= \mathbb{E}[X] \end{aligned}$$

proving the first bound. For an upper bound we look at

$$\begin{aligned}
\mathbb{E}[X] &\leq \sum_{k \in \mathbb{N}} (k+1) \mathbb{E}[\mathbb{1}_{\{k \leq X < k+1\}}] = \sum_{k \in \mathbb{N}} (k+1) \mathbb{P}(k \leq X < k+1) \\
&= \sum_{k \in \mathbb{N}} k \mathbb{P}(k \leq X < k+1) + \sum_{k \in \mathbb{N}} \mathbb{P}(k \leq X < k+1) \\
&= \sum_{n \in \mathbb{N}} \sum_{k \geq n} \mathbb{P}(k \leq X < k+1) + \sum_{k \in \mathbb{N}} \mathbb{P}(k \leq X < k+1) \\
&= \sum_{n \in \mathbb{N}} \mathbb{P}(X \geq n) + \underbrace{\sum_{k \in \mathbb{N}} \mathbb{P}(k \leq X < k+1)}_{=1} = \sum_{n \in \mathbb{N}} \mathbb{P}(X \geq n) + 1
\end{aligned}$$

Plugging (7.6) into (7.7) we get that $\mathbb{E}[|X_1|] < \infty$ so we can use the strong law of large numbers (theorem 6.1) and get $\frac{S_n}{n} \xrightarrow{n \rightarrow \infty} \mathbb{E}[X_1]$ $\mathbb{P}$ -a.s. so $\mathbb{E}[X_1] = C$. $\square$

Remark 7.10. If $\mathbb{E}[X_1]$ is not finite then the strong law of large number still holds true with $\frac{S_n}{n} \xrightarrow{n \rightarrow \infty} \infty$ $\mathbb{P}$ -a.s. For example take $\mathbb{E}[X_1^+] = \infty$ and $\mathbb{E}[X_1^-] < \infty$. For $C > 0$ define

$$S_n^{(C)} := \sum_{k=1}^n X_k \mathbb{1}_{\{X_k \leq C\}}.$$

Then

$$\liminf_{n \rightarrow \infty} \frac{S_n}{n} \geq \liminf_{n \rightarrow \infty} \frac{S_n^{(C)}}{n} = \liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n X_k \mathbb{1}_{\{X_k \leq C\}} = \mathbb{E}[X_1 \mathbb{1}_{\{X_1 \leq C\}}]$$

using the strong law of large numbers at the last equality because $[X_k \mathbb{1}_{\{X_k \leq C\}}] < \infty$.

Taking the limit $C \rightarrow \infty$ we have

$$\liminf_{n \rightarrow \infty} \frac{S_n}{n} \geq \lim_{C \rightarrow \infty} \mathbb{E}[X_1 \mathbb{1}_{\{X_1 \leq C\}}] = \mathbb{E}[X_1] = \infty.$$

$\triangle$

Remember that the strong law of large numbers concerns $\frac{S_n}{n} = \sum_{k=1}^n \frac{X_k}{n}$. Question: Is there a link between the convergence of $S_n = \sum_{k=1}^n X_k$ and the convergence of $\frac{S_n}{n}$?

Lemma 7.11 (Kronecker's lemma). Let $a_n \uparrow \infty$ and $\sum_{n \in \mathbb{N}} \frac{x_n}{a_n} < \infty$ for $\{x_n\}_{n \in \mathbb{N}} \subset \mathbb{R}$ then

$$\lim_{n \rightarrow \infty} \frac{1}{a_n} \sum_{k=1}^n x_k = 0.$$

Proof. Let us fix $a_0 := 0$ and define $\{b_k\}_{k \in \mathbb{N}}$ with $b_0 := 0$ and $b_k := \sum_{i=1}^k \frac{x_i}{a_i}$. Then

$x_k = a_k(b_k - b_{k-1})$. Consider the following sequence:

$$\begin{aligned}
 \frac{1}{a_n} \sum_{k=1}^n x_k &= \frac{1}{a_n} \sum_{k=1}^n a_k(b_k - b_{k-1}) = \frac{1}{a_n} \left(\sum_{k=1}^n a_k b_k - \sum_{k=1}^n a_k b_{k-1} \right) \\
 &= \frac{1}{a_n} \left(a_n b_n + \sum_{k=1}^{n-1} a_k b_k - \sum_{k=1}^n a_k b_{k-1} \right) \\
 &= \frac{1}{a_n} \left(a_n b_n + \sum_{k=2}^n a_{k-1} b_{k-1} - \sum_{k=1}^n a_k b_{k-1} \right) \\
 &\stackrel{(a_0=0)}{=} b_n + \sum_{k=1}^n \frac{a_k - a_{k-1}}{a_n} b_{k-1}.
 \end{aligned}$$

Observe that

$$\sum_{k=1}^n (a_k - a_{k-1}) = a_n \xrightarrow{n \rightarrow \infty} \infty.$$

Thanks to our assumption we know that $\{b_n\}_{n \in \mathbb{N}}$ is a convergent sequence so there is b_∞ s.t. $\lim_{n \rightarrow \infty} b_n = b_\infty$. Let $B := \sup_n |b_n|$ and fix $\varepsilon, M > 0$ such that for every $n \geq M$ it holds that $|b_n - b_\infty| < \frac{\varepsilon}{2}$. Moreover, pick $n \in \mathbb{N}$ such that $a_n/a_n < \varepsilon/(4B)$. Then for $n \geq M$

$$\begin{aligned}
 \left| \sum_{k=1}^n \frac{a_k - a_{k-1}}{a_n} b_{k-1} - b_\infty \right| &\leq \sum_{k=1}^n \left| \frac{a_k - a_{k-1}}{a_n} b_{k-1} - b_\infty \right| \\
 &= \sum_{k=1}^n \left| \frac{a_k - a_{k-1}}{a_n} b_{k-1} - \frac{a_k - a_{k-1}}{a_n} b_\infty + \frac{a_k - a_{k-1}}{a_n} b_\infty - b_\infty \right| \\
 &\leq \sum_{k=1}^n \frac{a_k - a_{k-1}}{a_n} |b_{k-1} - b_\infty| + \underbrace{\sum_{k=1}^n \left| \frac{a_k - a_{k-1}}{a_n} - 1 \right| \cdot |b_\infty|}_{=0} \\
 &\leq \sum_{k=1}^M \frac{a_k - a_{k-1}}{a_n} \underbrace{|b_k - b_\infty|}_{\leq 2B} + \sum_{k>M} \frac{a_k - a_{k-1}}{a_n} |b_k - b_\infty| \\
 &\leq 2B \cdot \underbrace{\frac{a_M}{a_n}}_{\leq \varepsilon/(4B)} + \frac{\varepsilon}{2} \underbrace{\frac{a_n - a_M}{a_n}}_{\rightarrow 1} < \varepsilon.
 \end{aligned}$$

hence $\sum_{k=1}^n \frac{a_k - a_{k-1}}{a_n} b_{k-1} \xrightarrow{n \rightarrow \infty} \lim_{n \rightarrow \infty} b_n$. Using the definition of b_k we have

$$\frac{1}{a_n} \sum_{k=1}^n x_k = b_n - \sum_{k=1}^n \frac{a_k - a_{k-1}}{a_n} b_{k-1} \xrightarrow{n \rightarrow \infty} 0$$

thanks to the calculations above proving the point. □

Theorem 7.12 (Revised version of the strong law of large numbers). Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of i.i.d. random variable with $\mathbb{E}[X_n] = 0$ and $\mathbb{E}[X_n^2] = \sigma^2 \in (0, \infty)$ for every $n \in \mathbb{N}$. Then for $\varepsilon > 0$ it holds

$$\frac{S_n}{\sqrt{n}(\log(n))^{1/2+\varepsilon}} \xrightarrow{n \rightarrow \infty} 0 = \mathbb{E}[X_1] \quad \mathbb{P}\text{-a.s.}$$

Proof. Let $a_n := \sqrt{n}(\log(n))^{1/2+\varepsilon}$ for $n \geq 2$ and assume that $a_1 > 0$. Then

$$\begin{aligned} \sum_{n \in \mathbb{N}} \text{Var} \left(\frac{X_n}{a_n} \right) &= \sum_{n \in \mathbb{N}} \frac{1}{a_n^2} \text{Var}(X_n) = \sum_{n \in \mathbb{N}} \frac{1}{a_n^2} \mathbb{E}[X_n^2] = \sigma^2 \sum_{n \in \mathbb{N}} \frac{1}{a_n^2} \\ &= \sigma^2 \left[\frac{1}{a_1^2} + \sum_{n=2}^{\infty} \frac{1}{a_n^2} \right] = \sigma^2 \left[\frac{1}{a_1^2} + \sum_{n=2}^{\infty} \frac{1}{n(\log(n))^{1+2\varepsilon}} \right] < \infty. \end{aligned}$$

This allows us to use theorem 7.3 implying $\mathbb{P}(\bar{\Omega}) = 1$, where $\bar{\Omega}$ is the convergence set of $\sum_{n=2}^k \frac{1}{n^{1/2}(\log(n))^{1/2+\varepsilon}}$. Then

$$\sum_{n \in \mathbb{N}} \frac{X_n}{n^{1/2}(\log(n))^{1/2+\varepsilon}} < \infty \quad \mathbb{P}\text{-a.s.}$$

and $a_n = n^{1/2}(\log(n))^{1/2+\varepsilon} \uparrow \infty$ as $n \rightarrow \infty$ so we can use lemma 7.11 obtaining that

$$\lim_{n \rightarrow \infty} \frac{1}{a_n} \sum_{k=1}^n X_k = \lim_{n \rightarrow \infty} \frac{S_n}{n^{1/2}(\log(n))^{1/2+\varepsilon}} = 0 \quad \mathbb{P}\text{-a.s.}$$

hence the result. □

8 Central limit theorem and its consequences

For this chapter there are a number of preliminary notions which we will first see and prove later.

8.1 Preliminary notions

Recall the definition of the distribution function (definition 2.18): The *distribution* of a random variable X is the measure satisfying

$$\mu_X((-\infty, x]) = \mathbb{P}(X \leq x) =: F_X(x)$$

where F_X is called the *distribution function*. We can also write for every $B \in \mathcal{S}$

$$\mu_X(B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B)$$

for a random variable $X : \Omega \rightarrow S$ where $(\Omega, \mathcal{A}, \mathbb{P})$ and $(S, \mathcal{S}, \mu_X)$ are probability spaces.

Definition 8.1 (Convergence in distribution). Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of real valued random variables. We say that X_n **converges in distribution** as $n \rightarrow \infty$ (Notation: $X_n \xrightarrow{d} X$ as $n \rightarrow \infty$) if

$$F_{\mu_{X_n}}(x) \xrightarrow{n \rightarrow \infty} F_{\mu_X}(x)$$

for every x point of continuity of F_{μ_X} .

This is in fact a special case of *weak convergence* or *weak $\star$ -convergence* which we will introduce later (be careful it is not the same as in analysis).

Definition 8.2 (Characteristic function). Let μ be a probability measure on $\mathbb{R}$. The **characteristic function of μ** is the function

$$\hat{\mu} : \mathbb{R} \rightarrow \mathbb{C}, \quad t \mapsto \int_{\mathbb{R}} e^{itx} d\mu(x).$$

The **characteristic function of a random variable X** is the characteristic function of its distribution which we will denote with φ_X and it satisfies $\varphi_X(t) := \mathbb{E}[e^{itX}] =: \hat{\mu}_X(t)$.

The characteristic function is like the *Fourier transform* of X . Next we will see two properties of these characteristic functions:

Remark 8.3. φ_X as defined in definition 8.2 satisfies:

1. Let X be a random variable and $a, b \in \mathbb{R}$ then

$$\varphi_{aX+b}(t) = e^{ibt} \varphi_X(at).$$

2. Let X, Y be two independent random variables then

$$\varphi_{X+Y}(t) = \varphi_X(t) \varphi_Y(t).$$

△

Lemma 8.4. Let X be a random variable with distribution μ_X . Assume that

$$\mathbb{E}[|X|^k] = \int |x|^k d\mu(x) < \infty$$

for some $k \in \mathbb{N}$. Then φ_X is k times differentiable. In particular

$$\frac{d^l}{dt^l} \hat{\mu}(t) = \int_{\mathbb{R}} (ix)^l e^{ixt} d\mu(x)$$

for $l = 0, \dots, k$. Moreover

$$\frac{d^l}{dt^l} \hat{\mu}(0) = i^l \mathbb{E}[X^l].$$

We will prove this later.

Lemma 8.5. If $\hat{\mu}_{X_n}(t) \xrightarrow{n \rightarrow \infty} \varphi(t)$ for every $t \in \mathbb{R}$ and φ is continuous at $t = 0$ then there is a probability measure such that $\hat{\mu} = \varphi$ and $\mu_{X_n} \xrightarrow{w} \mu$ (weak convergence) i.e. for every $\psi \in \mathcal{C}_b(\mathbb{R})$,

$$\int_{\mathbb{R}} f(x) d\mu_{X_n}(x) \xrightarrow{n \rightarrow \infty} \int_{\mathbb{R}} f(x) d\mu(x)$$

where $\mathcal{C}_b(\mathbb{R}) = \{f \in \mathcal{C}^0(\mathbb{R}) : f \text{ is bounded}\}$.

8.2 Central limit theorem and its proof

Theorem 8.6 (Central limit theorem). Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of identically distributed independent random variables s.t. $\sigma^2 = \text{Var}(X_i) \in (0, \infty)$ for every $i \in \mathbb{N}$. Furthermore define $m := \mathbb{E}[X_i] = \mathbb{E}[X_1]$. Let

$$Z_n := \frac{\sum_{i=1}^n X_i - n \cdot m}{\sqrt{\sigma^2 n}}.$$

Then as $n \rightarrow \infty$

$$Z_n \xrightarrow{d} \mathcal{N}(0, 1).$$

In the proof we will use the following fact whose proof is left as an exercise to the reader:

Exercise 8.7. The characteristic function of a normal distribution $\mathcal{N}(m, \sigma^2)$ is given by

$$\varphi_X(t) = e^{imt - \frac{t^2 \sigma^2}{2}}$$

where $X \sim \mathcal{N}(m, \sigma^2)$. Take $m = 0$ and $\sigma^2 = 1$ then $\varphi_X(t) = e^{-\frac{t^2}{2}}$.

Proof of theorem 8.6. We want to use lemma 8.5 and show convergence of the characteristic functions. Define $S_n := X_1 + \cdots + X_n$. Consider the centered random variable $\tilde{X}_i = X_i - \mathbb{E}[X_i]$ and $\tilde{S}_n := \tilde{X}_1 + \cdots + \tilde{X}_n = S_n - n \cdot m$ its centered partial sum. Then

$$\varphi_{\frac{\tilde{S}_n}{\sqrt{\sigma^2 n}}}(t) = \varphi_{Z_n}(t) = \mathbb{E}[e^{itZ_n}] = \mathbb{E}\left[\exp\left(it \frac{\tilde{S}_n}{\sqrt{\sigma^2 n}}\right)\right] = \mathbb{E}\left[\exp\left(it \frac{\sum_{i=1}^n X_i - n \cdot m}{\sqrt{\sigma^2 n}}\right)\right].$$

Using independence (see remark 8.3) we have

$$\begin{aligned} \mathbb{E}\left[\exp\left(it \frac{\sum_{i=1}^n X_i - n \cdot m}{\sqrt{\sigma^2 n}}\right)\right] &= \prod_{i=1}^n \mathbb{E}\left[\frac{it \tilde{X}_i}{\sqrt{\sigma^2 n}}\right] = \prod_{i=1}^n \mathbb{E}\left[\exp\left(\frac{it \tilde{X}_1}{\sqrt{\sigma^2 n}}\right)\right] \\ &= \mathbb{E}\left[\exp\left(\frac{it \tilde{X}_1}{\sqrt{\sigma^2 n}}\right)\right]^n = \left(\varphi_{\tilde{X}_1}\left(\frac{t}{\sqrt{\sigma^2 n}}\right)\right)^n \end{aligned}$$

because the random variables are identically distributed by assumption. By assumption $\text{Var}(X_1) < \infty$ and therefore $\text{Var}(\tilde{X}_1) < \infty$ which implies $\mathbb{E}[\tilde{X}_1^2] < \infty$. By lemma 8.4 $\varphi_{\tilde{X}_1}$ is twice-differentiable so

$$\frac{d}{dt} \varphi_{\tilde{X}_1}(0) = \mathbb{E}[i \tilde{X}_1 \exp(i \cdot 0 \cdot \tilde{X}_1)] = i \cdot \mathbb{E}[\tilde{X}_1] = 0$$

and

$$\frac{d^2}{dt^2} \varphi_{\tilde{X}_1}(0) = -\mathbb{E}[\tilde{X}_1^2] = -\text{Var}(\tilde{X}_1) = -\sigma^2$$

where we used dominated convergence to take the derivative inside the expectation. Now we use Taylor expansion around $t = 0$:

$$\begin{aligned} \varphi_{Z_n}(t) &= \left(\varphi_{\tilde{X}_1}\left(\frac{t}{\sqrt{\sigma^2 n}}\right)\right)^n \\ &= \left(\varphi_{\tilde{X}_1}(0) + \frac{d}{dt} \varphi_{\tilde{X}_1}(0) \frac{t}{\sigma \sqrt{n}} + \frac{\frac{d^2}{dt^2} \varphi_{\tilde{X}_1}(0)}{2!} \cdot \left(\frac{t}{\sigma \sqrt{n}}\right)^2 + \varepsilon\left(\frac{t}{\sigma \sqrt{n}}\right)\right)^n \\ &= \left(1 - \frac{t^2}{2n} + \varepsilon\left(\frac{t}{\sigma \sqrt{n}}\right)\right)^n = \exp\left(n \cdot \log\left(1 - \frac{t^2}{2n} + \varepsilon\left(\frac{t}{\sigma \sqrt{n}}\right)\right)\right) \\ &\xrightarrow{n \rightarrow \infty} e^{-\frac{t^2}{2}} = \varphi_X(t). \end{aligned}$$

So we have shown that $\varphi_{Z_n}(t) \xrightarrow{n \rightarrow \infty} e^{-\frac{t^2}{2}} = \varphi_X(t)$ where $X \sim \mathcal{N}(0, 1)$. Furthermore $\varphi_X(t)$ is continuous at $t = 0$ because $e^{-\frac{t^2}{2}} \xrightarrow{t \rightarrow 0} 1 = e^0$. So we are in the position of applying lemma 8.5 so for every $\psi \in \mathcal{C}_b(\mathbb{R})$,

$$\int_{\mathbb{R}} \psi(x) d\mu_{Z_n}(x) \xrightarrow{n \rightarrow \infty} \int_{\mathbb{R}} \psi(x) d\mu(x)$$

holds where μ is the normal distribution. Recall that $Z_n \xrightarrow{d} X$ if $F_{\mu_X}(x) \xrightarrow{n \rightarrow \infty} F_{\mu_X}(x)$ for every x point of continuity of F_{μ_X} . We pick $\psi \in \mathcal{C}_b(\mathbb{R})$ (continuous and bounded). Let us restrict on the set $S \subset \mathbb{R}$ of points of continuity of F_{μ_X} .

We can pick $\psi \in \mathcal{C}_b(S)$ s.t.

$$\psi(x) = \mathbf{1}_{(-\infty, x]}$$

and conclude that for every $x \in S$,

$$F_{\mu_{Z_n}}(x) = \int_{\mathbb{R}} \mathbf{1}_{(-\infty, x]} d\mu_{Z_n}(x) \xrightarrow{n \rightarrow \infty} \int_{\mathbb{R}} \mathbf{1}_{(-\infty, x]} d\mu_X(x) = F_{\mu_X}(x)$$

by definition 8.1 we have that $Z_n \xrightarrow{d} X$ with $X \sim \mathcal{N}(0, 1)$. $\square$

Remark 8.8 (Berry-Essen theorem). It is a quantitative version of the central limit theorem: let $\{X_n\}_{n \in \mathbb{N}}$ be identically distributed independent random variables with $\mathbb{E}[X_n] = 0$, $\mathbb{E}[X_n^2] = \sigma^2$ for every $n \in \mathbb{N}$. Assume moreover $\mathbb{E}[|X_n^3|] = \rho < \infty$ then $F_{\mu_{S_n}/\sigma\sqrt{n}}$ is the distribution function of $S_n/\sigma\sqrt{n}$ with

$$\left| F_{\frac{\mu_{S_n}}{\sigma\sqrt{n}}}(x) - F_{\mu}(x) \right| \leq \frac{3\rho}{\sigma^2\sqrt{n}},$$

where μ is $\mathcal{N}(0, 1)$. $\triangle$

Remark 8.9. The central limit theorem tells us informations about $S_n \in (a\sqrt{n}, b\sqrt{n})$ for n large but we don't get local information i.e. on intervals smaller than $\sqrt{n}$. $\triangle$

Theorem 8.10 (Local CLT). Let $\{X_n\}_{n \in \mathbb{N}}$ be a i.i.d. sequence of random variables with $\mathbb{E}[X_n] = 0$, $\mathbb{E}[X_n^2] = \sigma^2 < \infty$. Assume that the characteristic function $\varphi(t)$ s.t. $|\varphi(t)| < 1$ for every $t \in \mathbb{R} \setminus \{0\}$. Then if $\frac{x_n}{\sigma\sqrt{n}} \xrightarrow{n \rightarrow \infty} x$, $a < b$,

$$\sqrt{n}\mathbb{P}(S_n \in (x_n + a, x_n + b)) \xrightarrow{n \rightarrow \infty} \frac{b - a}{\sqrt{2\pi}}.$$

Proof. The proof of this is long and quite technical. The interested reader can find it in the literature. The theorem is not really important for the course it is more meant as a remark to see what is possible. $\square$

Remark 8.11 (Comments on the strong law of large numbers). Recall the weak law of large numbers from “Einführung in die Statistik”: Consider it under the independence assumption (as for the strong law of large numbers). Then the weak law of large numbers tells us

$$\frac{S_n}{n} = \frac{\sum_{k=1}^n X_k}{n} \xrightarrow{\mathbb{P}} \mathbb{E}[X_1] \quad \text{as } n \rightarrow \infty$$

i.e.

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(\left|\frac{S_n}{n} - \mathbb{E}[X_1]\right| < \varepsilon\right) = 1 \quad \text{for every } \varepsilon > 0.$$

For the strong law of large numbers we have instead convergence almost everywhere that is

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \frac{S_n}{n} = \mathbb{E}[X_1]\right) = 1.$$

The convergence in the weak law of large numbers means that for n large $\frac{S_n}{n} \sim \mathbb{E}[X_1]$ likely close. Thus it might happen that

$$\left|\frac{S_n}{n} - \mathbb{E}[X_1]\right| > \varepsilon \quad \text{infinitely often,}$$

i.e. for every $n \in \mathbb{N}$,

$$\left|\frac{S_n}{n} - \mathbb{E}[X_1]\right| \neq 0.$$

On the contrary in the strong law of large numbers this option is not admissible because the convergence is a.e. However it is not uniform convergence. In particular the strong law of large number generally does *not* imply that

$$\forall \varepsilon > 0 \quad \exists \bar{n} \in \mathbb{N} \text{ s.t. } \mathbb{P}\left(\left|\frac{S_n}{n} - \mathbb{E}[X_1]\right| < \varepsilon\right) = 1 \quad \forall n \geq \bar{n}.$$

△

9 Weak convergence of measures and random variables

In the following let (S, d) be a complete metric space and $\mathcal{S} = \mathcal{B}((S, d))$, the Borel σ -algebra generated by the topology of (S, d) , $\mathcal{C}_b(S) = \{f : S \rightarrow \mathbb{R} \mid f \in \mathcal{C}^0 \text{ and bounded}\}$ and let μ be a probability measure on $\mathcal{S}$.

9.1 Definition and basic properties

We already made use of this definition in the last chapter but we will state the basic properties again for clearness

Definition 9.1 (Weak convergence of measures). let μ_n be a sequence of prob. measures on $(S, \mathcal{S})$. We say that μ_n **converges weakly** to μ if

$$\int_S \varphi(x) d\mu_n(x) \xrightarrow{n \rightarrow \infty} \int_S \varphi(x) d\mu(x), \quad \forall \varphi \in \mathcal{C}_b.$$

We will write $\mu_n \xrightarrow{w} \mu$ (or $\mu_n \xrightarrow{*} \mu$ in Analysis).

Remark 9.2. There are other possible notions of convergence for measures. For example $\lim_{n \rightarrow \infty} \mu_n(A) = \mu(A)$ for any $A \in \mathcal{S}$ (pointwise convergence). Or consider

$$\lim_{n \rightarrow \infty} \sup_{A \in \mathcal{S}} \underbrace{|\mu_n(A) - \mu(A)|}_{\text{total variation}} = 0$$

which would be uniform convergence. However those notion are too strong in many situations. $\triangle$

Example 9.3. Let $\{x_n\}_{n \in \mathbb{N}}$ with $x_n \xrightarrow{n \rightarrow \infty} x$. and $x_n \neq x$ for any $n \in \mathbb{N}$. and define $\mu_n = \delta_{x_n}$ and $\mu = \delta_x$. But then we don't have pointwise convergence because $\mu(\{x\}) = 1$ and $\mu_n(\{x\}) = 0$ for every $n \in \mathbb{N}$. But for every $\varphi \in \mathcal{C}_b(S)$ we have

$$\int_S \varphi(y) d\mu_n(y) = \varphi(x_n) \xrightarrow{n \rightarrow \infty} \varphi(x) = \int_S \varphi(y) d\mu(y).$$

So our definition of weak convergence would be satisfied. $\triangle$

Theorem 9.4 (Portmanteau). Let $\{\mu_n\}_{n \in \mathbb{N}}$ be a sequence of probability measures and μ a probability measure on $(S, \mathcal{S})$. The following conditions are equivalent:

1. $\mu_n \xrightarrow{n \rightarrow \infty} \mu$
2. $\limsup_{n \rightarrow \infty} \mu_n(F) \leq \mu(F)$ for every $F \subset S$ closed
3. $\liminf_{n \rightarrow \infty} \mu_n(G) \geq \mu(G)$ for every $G \subset S$ open
4. $\lim_{n \rightarrow \infty} \mu_n(A) = \mu(A)$ for every $A \in \mathcal{S}$ with $\mu(\partial A) = 0$.

Proof. • “(1) $\Rightarrow$ (2)”: Let $F \subseteq S$ not necessarily closed. Define for

$$d(x, F) = \inf_{y \in F} d(x, y) \quad \text{for } x \in S.$$

Note that $d(\cdot, F) : S \rightarrow [0, \infty)$ is a Lipschitz continuous function: Indeed, let $x, y \in S$ and look at

$$d(x, F) = \inf_{z \in F} d(x, z) \leq d(x, y) + \inf_{z \in F} d(y, z) \leq d(x, y) + d(y, F)$$

so

$$d(x, F) - d(y, F) \leq d(x, y).$$

If we change the role of x and y we have that

$$d(y, F) - d(x, F) \leq d(y, x) = d(x, y)$$

hence

$$|d(y, F) - d(x, F)| \leq d(x, y)$$

i.e. $d(\cdot, F)$ is Lipschitz continuous.

Now take $F \subset S$ closed and so $d(x, F) = 0$ if and only if $x \in \overline{F} = F$. Fix $\varepsilon > 0$ and define

$$\rho_\varepsilon(x) = \begin{cases} 1 & x \leq 0 \\ 1 - \frac{x}{\varepsilon} & x \in (0, \varepsilon) \\ 0 & x > \varepsilon \end{cases}.$$

Then consider

$$\varphi_\varepsilon : \mathbb{R} \rightarrow [0, \infty), \quad x \mapsto \rho_\varepsilon(d(x, F)),$$

which clearly satisfies $\varphi_\varepsilon \in \mathcal{C}_b(S)$ for every $\varepsilon > 0$.

Furthermore it holds that $\varphi_\varepsilon(x) = 1$ if and only if $x \in \overline{F} = F$. But $\varphi_\varepsilon(x) = 0$ implies that $d(x, F) > \varepsilon$ hence

$$\mathbb{1}_F(x) \leq \varphi_\varepsilon(x) \leq \mathbb{1}_{F_\varepsilon} \quad \text{for every } x \in S, \tag{9.1}$$

where $F_\varepsilon = \{x \in S : d(x, F) < \varepsilon\}$. Using the fact that $\mu_n \xrightarrow{w} \mu$ we see that

$$\lim_{n \rightarrow \infty} \int_S \underbrace{\varphi_\varepsilon(x)}_{\geq \mathbb{1}_F(x)} d\mu_n(x) = \int_S \varphi_\varepsilon(x) d\mu(x) \tag{9.2}$$

which gives

$$\begin{aligned} \limsup_{n \rightarrow \infty} \mu_n(F) &= \limsup_{n \rightarrow \infty} \int_S \mathbf{1}_F d\mu_n \stackrel{(9.1)}{\leq} \limsup_{n \rightarrow \infty} \int_S \varphi_\varepsilon d\mu_n \stackrel{(9.2)}{=} \int_S \varphi_\varepsilon d\mu \\ &\stackrel{(9.1)}{\leq} \int_S \mathbf{1}_{F_\varepsilon} d\mu = \mu(F_\varepsilon), \end{aligned}$$

for every $\varepsilon > 0$. If we now take a sequence $\{\varepsilon_n\}_{n \in \mathbb{N}}$ with $\varepsilon_n \downarrow 0$ then we get by taking the limit on both sides that

$$\lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} \mu_n(F) \leq \lim_{k \rightarrow \infty} \mu(F_{\varepsilon_k}) \stackrel{\text{lemma 2.2}}{=} \mu\left(\bigcap_{k=1}^{\infty} F_{\varepsilon_k}\right) = \mu(F)$$

hence $\limsup_{n \rightarrow \infty} \mu_n(F) \leq \mu(F)$ for every F closed.

- “(2) $\Rightarrow$ (3)”: Take $G \subset S$ open so G^C is closed. We know that

$$\limsup_{n \rightarrow \infty} \mu_n(G^C) \leq \mu(G^C)$$

by assumption. But using that $\mu_n(G^C) = 1 - \mu_n(G)$ we get that

$$1 - \liminf_{n \rightarrow \infty} \mu_n(G) \leq 1 - \mu(G).$$

- “(3) $\Rightarrow$ (2)” is the same as “(2) $\Rightarrow$ (3)”.
- “(2) & (3) $\Rightarrow$ (4)”: Let $A \in \mathcal{S}$. Take the interior part of A notated $\overset{\circ}{A}$ and the closure $\overline{A}$. then

$$\mu(\overset{\circ}{A}) \stackrel{2.}{\leq} \liminf_{n \rightarrow \infty} \mu_n(\overset{\circ}{A}) \leq \liminf_{n \rightarrow \infty} \mu_n(A) \leq \limsup_{n \rightarrow \infty} \mu_n(A) \leq \limsup_{n \rightarrow \infty} \mu_n(\overline{A}) \stackrel{3.}{\leq} \mu(\overline{A})$$

If $\mu(\partial A) = 0$ we can use that $\partial A = \overline{A} \setminus \overset{\circ}{A}$ hence

$$0 = \mu(\partial A) = \mu(\overline{A} \setminus \overset{\circ}{A}) = \mu(\overline{A}) - \mu(\overset{\circ}{A})$$

so $\mu(\overline{A}) = \mu(\overset{\circ}{A})$.

- “(4) $\Rightarrow$ (1)”: Assume $\mu_n(A) \xrightarrow{n \rightarrow \infty} \mu(A)$ for any $A \in \mathcal{S}$ with $\mu(\partial A) = 0$. Now we fix $f \in \mathcal{C}_b(S)$ and we decompose the range of f as follows:

$$[\inf f, \sup f] \subseteq \bigcup_{j=1}^N [c_j, c_{j+1})$$

with $c_1 \leq c_2 \leq \dots \leq c_{N+1}$ and $|c_{j+1} - c_j| < \varepsilon$ and $\mu(f^{-1}(c_j)) = 0$ for every j . To see that such points exist consider the map $t \in \mathbb{R} \mapsto \mu(f^{-1}((-\infty, t]))$ which is non-decreasing, so it has at most finitely many discontinuities. Note that $t_0 \in \mathbb{R}$ is a point of discontinuity if and only if $\mu(f^{-1}(t_0)) > 0$. Consider $g_\varepsilon = \sum_{j=1}^N c_j \mathbf{1}_{A_j}$ with

$A_j := f^{-1}([c_j, c_{j+1}))$. This implies $\partial A_j = f^{-1}(\{c_j\}) \cup f^{-1}(\{c_{j+1}\})$ so $\mu(\partial A_j) = 0$ by the choice of points. Looking at

$$\begin{aligned} \lim_{n \rightarrow \infty} \int_S g_\varepsilon d\mu_n &= \lim_{n \rightarrow \infty} \int_S \sum_{j=1}^N c_j \mathbb{1}_{A_j} d\mu_n = \lim_{n \rightarrow \infty} \sum_{j=1}^N \int_S c_j \mathbb{1}_{A_j} d\mu_n \\ &= \lim_{n \rightarrow \infty} \sum_{j=1}^N c_j \mu_n(A_j) \stackrel{(4)}{=} \sum_{j=1}^N c_j \mu(A_j) = \int_S g_\varepsilon d\mu \end{aligned}$$

gives us that the weak convergence holds for simple functions. Moreover

$$\left| \int_S f d\mu_n - \int_S f d\mu \right| \leq \underbrace{\left| \int_S f - g_\varepsilon d\mu_n \right|}_{\leq \varepsilon \cdot \mu_n(S) = \varepsilon} + \underbrace{\left| \int_S g_\varepsilon d\mu_n - \int_S g_\varepsilon d\mu \right|}_{\xrightarrow{n \rightarrow \infty} 0} + \underbrace{\left| \int_S g_\varepsilon - f d\mu \right|}_{\leq \mu(S) = \varepsilon}$$

so taking the limsup as $n \rightarrow \infty$ yields

$$\limsup_{n \rightarrow \infty} \left| \int_S f d\mu_n - \int_S f d\mu \right| \leq 2\varepsilon.$$

Since we chose $\varepsilon > 0$ arbitrarily, this limit has to be zero, thus validating the claim. $\square$

Now we are going to analyze weak convergence in $\mathbb{R}$:

Theorem 9.5. Consider a sequence of probability measures $\{\mu_n\}_{n \in \mathbb{N}}$ on $\mathbb{R}$. Let $F_n(t) = \mu_n((-\infty, t])$ and $F(t) = \mu((-\infty, t])$. Then the following conditions are equivalent:

1. $\mu_n \xrightarrow{w} \mu$
2. $F_n(t) \xrightarrow{n \rightarrow \infty} F(t)$ for every $t \in \mathbb{R}$ point of continuity of F .
3. There exists a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and random variables $Y_n, Y : \Omega \rightarrow \mathbb{R}$ s.t. μ_n is the distribution of Y_n and μ is the distribution of Y and $Y_n(\omega) \xrightarrow{n \rightarrow \infty} Y(\omega)$ for $\mathbb{P}$ -a.e $\omega \in \Omega$.

Remark 9.6. The last point can be generalized for any metric space. $\triangle$

Proof of theorem 9.5. Let

$$\text{Disc}(F) := \{t \in \mathbb{R} : F \text{ discontinuous at } t\} = \{t \in \mathbb{R} : \mu(\{t\}) > 0\}.$$

- "(1) $\Rightarrow$ (2)": Let $t_0 \in \mathbb{R} \setminus \text{Disc}(F)$. From theorem 9.4 we have that $\lim_{n \rightarrow \infty} \mu_n(A) = \mu(A)$ for every $A \in \mathcal{B}(\mathbb{R})$ with $\mu(\partial A) = 0$. So taking $A = (-\infty, t_0]$ hence $\partial A = \{t_0\}$ satisfies those conditions, thus

$$\lim_{n \rightarrow \infty} F_n(t_0) = \lim_{n \rightarrow \infty} \mu_n(A) = \mu(A) = F(t_0).$$

- "(3) $\Rightarrow$ (1)": Take $\varphi \in \mathcal{C}_b(\mathbb{R})$. Then we look at

$$\int_{\mathbb{R}} \varphi d\mu_n = \int_{\Omega} \varphi \circ Y_n d\mathbb{P} \xrightarrow{n \rightarrow \infty} \int_{\Omega} \varphi \circ Y d\mathbb{P} = \int_{\mathbb{R}} \varphi d\mu$$

which holds by dominated convergence. We can apply dominated convergence because $|\varphi(Y_n(\omega))| \leq \sup_{x \in \mathbb{R}} (\varphi(x)) < \infty$ since φ is bounded. Also $Y_n \rightarrow Y$ a.e. implies $\varphi \circ Y_n \rightarrow \varphi \circ Y$ a.e. because φ is continuous.

- "(2) $\Rightarrow$ (3)": Take $(\Omega, \mathcal{A}, \mathbb{P}) = ((0, 1), \mathcal{B}((0, 1)), \lambda|_{(0, 1)})$. Consider $Y_n(\omega) = \sup_{y \in \mathbb{R}} \{F_n(y) < \omega\}$ and $Y(\omega) = \sup_{y \in \mathbb{R}} \{F(y) < \omega\}$ for every $n \in \mathbb{N}$.¹ As in the proof of Lebesgue-Stieltjes theorem μ_n is the distribution of Y_n and μ is the distribution of Y . To check pointwise convergence, set

$$\tilde{Y}(\omega) := \inf_{y \in \mathbb{R}} \{F(y) > \omega\} \quad \text{and} \quad \Omega_0 := \{\omega \in (0, 1) : Y(\omega) = \tilde{Y}(\omega)\}.$$

The function $\omega \mapsto Y(\omega)$ is nondecreasing and the set $\text{Disc}(F)$ is at most countable so $\mathbb{P}(\text{Disc}(F)) = 0$. Moreover for any $\omega > \omega'$ we have that

$$Y(\omega) \leq \tilde{Y}(\omega) \leq Y(\omega') \leq \tilde{Y}(\omega').$$

Hence, if $\omega_0 \in \Omega \setminus \Omega_0$ then $\omega_0 \in \text{Disc}(F)$ i.e.

$$\Omega \setminus \Omega_0 \subseteq \text{Disc}(Y)$$

which implies thanks to the monotonicity of the probability measure $\mathbb{P}$,

$$\mathbb{P}(\Omega \setminus \Omega_0) \leq \mathbb{P}(\text{Disc}(Y)) = 0$$

therefore $\mathbb{P}(\Omega_0) = 1$.

So it is enough to show the convergence for every $\omega \in \Omega_0$: Fix $\omega_0 \in \Omega_0$ and claim that $Y_n(\omega_0) \xrightarrow{n \rightarrow \infty} Y(\omega_0)$. To validate this claim take $y \in \mathbb{R} \setminus \text{Disc}(F)$ such that $y < Y(\omega_0)$. Then $F(y) < \omega_0$ and $F_n(y) < \omega_0$ for n large enough so

$$y \leq \liminf_{n \rightarrow \infty} Y_n(\omega_0).$$

Take a sequence $y_k \uparrow Y(\omega_0)$ of points of continuity for F .² Then we have that

$$Y(\omega_0) = \sup_{k \in \mathbb{N}} y_k \leq \liminf_{n \rightarrow \infty} Y_n(\omega_0). \quad (9.3)$$

On the other hand, for y , any point of continuity of F with $Y(\omega_0) < y$ we have that $\tilde{Y}(\omega_0) < y$ since $\omega_0 \in \Omega_0$. Thus we deduce that $F_\mu(y) > \omega_0$. As before for n large enough we get that $F_{\mu_n}(y) > \omega_0$ yielding

$$\limsup_{n \rightarrow \infty} Y_n(\omega_0) \leq y.$$

Choosing $\{y_k\}_{k \in \mathbb{N}}$ such that F_μ is continuous at y_k and $y_k \downarrow Y(\omega_0)$ we have that

$$\limsup_{n \rightarrow \infty} Y_n(\omega_0) \leq \inf_{k \in \mathbb{N}} y_k \leq Y(\omega_0) \quad (9.4)$$

so (9.3) and (9.4) imply

$$\lim_{n \rightarrow \infty} Y_n(\omega_0) = Y(\omega_0).$$

¹We call those functions the *pseudo inverse* of F and F_n .

²this is possible since F_μ has at most countably many points of discontinuity.

□

There is actually a more general theorem:

Theorem 9.7 (Skorokhod's representation theorem). Let $\{\mu_n\}_{n \in \mathbb{N}}$, μ be a probability measures on $(S, \mathcal{S})$ such that $\mu_n \xrightarrow{w} \mu$. There exists $\{X_n\}_{n \in \mathbb{N}}$, X a sequence of random variables on $(\Omega, \mathcal{A}, \mathbb{P})$ s.t. μ_n is the distribution function of X_n for every $n \in \mathbb{N}$ and μ is the distribution of X and $X_n(\omega) \xrightarrow{\mathbb{P}\text{-a.s.}} X(\omega)$. It is possible to take $(\Omega, \mathcal{A}, \mathbb{P}) = ((0, 1), \mathcal{B}((0, 1)), \lambda|_{(0,1)})$.

Proof. We won't prove this in this lecture (nicht prüfungsrelevant). □

Definition 9.8. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, (S, d) a complete metric space and $X_n, X : \Omega \rightarrow S$ random variables. $X_n \rightarrow X$ in law (or distribution) if $\mu_n \xrightarrow{w} \mu$ where μ_n is the distribution of X_n for every $n \in \mathbb{N}$ and μ is the distribution of X .

Example 9.9. Consider $\{x_n\}_{n \in \mathbb{N}} \subset \mathbb{R}$ with $\lim_{n \rightarrow \infty} x_n = x$ and $x_n \neq x$ for every $n \in \mathbb{N}$. Let $\mu_n = \delta_{x_n}$ and $\mu = \delta_x$. The distribution functions would look like

$$F_{\mu_n}(t) = \begin{cases} 1 & t < x_n \\ 0 & t > x_n \end{cases}$$

and

$$F_{\mu}(t) = \begin{cases} 1 & t < x \\ 0 & t > x \end{cases}.$$

Assume that $x_n \downarrow x$ then $F_{\mu_n}(x) = 0$ and $F_{\mu}(x) = 1$ so there is no pointwise convergence in general at discontinuous points of F_{μ} . △

Example 9.10. Let $(X_i)_{i \geq 1}$ a sequence of i.i.d. random variabes with $\mathbb{E}[X_i] = m$ and $\mathbb{E}[X_i^2] = \sigma^2$. Then the central limit theorem implies that

$$\lim_{n \rightarrow \infty} \mathbb{P}(Z_n \leq u) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^u e^{-\frac{x^2}{2}} dx$$

where $Z_n = \sum_{i=1}^n \frac{(X_i - m)}{\sigma\sqrt{n}}$. So $Z_n \xrightarrow{l} Z$ with some $Z \sim \mathcal{N}(0, 1)$ but it does not converge in probability. △

9.2 Tightness

Example 9.11. Let $\mu_n = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_n$ where δ is the Dirac measure. We would like to write " $\mu_n \rightarrow \mu = \frac{1}{2}\delta_0$ ". Check that

$$\int_{\mathbb{R}} \varphi(x) d\mu_n(x) \xrightarrow{n \rightarrow \infty} \int_{\mathbb{R}} \varphi(x) d\mu(x) \quad \text{for every } \varphi \in \mathcal{C}_c^0(\mathbb{R}).$$

1. We cannot say that $\mu_n \xrightarrow{w} \mu$.
2. μ is not a probability measure.

△

Definition 9.12 (Tightness). A $\mathcal{F}$ family of probability measures on a metric space (X, d) is **tight** if for every $\varepsilon > 0$ there exists $K_\varepsilon \subseteq X$ compact such that $\sup_{\mu \in \mathcal{F}} \mu(X \setminus K_\varepsilon) \leq \varepsilon$.

Remark 9.13. $\sup_{\mu \in \mathcal{F}} \mu(X \setminus K_\varepsilon) \leq \varepsilon$ from definition 9.12 is equivalent to

$$\mu(K_\varepsilon) \geq 1 - \varepsilon \quad \text{for every } \mu \in \mathcal{F}.$$

A family of probability measures is tight whenever “most” of the mass of all $\mu \in \mathcal{F}$ is concentrated on a common compact set. $\triangle$

Definition 9.14 (Relative compactness). $\mathcal{F}$, a family of probability measures on a metric space (X, d) is **sequentially relatively compact with respect to the weak convergence** if for every sequence of probability measures $\{\mu_n\}_{n \in \mathbb{N}} \subset \mathcal{F}$, there is a subsequence $\{\mu_{n_k}\}_{k \in \mathbb{N}}$ and a probability measure μ such that $\mu_{n_k} \xrightarrow{w} \mu$.

Remark 9.15. The definition doesn't require $\mu \in \mathcal{F}$. $\triangle$

In fact we can prove that the two definitions stated above are equivalent. However the proof of this is quite long which is the reason we won't do it in this lecture.

Theorem 9.16 (Prokhorov). Let $\mathcal{F}$ be a family of probability measures on a (X, d) a complete metric space. Then the following are equivalent

1. $\mathcal{F}$ is tight
2. $\mathcal{F}$ is sequentially relatively weakly compact.

Proof. Omitted, see functional analysis. $\square$

Theorem 9.17 (Helly's selection). Let $\{F_n\}_{n \in \mathbb{N}}$ be a sequence of distribution functions on $\mathbb{R}$. There exists a subsequence $\{F_{n_k}\}_{k \in \mathbb{N}}$ and $F : \mathbb{R} \rightarrow [0, 1]$ non decreasing and right continuous such that $F(y) = \lim_{k \rightarrow \infty} F_{n_k}(y)$ for every $y \in \mathbb{R}$, point of continuity of F .

Remark 9.18. Note that the theorem doesn't require F to be a distribution function. If we would assume in addition that $\lim_{t \rightarrow -\infty} F(t) = 0$ and $\lim_{t \rightarrow \infty} F(t) = 1$ then there exists a probability measure μ on $\mathbb{R}$ such that $\mu_{n_k} \xrightarrow{w} \mu$ for some subsequence. $\triangle$

Proof of theorem 9.17. We take a numeration of the rational numbers $\mathbb{Q} = \{q_1, q_2, \dots\}$. For any $i \in \mathbb{N}$, $\{F_n(q_i)\}_{n \in \mathbb{N}}$ is a sequence in $[0, 1]$. Thus we find a subsequence $\{n_1(k)\}_{k \in \mathbb{N}}$ and a constant $G(q_1) \in [0, 1]$ such that

$$F_{n_1(k)}(q_1) \xrightarrow{k \rightarrow \infty} G(q_1).$$

Similarly we find a subsequence $\{n_2(k)\}_{k \in \mathbb{N}}$ of $\{n_1(k)\}_{k \in \mathbb{N}}$ and $G(q_2) \in [0, 1]$ such that

$$F_{n_2(k)}(q_2) \xrightarrow{k \rightarrow \infty} G(q_2).$$

Iterating this procedure, we find a diagonal subsequence $\{n(k)\}_{k \in \mathbb{N}}$ and a constant $G(q_i) \in [0, 1]$ such that

$$F_{n(k)}(q_i) \xrightarrow{k \rightarrow \infty} G(q_i)$$

for any $i \in \mathbb{N}$.

If we think of G as a function $G : \mathbb{Q} \rightarrow [0, 1]$ then this is a non decreasing function. Thus, defining

$$F : \mathbb{R} \rightarrow [0, 1], \quad t \mapsto \inf_{t < q, q \in \mathbb{Q}} \{G(q)\},$$

then F is non decreasing and right-continuous.

Now we prove that $F_{n(k)} \rightarrow F(t)$ for every $t \in \mathbb{R}$, point of continuity of F : Let $y \in \mathbb{R}$ be a point of continuity for F . Then, for any $\varepsilon > 0$ we find $r_1, r_2, s \in \mathbb{Q}$ with $r_1 < r_2 < y < s$ such that

$$F(y) - \varepsilon < F(r_1) \leq F(r_2) \leq F(y) \leq F(s) \leq F(y) + \varepsilon.$$

Then $F_{n(k)}(r_2) \xrightarrow{k \rightarrow \infty} G(r_2) \geq F(r_1)$ and $F_{n(k)}(s) \xrightarrow{k \rightarrow \infty} G(s) \leq F(s)$. Thus for k large enough, we have that

$$F(y) - \varepsilon \leq F_{n(k)}(r_2) \leq F_{n(k)}(y) \leq F_{n(k)}(s) \leq F(y) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary we have that

$$\lim_{k \rightarrow \infty} F_{n(k)}(y) = F(y),$$

thus proving the theorem. □

Corollary 9.19. Let $\{F_n\}_{n \geq 1}$ be a family of distribution functions and let μ_n be the associated probability measures. Assume that $\{\mu_n\}_{n \geq 1}$ is tight. Then $\{\mu_n\}_n$ is weakly relatively compact.

Proof. Fix a subsequence $\{n_k\}_{k \geq 1}$. We have to show that there exists a further subsequence $\{n_{k_l}\}_{l \in \mathbb{N}}$ such that $\mu_{n_{k_l}} \xrightarrow{w} \mu$ for some probability measure μ .

By theorem 9.17 we find a subsequence $\{\mu_{k_l}\}_{l \in \mathbb{N}}$ and F such that

- $F_{n_{k_l}}(t) \rightarrow F(t)$ for every $t \in \mathbb{R}$, point of continuity of F .
- F is right-continuous and non decreasing.

If we show that F is the distribution function of a probability measure μ . Then we conclude that $\mu_{n_{k_l}} \xrightarrow{w} \mu$ by theorem 9.5.

Indeed, it is enough to check that $\lim_{t \rightarrow -\infty} F(t) = 0$ and $\lim_{t \rightarrow \infty} F(t) = 1$: Fix $\varepsilon > 0$. By tightness, we find $M > 0$ such that

$$\mu_{n_k}([-M, M]^C) \leq \varepsilon \quad \text{for every } k \in \mathbb{N}.$$

Thus, for any $y_1 < -M$ and $y_2 > M$ continuity points of F , we have that

$$F(y_1) = \lim_{l \rightarrow \infty} F_{n_{k_l}}(y_1) \leq \limsup_{l \rightarrow \infty} \mu_{n_{k_l}}((-\infty, -M)) \leq \varepsilon$$

and

$$F(y_2) = \lim_{l \rightarrow \infty} F_{n_{k_l}}(y_2) \geq \liminf_{l \rightarrow \infty} \mu_{n_{k_l}}((-\infty, M)) \geq 1 - \varepsilon$$

yielding

$$\lim_{t \rightarrow -\infty} F(t) = 0 \quad \text{and} \quad \lim_{t \rightarrow \infty} F(t) = 1$$

thanks to the fact that we chose $\varepsilon > 0$ arbitrarily. □

Remark 9.20. Indeed, corollary 9.19 is a first step to prove theorem 9.16. △

10 Characteristic function

We have already seen this definition but we will repeat for clearness.

Definition 10.1 (Characteristic function). Let μ be a probability measure on $\mathbb{R}$. We define the **characteristic function** of μ as

$$\hat{\mu} : \mathbb{R} \rightarrow \mathbb{C}, \quad t \mapsto \int_{\mathbb{R}} e^{itx} d\mu(x)$$

and if X is a random variable we denote

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = \hat{\mu}_X(t).$$

Remark 10.2. If μ has a density f , then we can write the characteristic function as

$$\hat{\mu}(t) = \int_{\mathbb{R}} e^{itx} f(x) dx$$

the *Fourier transform* of f . △

Example 10.3. • Poisson Distribution: Let $X \sim \text{Pois}(\lambda)$ hence $\mathbb{P}(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$ for $k \geq 0$. Then we can compute

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = \sum_{k=0}^{\infty} e^{itk} \frac{\lambda^k}{k!} e^{-\lambda} = e^{\lambda(e^{it} - 1)}.$$

• Uniform distribution: Let $X \sim U[a, b]$. We again compute

$$\varphi_X(t) = \int_a^b \frac{e^{itx}}{b-a} dx = \frac{1}{it} \frac{(e^{itb} - e^{ita})}{(b-a)}.$$

These are two important cases. △

Lemma 10.4 (Properties of the characteristic function). Let μ be a probability measure on $\mathbb{R}$. Then

1. $\hat{\mu}(0) = 1$,
2. $|\hat{\mu}(t)| \leq 1$ for every $t \in \mathbb{R}$,
3. $\hat{\mu}(-t) = \overline{\hat{\mu}(t)}$ for every $t \in \mathbb{R}$,

4. $\hat{\mu}$ is uniformly continuous,
5. if X is uniformly distributed and $a, b \in \mathbb{R}$ then

$$\varphi_{aX+b}(t) = e^{ibt} \varphi_X(at)$$

and

6. If X, Y are independent random variables then

$$\varphi_{X+Y} = \varphi_X(t) \varphi_Y(t).$$

Proof. The easy proof is left as an exercise to the reader. $\square$

Lemma 10.5. Let X be a random variable with distribution μ_X such that $\mathbb{E}[|X|^k] < \infty$ for some $k \in \mathbb{N}_1$. Then $\varphi_X = \hat{\mu}$ is k -times differentiable and

$$\frac{d^l}{dt^l} \hat{\mu}(t) = \int_{\mathbb{R}} (ix)^l e^{itx} d\mu(x), \quad l = 0, \dots, k.$$

In particular

$$\frac{d^l}{dt^l} \hat{\mu}(t)|_{t=0} = i^l \mathbb{E}[X^l], \quad l = 0, \dots, k$$

Proof. We prove this result by induction on l . For $l = 0$ note that this is just the definition of the characteristic function. For the induction step $l \mapsto l+1$ assume that the proposition holds for some $l < k$. For $x, t \in \mathbb{R}$ and $h > 0$ consider

$$\left| \frac{e^{i(t+h)x} - e^{itx}}{h} \right| = \left| \frac{e^{ihx} - 1}{h} \right| = \left| \frac{1}{h} \int_0^{hx} ie^{iz} dz \right| \leq \frac{1}{h} \int_0^{hx} \underbrace{|ie^{iz}|}_{=1} dz = \frac{1}{h} |hx| = |x|.$$

and

$$|(ix)^l| \cdot |x| = |x|^{l+1} \in L^1(\mu_x)$$

because we have by assumption that for every $l \leq k$ that

$$\mathbb{E}[|X|^l] = \int_{\mathbb{R}} |x|^l d\mu(x) < \infty.$$

So

$$\begin{aligned} \frac{d^{l+1}}{dt^{l+1}} \hat{\mu}_X(t) &= \lim_{h \downarrow 0} \frac{1}{h} \left(\frac{d^l}{dt^l} \hat{\mu}_X(t+h) - \frac{d^l}{dt^l} \hat{\mu}_X(t) \right) \\ &\stackrel{\text{ind.}}{=} \lim_{h \downarrow 0} \int_{\mathbb{R}} (ix)^l \frac{e^{i(t+h)x} - e^{itx}}{h} d\mu_X(x) \\ &\stackrel{\text{dom. conv.}}{=} \int_{\mathbb{R}} (ix)^{l+1} e^{ixt} d\mu_X(x) \end{aligned}$$

which concludes the proof of the first part. To see the second part we just plug in $t = 0$ and one gets the result easily. $\square$

Proposition 10.6. For $a < b$,

$$\lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_{-T}^T \frac{e^{ita} - e^{itb}}{it} \hat{\mu}(t) dt = \mu((a, b)) + \frac{1}{2} \mu(\{a, b\}). \quad (10.1)$$

Proof. We rewrite the term in the following way:

$$\begin{aligned} \frac{1}{2\pi} \int_{-T}^T \frac{e^{ita} - e^{itb}}{it} \hat{\mu}(t) dt &= \frac{1}{2\pi} \int_{-T}^T \int_{\mathbb{R}} \frac{e^{ita} - e^{itb}}{it} e^{itx} d\mu(x) dt \\ &\stackrel{\text{Fubini}}{=} \frac{1}{2\pi} \int_{\mathbb{R}} \int_{-T}^T \frac{e^{ita} - e^{itb}}{it} e^{itx} dt d\mu(x) \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} \int_{-T}^T \frac{\sin(t(x-a))}{t} dt - \int_{-T}^T \frac{\sin(t(x-b))}{t} dt d\mu(x). \end{aligned}$$

Let $J(\alpha, t) := \int_{-T}^T (\sin(\alpha t)/t) dt$. One can show (appendix in Cerny's notes, computations like this won't be asked for in the exam) that

$$\lim_{T \rightarrow \infty} J(\alpha, t) = \pi \operatorname{sign}(\alpha).$$

In our case

$$\lim_{T \rightarrow \infty} J(x-a, t) - J(x-b, t) = \begin{cases} 2\pi, & x \in (a, b) \\ \pi, & x \in \{a, b\} \\ 0, & \text{otherwise} \end{cases}.$$

Observe that $\sup_{\alpha} J(\alpha, t) < \infty$ for every $t \in [-T, T]$. That together with μ being a probability measure allows us to use the dominated convergence theorem and pass the limit inside so

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_{-T}^T \frac{e^{ita} - e^{itb}}{it} \hat{\mu}(t) dt &= \frac{1}{2\pi} \int_{\mathbb{R}} \lim_{T \rightarrow \infty} J(x-a, t) - J(x-b, t) d\mu(x) \\ &= \mu((a, b)) + \frac{1}{2} \mu(\{a, b\}). \end{aligned}$$

□

Theorem 10.7 (Uniqueness). Let μ, ν be two probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Then if $\hat{\mu} = \hat{\nu}$, it follows that $\mu = \nu$.

Proof. This is a direct consequence of proposition 10.6: Said proposition gives us the result for all intervals (a, b) , $a < b$. Using the fact that $\mathcal{D} := \{(a, b) \mid a < b, a, b \in \mathbb{R}\}$ is a Dynkin system we can use Dynkin's lemma to extend it to $\mathcal{B}(\mathbb{R})$. □

We now want to prove tightness to show convergence of a sequence of measures.¹ In particular we aim at showing that the behaviour at ∞ of μ is determined by the behaviour of $\hat{\mu}$ close to 0.

¹See Lévy continuity theorem below, (thm 10.9).

In the proof of the central limit theorem we used that

$$\hat{\mu}_n(t) \rightarrow \varphi(t)$$

with $\varphi(t)$ continuous at 0. Then there exists a probability measure μ such that $\hat{\mu} = \varphi$ and $\mu_n \xrightarrow{w} \mu$. To prove this we need the following lemma:

Lemma 10.8. Let μ be a probability measure on $\mathbb{R}$. For $y > 0$ holds

$$\mu \left(\left\{ x : |x| \geq \frac{2}{y} \right\} \right) \leq \frac{1}{y} \int_{-y}^y (1 - \hat{\mu}(t)) dt.$$

Proof. Let $x \neq 0$,

$$\begin{aligned} \int_{-y}^y (1 - e^{itx}) dt &= 2y - \frac{e^{iyx} - e^{-iyx}}{ix} \\ &\stackrel{(!)}{=} 2y \left(1 - \frac{\sin(yx)}{yx} \right) \geq 0 \end{aligned}$$

using the Euler identity at (!). Now looking at

$$\begin{aligned} \frac{1}{y} \int_{-y}^y 1 - \hat{\mu}(t) dt &\stackrel{(\text{Fubini})}{=} \int_{\mathbb{R}} \frac{1}{y} \int_{-y}^y (1 - e^{itx}) dt d\mu(x) = 2 \int_{\mathbb{R}} \underbrace{\left(1 - \frac{\sin(yx)}{yx} \right)}_{\geq 0} d\mu(x) \\ &\geq 2 \int_{|x| \geq \frac{2}{y}} 1 - \frac{1}{|xy|} d\mu(x) \geq 2 \int_{|x| \geq \frac{2}{y}} \frac{1}{2} d\mu(x) = \mu \left(\left\{ x : |x| \geq \frac{2}{y} \right\} \right) \end{aligned}$$

yields the desired result. $\square$

Theorem 10.9 (Lévy continuity theorem). Let $\{\mu_n\}_{n \in \mathbb{N}}$ be a sequence of probability measures on $\mathbb{R}$.

1. If $\mu_n \xrightarrow{w} \mu$ then $\hat{\mu}_n(t) \rightarrow \hat{\mu}(t)$ for every $t \in \mathbb{R}$
2. If $\hat{\mu}_n(t) \xrightarrow{n \rightarrow \infty} \varphi(t)$ for every $t \in \mathbb{R}$ with $\varphi(t)$ continuous at $t = 0$ then there exists a probability measure μ such that $\hat{\mu}(t) = \varphi(t)$ and $\mu_n \xrightarrow{w} \mu$.

Proof. 1. Follows basically from the definition: Using that $x \mapsto e^{ixt}$ is continuous and bounded for every $t \in \mathbb{R}$ we compute

$$\hat{\mu}_n(t) = \int_{\mathbb{R}} e^{ixt} d\mu_n(x) \xrightarrow{n \rightarrow \infty} \int_{\mathbb{R}} e^{ixt} d\mu(x) = \hat{\mu}(t).$$

2. First prove that $\{\mu_n\}_{n \in \mathbb{N}}$ is tight. For $\varepsilon > 0$ there exists $y > 0$ such that

$$\varepsilon > \frac{1}{y} \int_{-y}^y 1 - \varphi(t) dt$$

because $\varphi(t)$ is continuous at $t = 0$. Using the dominated convergence theorem we can rewrite this as

$$\frac{1}{y} \int_{-y}^y 1 - \varphi(t) \, dt = \lim_{n \rightarrow \infty} \frac{1}{y} \int_{-y}^y 1 - \hat{\mu}_n(t) \, dt.$$

By lemma 10.8 for n sufficiently large

$$\varepsilon > \frac{1}{y} \int_{-y}^y 1 - \hat{\mu}_n(t) \, dt \geq \mu \left(\left[-\frac{2}{y}, \frac{2}{y} \right]^C \right).$$

Then by definition of tightness we have that $\{\mu_n\}_{n \in \mathbb{N}}$ is tight. So there exists a subsequence $\{\mu_{n_k}\}_{k \in \mathbb{N}}$ such that $\mu_{n_k} \xrightarrow{w} \nu$ such that $\hat{\nu} = \varphi$ from point 1. This implies that weak convergence does not depend on the choice of the subsequence $\{\mu_{n_k}\}_{k \in \mathbb{N}}$. Therefore $\mu_n \xrightarrow{w} \nu$ so $\nu = \mu$.

□

11 Conditional expectation

We've seen in "Einführung in die Statistik" that $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$ for $\mathbb{P}(B) > 0$. The *conditional expectation* is similar in spirit and is basically the generalization of the expectation. For example we will see that $\mathbb{E}[X|\{\emptyset, \Omega\}] = \mathbb{E}[X]$ so the normal expectation is just a special case of this.

Definition 11.1 (Conditional expectation). Let $X \in L^1$ a random variable on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and let $\mathcal{G}$ be a sub- σ -algebra of the σ -algebra $\mathcal{A}$. Then we define the **conditional expectation** of X given $\mathcal{G}$ as $\mathbb{E}[X|\mathcal{G}]$ as follows: it is any random variable Z such that

1. Z is $\mathcal{G}$ measurable.
2. $\mathbb{E}[X\mathbb{1}_G] = \mathbb{E}[Z\mathbb{1}_G]$ for every $G \in \mathcal{G}$.

Furthermore we define

$$\mathbb{P}(A|\mathcal{G}) := \mathbb{E}[\mathbb{1}_A|\mathcal{G}]$$

Theorem 11.2. The conditional expectation $\mathbb{E}[X|\mathcal{G}]$ exists and is unique almost everywhere. Moreover if X is non-negative $\mathbb{E}[X|\mathcal{G}] \geq 0$ $\mathbb{P}$ -a.s.

To prove the existence of the conditional expectation we need some measure theory results. Namely we require the theorem of Radon-Nykodym. The proof of the latter is very long which is the reason we won't do it in this course.

Theorem 11.3 (Radon-Nikodym). Let μ, ν be two σ -finite nonnegative measures on $(\Omega, \mathcal{A})$ a measurable space. Let ν be absolutely continuous with regards to μ (meaning $\mu(A) = 0 \Rightarrow \nu(A) = 0$). Then there exists f , a non-negative $\mathcal{A}$ -measurable, integrable function satisfying

$$\nu(A) = \int_A f \, d\mu \quad \forall A \in \mathcal{A}.$$

In fact we call f the Radon-Nikodym derivative of ν with regards to μ , notated as

$$f = \frac{d\nu}{d\mu}$$

Proof of theorem 11.2. We start by proving the existence:

Assume that $X \geq 0$ and take $\mathbb{O}$ a probability measure on $(\Omega, \mathcal{A})$ with

$$\mathbb{O} : \mathcal{A} \rightarrow [0, 1], \quad A \mapsto \int_A X \, d\mathbb{P} = \mathbb{E}[X\mathbb{1}_A].$$

Because of the definition of Radon-Nykodym derivative we have $X = \frac{d\mathbb{O}}{d\mathbb{P}}$. Then we consider $\tilde{\mathbb{P}}$ and $\tilde{\mathbb{O}}$ the restrictions of $\mathbb{P}$ and $\mathbb{O}$ to the sub- σ -algebra $\mathcal{G}$ i.e.

$$\tilde{\mathbb{P}} = \mathbb{P}|_{\mathcal{G}}, \quad \tilde{\mathbb{O}} = \mathbb{O}|_{\mathcal{G}}.$$

Now we check if $\tilde{\mathbb{O}}$ is σ -finite: Clearly

$$\tilde{\mathbb{O}}(\Omega) = \mathbb{O}(\Omega) = \int_{\Omega} X \, d\mathbb{P} = \mathbb{E}[X] < \infty$$

because X is integrable by assumption thus $\tilde{\mathbb{O}}$ is finite and moreover for every $G \in \mathcal{G}$ with $\tilde{\mathbb{P}}(G) = 0$ we have that $\tilde{\mathbb{P}}(G) = \mathbb{P}(G) = 0$. This implies by the Radon-Nykodym theorem that $0 = \mathbb{P}(G) = \mathbb{O}(G) \stackrel{G \in \mathcal{G}}{=} \tilde{\mathbb{O}}(G)$. So it follows that $\tilde{\mathbb{O}}$ is absolutely continuous with respect to $\tilde{\mathbb{P}}$. By the Radon-Nykodym theorem there exists a non-negative, $\mathcal{G}$ -measurable random variable Z such that

$$\mathbb{E}[X\mathbf{1}_G] = \tilde{\mathbb{O}}(G) = \int_G Z \, d\tilde{\mathbb{P}} = \int_G Z \, d\mathbb{P} = \mathbb{E}[Z\mathbf{1}_G]$$

for every $G \in \mathcal{G}$. So we checked both parts of definition 11.1 hence result is proven for non-negative random variables.

Now we consider X a general real-valued random variable. As always we can write $X = X^+ - X^-$ and work separately with X^+ and X^- . The same construction we did for X works for X^+ and X^- as they are both non-negative random variables.

$$\mathbb{O}^+(A) = \int_A X^+ \, d\mathbb{P} \quad \text{for every } A \in \mathcal{A}$$

and similarly for the other measures. Proceed as before and obtain that there exists Z^+ and Z^- both integrable and such that $\mathbb{E}[Z^{\pm}\mathbf{1}_G] = \mathbb{E}[Z^{\pm}\mathbf{1}_G]$ for every $G \in \mathcal{G}$. Using that we have

$$\mathbb{E}[X\mathbf{1}_G] = \mathbb{E}[X^+\mathbf{1}_G] - \mathbb{E}[X^-\mathbf{1}_G] = \mathbb{E}[Z^+\mathbf{1}_G] - \mathbb{E}[Z^-\mathbf{1}_G] = \mathbb{E}[Z\mathbf{1}_G] \quad \forall G \in \mathcal{G}.$$

so we get the existence.

We now prove uniqueness: Note that the uniqueness is in a “weak sense” meaning the measures coincide for non-negligible sets. Let Z_1 and Z_2 be two conditional expectations given by $D := Z_1 - Z_2$

- D is $\mathcal{G}$ -measurable because Z_1 and Z_2 are $\mathcal{G}$ measurable.
- $\mathbb{E}[D\mathbf{1}_G] = \mathbb{E}[Z_1\mathbf{1}_G] - \mathbb{E}[Z_2\mathbf{1}_G] = \mathbb{E}[X\mathbf{1}_G] - \mathbb{E}[X\mathbf{1}_G] = 0$ for every $G \in \mathcal{G}$.
- $\mathbb{E}[|D|] = \mathbb{E}[D^+] - \mathbb{E}[D^-] = \mathbb{E}[D\mathbf{1}_{D \geq 0}] + \mathbb{E}[D\mathbf{1}_{D < 0}] = 0$ for the same reasoning.

This implies $D = 0$ $\mathbb{P}$ -a.s. hence the result. $\square$

The following property should help to understand the meaning of the conditional expectation:

Lemma 11.4 (Trivial conditioning). 1. $\mathbb{E}[X|\mathcal{A}] = X$ $\mathbb{P}$ -a.s.

2. If $\mathcal{G} = \{\emptyset, \Omega\}$ then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$.

3. If X a random variable and $\mathcal{G}$ a sub- σ -algebra are independent i.e. $\sigma(X)$ and $\mathcal{G}$ are independent then $\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$.

Proof. The easy proof is left as an exercise to the reader (or see Cerny's notes). Observe that the last formula follows from

$$\mathbb{E}[X\mathbf{1}_G] = \mathbb{E}[X]\mathbb{P}(G) \quad \forall G \in \mathcal{G}.$$

□

Lemma 11.5. If X, Y are integrable random variables the following holds

1. For every $a, b \in \mathbb{R}$, $\mathbb{E}[aX + bY|\mathcal{G}] = a\mathbb{E}[X|\mathcal{G}] + b\mathbb{E}[Y|\mathcal{G}]$.

2. If $X \leq Y$ then $\mathbb{E}[X|\mathcal{G}] \leq \mathbb{E}[Y|\mathcal{G}]$ $\mathbb{P}$ -a.s.

3. If $\{X_n\}_{n \in \mathbb{N}}$ is a sequence of random variables s.t. $X_n \uparrow X$ as $n \rightarrow \infty$ then

$$\mathbb{E}[X_n|\mathcal{G}] \xrightarrow{n \rightarrow \infty} \mathbb{E}[X|\mathcal{G}] \quad \mathbb{P}\text{-a.s.}$$

Proof. 1. The linearity is obvious from the definition.

2. This is an imediate consequence of the last claim in theorem 11.2 and the linearity.

3. Let $Y_n := X - X_n$. It then suffices to show that $\mathbb{E}[Y_n|\mathcal{G}] \downarrow 0$. Since Y_n is decreasing, 2. implies that $Z_n = \mathbb{E}[Y_n|\mathcal{G}]$ is decreasing and therefore a limit Z exists $\mathbb{P}$ -a.s. If $G \in \mathcal{G}$ then $\mathbb{E}[Z_n\mathbf{1}_G] = \mathbb{E}[Y_n\mathbf{1}_G]$. Since $0 \leq Y_n \leq X$ we see using the dominated convergence theorem that $\mathbb{E}[Z\mathbf{1}_G] = 0$. The same argument as the one for the uniqueness of the conditional expectation applies here implying that $Z = 0$ $\mathbb{P}$ -a.s.

□

Proposition 11.6. Let X be a $\mathcal{G}$ -measurable random variable and X, Y integrable. Then

$$\mathbb{E}[XY|\mathcal{G}] = X\mathbb{E}[Y|\mathcal{G}] \quad \mathbb{P}\text{-a.s.}$$

Remark 11.7. For this proposition we don't need any independence △

Proof. Again, the easy proof is left as an exercise to the reader. Hint: Prove it for indicator functions, then linear combinations of indicator functions. After that use the density of indicator functions to prove it for non-negative functions and so on. □

Proposition 11.8 (Tower property). Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be sub- σ -algebras of $\mathcal{A}$ with $\mathcal{G}_1 \subset \mathcal{G}_2 \subset \mathcal{A}$. Then

1. $\mathbb{E}[\mathbb{E}[Z|\mathcal{G}_1]|\mathcal{G}_2] = \mathbb{E}[Z|\mathcal{G}_1]$,
2. $\mathbb{E}[\mathbb{E}[X|\mathcal{G}_2]|\mathcal{G}_1] = \mathbb{E}[X|\mathcal{G}_1]$ and
3. $\mathbb{E}[\mathbb{E}[X|\mathcal{G}_2]] = \mathbb{E}[X]$.

Proof. 1. $\mathbb{E}[X|\mathcal{G}_1]$ is $\mathcal{G}_2$ -measurable because $\mathcal{G}_1 \subset \mathcal{G}_2$. Apply proposition 11.6 (with $X = \mathbb{E}[Z|\mathcal{G}_1]$ and $Y = 1$, that is X, Y in said proposition). So we get

$$\mathbb{E}[\mathbb{E}[Z|\mathcal{G}_1]|\mathcal{G}_2] = \mathbb{E}[Z|\mathcal{G}_1] \underbrace{\mathbb{E}[1|\mathcal{G}_2]}_{=1} = \mathbb{E}[Z|\mathcal{G}_1].$$

2. Pick $G \in \mathcal{G}_1$ and consider

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}_2]\mathbf{1}_G] \stackrel{\mathcal{G}_1 \subset \mathcal{G}_2}{=} \mathbb{E}[X\mathbf{1}_G] = \mathbb{E}[\mathbb{E}[X|\mathcal{G}_1]\mathbf{1}_G] = \mathbb{E}[\mathbb{E}[X|\mathcal{G}_1]|\mathcal{G}_2].$$

Since this holds for any $G \in \mathcal{G}_1$ this implies

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}_2]|\mathcal{G}_1] = \mathbb{E}[\mathbb{E}[X|\mathcal{G}_1]|\mathcal{G}_2] \stackrel{(1)}{=} \mathbb{E}[X|\mathcal{G}_1].$$

3. Choose $\mathcal{G}_1 = \{\emptyset, \Omega\}$. Then

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}_2]] \stackrel{(2)}{=} \mathbb{E}[X|\mathcal{G}_1] \stackrel{\text{lemma 11.4}}{=} \mathbb{E}[X].$$

□

Lemma 11.9 (Jensen's inequality). Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function and X an integrable random variable with $\mathbb{E}[\varphi \circ X] < \infty$. Then

$$\varphi(\mathbb{E}[X|\mathcal{G}]) \leq \mathbb{E}[\varphi \circ X|\mathcal{G}].$$

Proof. Let $\varphi(x) = ax + b$ for $a, b \in \mathbb{R}$. Then the claim directly follows by linearity of the conditional expectation. For general convex functions φ observe that φ can be written as

$$\varphi(x) = \sup_{n \geq 0} \varphi_n(x)$$

with

$$\varphi_n(x) = a_n x + b_n$$

all tangents of φ at rational points. By monotonicity of the conditional expectation

$$\mathbb{E}[\varphi \circ X|\mathcal{G}] \geq \mathbb{E}[\varphi_n \circ X|\mathcal{G}] = \varphi_n(\mathbb{E}[X|\mathcal{G}])$$

so taking the sup on both sides gives us the desired result since the LHS is independent from n . □

For the next corollary we first need the following definition:

Definition 11.10. Let $T : (X, \|\cdot\|_X) \rightarrow (Y, \|\cdot\|_Y)$. We call T a **contraction** if

$$\|T(x)\|_Y \leq \|x\|_X \quad \text{for every } x \in X$$

Corollary 11.11 (Conditional expectation is a contraction in L^p). Let $X \in L^p(\Omega, \mathcal{A}, \mathbb{P})$ with $p \in [1, \infty]$. Let $\mathcal{G} \subset \mathcal{A}$. Then $\mathbb{E}[X|\mathcal{G}] \in L^p(\Omega, \mathcal{A}, \mathbb{P})$ and

$$\|\mathbb{E}[X|\mathcal{G}]\|_{L^p} \leq \|X\|_{L^p}. \quad (11.1)$$

Proof. We first consider the case $p \in [1, \infty)$. It is enough to prove (11.1). Clearly $x \mapsto |x|^p$ is a convex function so using lemma 11.9 we get

$$\|\mathbb{E}[X|\mathcal{G}]\|_{L^p}^p = \mathbb{E}[\|\mathbb{E}[X|\mathcal{G}]\|^p] \leq \mathbb{E}[\mathbb{E}[|X|^p|\mathcal{G}]] \stackrel{\text{prop 11.8}}{=} \mathbb{E}[|X|^p] = \|X\|_{L^p}^p$$

so the result holds for finite p .

Finally let $p = \infty$. Then the monotonicity of the conditional expectation together with $-\|X\|_{L^\infty} \leq X \leq \|X\|_{L^\infty}$ $\mathbb{P}$ -a.e implies

$$\|\mathbb{E}[X|\mathcal{G}]\|_{L^\infty} \leq \mathbb{E}[\|X\|_{L^\infty}|\mathcal{G}] = \|X\|_{L^\infty}.$$

□

Theorem 11.12. Let $\mathcal{G} \subset \Omega$ be a sub- σ -algebra of $\mathcal{A}$, where $(\Omega, \mathcal{A}, \mathbb{P})$ is a probability space. Then

$$\mathbb{E}[\cdot|\mathcal{G}] : L^2(\Omega, \mathcal{A}, \mathbb{P}) \rightarrow L^2(\Omega, \mathcal{G}, \mathbb{P}), \quad X \mapsto \mathbb{E}[X|\mathcal{G}]$$

is an orthogonal projection of X from $L^2(\Omega, \mathcal{A}, \mathbb{P})$ onto $L^2(\Omega, \mathcal{G}, \mathbb{P})$.

Proof. Let $Z \in L^2(\Omega, \mathcal{G}, \mathbb{P})$, implying that $Z \in L^1(\Omega, \mathcal{G}, \mathbb{P})$ by Cauchy Schwartz so Z is $\mathcal{G}$ -measurable. Using prop 11.6 we have that $\mathbb{E}[ZX|\mathcal{G}] = Z\mathbb{E}[X|\mathcal{G}]$ $\mathbb{P}$ -a.s. Consider

$$\mathbb{E}[Z\mathbb{E}[X|\mathcal{G}]] = \mathbb{E}[\mathbb{E}[ZX|\mathcal{G}]] \stackrel{\text{prop 11.8}}{=} \mathbb{E}[ZX]$$

and set $Z := Y - \mathbb{E}[X|\mathcal{G}]$ with $Y \in L^2(\Omega, \mathcal{G}, \mathbb{P})$. Then

$$\begin{aligned} \|X - Y\|_{L^2}^2 &= \mathbb{E}[|X - Y|^2] = \mathbb{E}[(X - \mathbb{E}[X|\mathcal{G}] - Z)^2] \\ &= \mathbb{E}[(X - \mathbb{E}[X|\mathcal{G}])^2] + \mathbb{E}[Z^2] + 2\underbrace{\mathbb{E}[Z(X - \mathbb{E}[X|\mathcal{G}])]}_{=0} \\ &\stackrel{(!)}{\geq} \mathbb{E}[(X - \mathbb{E}[X|\mathcal{G}])^2] \end{aligned}$$

If $Y = \mathbb{E}[X|\mathcal{G}]$ then (!) is an identity. This is an important fact as the orthogonal projection minimizes the distance between two points i.e.

$$\inf_{Y \in L^2(\Omega, \mathcal{G}, \mathbb{P})} \|X - Y\|_{L^2}^2 \geq \|X - \mathbb{E}[X|\mathcal{G}]\|_{L^2}^2$$

implies

$$\min_{Y \in L^2(\Omega, \mathcal{G}, \mathbb{P})} \|X - Y\|_{L^2}^2 = \|X - \mathbb{E}[X|\mathcal{G}]\|_{L^2}^2.$$

□

Definition 11.13. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. The sequence $\{\mathcal{F}_n\}_{n \in \mathbb{N}}$ is a **filtration** if $\mathcal{F}_n$ is a non-decreasing sequence of sub- σ -algebras of $\mathcal{A}$ i.e.

$$\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{A}.$$

Definition 11.14. Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of random variables and $\{\mathcal{F}_n\}_{n \in \mathbb{N}}$ a filtration. We say that $\{X_n\}_{n \in \mathbb{N}}$ is **adapted** to the filtration $\{\mathcal{F}_n\}_{n \in \mathbb{N}}$ if X_k is $\mathcal{F}_k$ measurable for every $k \in \mathbb{N}$.

12 Markov chains

The problem with a lot of the previous results is that they require the random variables to be independent. This doesn't mimick the real life behaviour of random variables. For example for $\{X_n\}_{n \in \mathbb{N}}$ i.i.d. the partial sums $\{S_n\}_{n \in \mathbb{N}}$ already are not independent.

12.1 Definiton and basics of Markov chains

A Markov chain is a sequence of random variables where the random variable at the point n is only dependent on the value at $n-1$. The Markov chain is formally defined as follows:

Definition 12.1 (Markov chain). Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space with a filtration $\{\mathcal{F}_n\}_{n \in \mathbb{N}}$. Let $(S, \mathcal{S})$ be a measurable space. A sequence of $\mathcal{S}$ -measurable random variables $\{X_n\}_{n \in \mathbb{N}}$ is a **Markov chain** with respect to the $\mathcal{F}$ if

1. $\{X_n\}_{n \in \mathbb{N}}$ is adapted to $\{\mathcal{F}_n\}_{n \in \mathbb{N}}$
2. For every $A \in \mathcal{S}$, $n \in \mathbb{N}$

$$\mathbb{P}(X_{n+1} \in A | \mathcal{F}_n) = \mathbb{P}(X_{n+1} \in A | \sigma(X_n))$$

where for a σ -algebra $\mathcal{G} \subset \mathcal{A}$,

$$\mathbb{P}(A | \mathcal{G}) := \mathbb{E}[\mathbf{1}_A | \mathcal{G}]$$

There is an equivalent definition for the discrete-time Markov chain: Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of S -valued random variables defined on $(\Omega, \mathcal{A}, \mathbb{P})$. $\{X_n\}_{n \in \mathbb{N}}$ is a Markov chain with state space S if for every $n \in \mathbb{N}$

$$\mathbb{E}[f \circ X_{n+1} | \mathcal{F}_n] = \mathbb{E}[f \circ X_{n+1} | \sigma(X_n)], \quad \mathbb{P}\text{-a.s.}$$

for any bounded function $f : S \rightarrow \mathbb{R}$.

- Remark 12.2.**
1. We call the Markov chain *memoryless* because of the last property
 2. One can also consider Markov chains with uncountably many indecees so it is not a sequence in the usual sense. However we will only cover the countable case in this lecture.
 3. When we say that $\{X_n\}_{n \in \mathbb{N}}$ is a Markov chain without specifying a filtration this will refer to $\{X_n\}_{n \in \mathbb{N}}$ adapted to $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$

△

Alternatively one can define a Markov chain through the transition probability:

Definition 12.3 (Transition probability). Let $(S, \mathcal{S})$ be measurable space. We call a function $p : S \times \mathcal{S} \rightarrow [0, 1]$ **transition probability** if

1. For every $x \in S$, $p(x, \cdot) : A \mapsto p(x, A)$ is a probability measure on $(S, \mathcal{S})$ and
2. for every $A \in \mathcal{S}$, $p(\cdot, A) : x \mapsto p(x, A)$ is $\mathcal{S}$ -measurable.

A Markov chain $\{X_n\}_{n \in \mathbb{N}}$ has transition probabilities $\{p_n\}_{n \in \mathbb{N}}$ if for every $n \in \mathbb{N}$

$$\mathbb{P}(X_{n+1} \in A | \mathcal{F}_n) = p_{n+1}(X_n, A) \quad \text{for } A \in \mathcal{S}.$$

One example for a transition probability would be the map

$$p : \Omega \times \mathcal{A} \rightarrow \mathbb{R}, \quad (\omega, A) \mapsto \mathbb{P}(A | \mathcal{G})(\omega)$$

for some sub- σ -algebra $\mathcal{G} \subseteq \mathcal{A}$.

Remark 12.4. If $\{p_n\}_{n \in \mathbb{N}}$ transition probabilities for $\{X_n\}_{n \in \mathbb{N}}$ a Markov chain are such that $p_n = p$ for some p for every $n \in \mathbb{N}$ then $\{X_n\}_{n \in \mathbb{N}}$ is a time-homogenous Markov chain. $\triangle$

Definition 12.5. If $\{p_n\}_{n \in \mathbb{N}}$ transition probabilities for $\{X_n\}_{n \in \mathbb{N}}$ a Markov chain are such that $p_n = p$ for some p for every $n \in \mathbb{N}$ then $\{X_n\}_{n \in \mathbb{N}}$ is a **time-homogenous** Markov chain.

Definition 12.6 (Stochastic kernels). Given two measurable spaces $(S_1, \mathcal{S}_1)$ and $(S_2, \mathcal{S}_2)$, a stochastic kernel is a map $K : S_1 \times \mathcal{S}_2 \rightarrow [0, 1]$ such that

1. For every $A \in \mathcal{S}_2$, the map

$$x \in S_1 \mapsto K(x, A)$$

is $\mathcal{S}_1$ -measurable and

2. For every $x \in S_1$, the map

$$x \in \mathcal{S}_2 \mapsto K(x, A)$$

is a probability measure.

From now on we will consider time-homogenous Markov chains. Moreover we will consider the transition probability $\{p_{x,y}\}_{x,y \in S}$ with

1. $p_{x,y} \geq 0$ for every $x, y \in S$
2. $\sum_{y \in S} p_{x,y} = 1$ for every $x \in S$

which hold by definition. If S is finite, e.g. $S = \{1, \dots, k\}$ one can consider the *stochastic matrix* $\mathbf{P} = [p_{x,y}]_{x,y \in S}$ with $p_{x,y}$ satisfying the above points. This matrix therefore corresponds to a transition probability

$$p(x, A) = \sum_{y \in A} p_{x,y}.$$

Remark 12.7. Transition probabilities are stochastic kernels (the proof of this fact is left as an exercise to the reader). $\triangle$

Definition 12.8. A sequence of random variables $\{X_n\}_{n \in \mathbb{N}}$ all S -valued on $(\Omega, \mathcal{A}, \mathbb{P})$ is a Markov chain with state space S and transition probability $\{p_{x,y}\}_{x,y \in S}$ if for every $n \in \mathbb{N}$, $\mathbb{E}[f \circ X_{n+1} | \mathcal{F}_n] = \sum_{y \in S} p_{X_n, y} f(y)$ $\mathbb{P}$ -a.s. for every $f : S \rightarrow \mathbb{R}$ bounded.

Remark 12.9. Given $\{p_{x,y}\}$ transition probability, $p_{x,y}(n)$ for every $x, y \in S$ and $n \in \mathbb{N}$ we can define $p_{x,y}(n)$ inductively as

$$\begin{aligned} p_{x,y}(0) &= \delta_{x,y} \\ p_{x,y}(1) &= p_{x,y} \\ &\vdots \\ p_{x,y}(n+1) &= \sum_{z \in S} p_{x,z}(n) p_{z,y}. \end{aligned}$$

$\triangle$

Proposition 12.10. Consider a sequence of random variables $\{X_n\}_{n \in \mathbb{N}}$ that is a Markov chain with state space S and transition probability $\{p_{x,y}\}_{x,y \in S}$ and initial distribution

$$\mu : S \rightarrow [0, 1], \quad x \mapsto \mathbb{P}(X_0 = x)$$

then for every $n, m \in \mathbb{N}$ we have

1. (Chapman-Kolmogorov equality)

$$p_{x,y}(n+m) = \sum_{z \in S} p_{x,z}(n) p_{z,y}(m)$$

- 2.

$$p_{x,y}(n) = (R^n \mathbf{1}_{\{y\}})(x),$$

where R is a linear operator on the set of bounded functions $f : S \rightarrow \mathbb{R}$ which is defined as follows

$$(Rf)(x) = \sum_{z \in S} p_{x,z} f(z)$$

3. For $x_0, x_1, \dots, x_n \in S$ we have

$$\mathbb{P}(X_0 = x_0, X_1 = x_1, \dots, X_n = x_n) = \mu(x_0) \prod_{i=1}^n p_{x_{i-1}, x_i}.$$

4. For $y \in S$ and $n \in \mathbb{N}$

$$\mathbb{P}(X_n = y) = (\mu R^n)(y)$$

where

$$(\mu R^n)(y) = \sum_{z \in S} \mu(z) (R^n \mathbf{1}_{\{y\}})(z).$$

Proof. The proof is by induction and is left as an exercise to the reader. $\square$

Example 12.11. 1. Simple random walk on $\mathbb{Z}^d$: Then $S = \mathbb{Z}^d$ and the transition probability would be

$$p_{x,y} = \begin{cases} \frac{1}{2d}, & |x - y| = 1 \\ 0, & \text{otherwise} \end{cases}$$

and it clearly satisfies

$$\sum_{y \in \mathbb{Z}^d} p_{x,y} = 1.$$

2. Reflected random walk: In this case we have the state space $S = \mathbb{N}$. Let $p, q > 0$ such that $p + q = 1$. Then

$$p_{x,y} = \begin{cases} 1, & x = 0, y = 1 \\ p, & x \geq 1, y = x + 1 \\ q, & x \geq 1, y = x - 1 \\ 0, & \text{otherwise} \end{cases}.$$

This would model the situation where you are on the natural numbers and your position is n then with probability p you jump to $n + 1$ and with probability q you jump to $n - 1$ (Obviously if you're at 1 you can only jump to the right).

3. Ehrenfest Model of diffusion: Consider two boxes A and B with a certain amount of molecules in it. Let N denote the total number of molecules in the two boxes. At each time step pick a random molecule from one of the two boxes and move it to the other box. So let $S = \{0, 1, 2, \dots, N\}$ be the state space representing the number of molecules in Box A . In this case we would have

$$p_{x,y} = \begin{cases} 1 - \frac{x}{N}, & x < N, y = x + 1 \\ \frac{x}{N}, & x > 0, y = x - 1 \\ 0, & \text{otherwise} \end{cases}.$$

$\triangle$

For the next theorem we need the definition of a “canonical Markov chain” (time-homogenous):

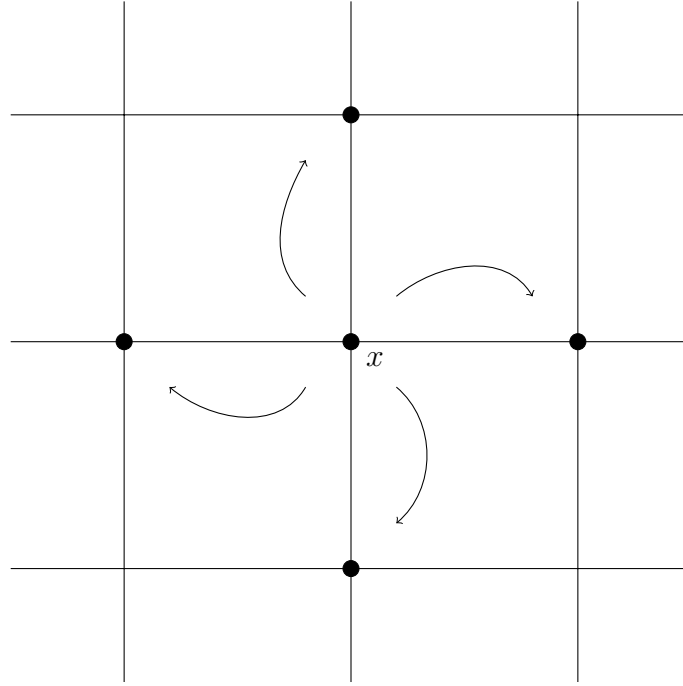
Definition 12.12. Let $\Omega_0 = S^{\mathbb{N}} = \{\text{sequences } \{x_n\}_{n \in \mathbb{N}}, x_n \in S, n \in \mathbb{N}\}$. Then $X_n(\omega) := \omega(n)$ are called **canonical coordinates** for every $n \in \mathbb{N}$ and let

$$\mathcal{A}_0 := \sigma(X_0, X_1, \dots)$$

the **canonical σ -algebra** and

$$\mathcal{F}_n := \sigma(X_0, \dots, X_n)$$

the **canonical filtration**

Figure 12.1: Random walk on $\mathbb{Z}^2$

Proposition 12.13. Let $\{p_{x,y}\}$ be a transition probability on S . For each $z \in S$ there exists a unique probability measure $\mathbb{P}_z$ on $(\Omega_0, \mathcal{A}_0) = (S^{\mathbb{N}}, \sigma(X_0, X_1, \dots))$ such that $\{X_n\}_{n \in \mathbb{N}}$ is a Markov chain on S with transition probabilities $\{p_{x,y}\}_{x,y \in S}$ and initial distribution δ_z with respect to the probability measure $\mathbb{P}_z$.

Furthermore, for μ a probability measure on S , under

$$\mathbb{P}_\mu := \sum_{x \in S} \mu(x) \mathbb{P}_x$$

$\{X_n\}_{n \in \mathbb{N}}$ is a Markov chain with transition probability $\{p_{x,y}\}$ and initial distribution μ .

Notation 12.14. From now on, $\mathbb{P}_z$ will denote this unique probability measure described in proposition 12.13. Furthermore we will use the notation $\mathbb{E}_z$ for the expectation with respect to said probability measure i.e.

$$\mathbb{E}_z[X] := \int_{\Omega} X \, d\mathbb{P}_z$$

for every $X : \Omega \rightarrow S$, random variable.

Proof of proposition 12.13. Given $z \in S$, prop 12.10 allows us to prove only that there exists a unique $\mathbb{P}_z$ such that

$$\mathbb{P}_z(X_0 = x_0, \dots, X_n = x_n) = \delta_z(x_0) \prod_{i=1}^n p_{x_{i-1}, x_i}.$$

For the uniqueness one can apply Dynkin's lemma and it is left as an exercise to the reader. We will prove the existence: Take λ the Lebesgue measure on $((0, 1), \mathcal{B}((0, 1)))$. For each

$x \in S$ let $\Phi : (x, \cdot) : (0, 1) \rightarrow S$ be a measurable function such that the image of λ under Φ is $p_{x, \cdot} = \Phi \circ \lambda$ on S . Let $(\tilde{\Omega}, \tilde{\mathcal{A}}, \tilde{\mathbb{P}})$ be a probability space and $\{U_n\}_{n \in \mathbb{N}}$ an i.i.d $(0, 1)$ -uniformly distributed random variables. Let $\tilde{X}_0 = z$, $\tilde{X}_1 = \Phi(z, U_1)$, $\tilde{X}_2 = \Phi(\tilde{X}_1, U_2), \dots$. So we constructed the sequence $\{\tilde{X}_i\}_{i \in \mathbb{N}}$ such that for any $n \in \mathbb{N}$, $x_0, \dots, x_n \in S$, $\tilde{\mathbb{P}}(\tilde{X}_0 = x_0, \dots, \tilde{X}_n = x_n) = \delta_z(x_0) \prod_{i=1}^n p_{x_{i-1}, x_i}$, which can be checked by induction and holds thanks to the construction of Φ . We then can define

$$\mathbb{P}_z = \text{law of } (\tilde{X}_0, \tilde{X}_1, \dots, \tilde{X}_n)$$

on $(\Omega_0, \mathcal{A}_0)$ and by construction $\mathbb{P}_z$ satisfies Prop 12.10.3.

For the last part let μ be a probability measure on S and let $f : S \rightarrow \mathbb{R}$ be a bounded function. For every $n \in \mathbb{N}$ and $x_0, x_1, \dots, x_n \in S$ we have

$$\begin{aligned} \mathbb{E}_\mu[f \circ X_{n+1} | X_0 = x_0, \dots, X_n = x_n] &= \sum_{x \in S} \mu(x) \mathbb{E}_x[f \circ X_{n+1} | X_0 = x_0, \dots, X_n = x_n] \\ &\stackrel{(!)}{=} \sum_{x \in S} \mu(x) \mathbb{P}_x(X_0 = x_0, \dots, X_n = x_n) (Rf)(x_n) \\ &= \sum_{x \in S} \mu(x) \mathbb{E}[X_0 = x_0, \dots, X_n = x_n, (Rf)(x_n)] \\ &= \mathbb{E}_\mu[(Rf)(X_n), X_0 = x_0, \dots, X_n = x_n] \end{aligned}$$

where we used at (!) that $\{X_n\}_n$ is a Markov chain under $\mathbb{P}_x$. So Markov's property under $\mathbb{P}_x$ follows and $\mathbb{P}_\mu(X_0 = x_0) = \mu(x_0)$ where μ is the initial distribution. This concludes this proof. $\square$

12.2 Markov Property and Strong Markov property

Let

$$\theta_n : \Omega_0 \rightarrow \Omega_0, \quad \theta_n(\omega)(\cdot) = \omega(n + \cdot)$$

with $\Omega_0 = S^\mathbb{N} = \{\{s_n\}_{n \in \mathbb{N}} \subset S : \{s_n\}_n \text{ is a sequence}\}$ so θ_n shifts the sequence by n in time.

Proposition 12.15 (Simple Markov property). If μ is a probability on S and Y a bounded $\mathcal{A}_0$ measurable function on Ω_0 . Then, for every $n \in \mathbb{N}$,

$$\mathbb{E}_\mu[Y \circ \theta_n | \mathcal{F}_n] = \mathbb{E}_{X_n}[Y], \quad \mathbb{P}_\mu\text{-a.s.}$$

where $\mathcal{F}_n$ is the filtration.

Remark 12.16. 1. $\mathbb{E}_{X_n}$ denotes the expectation with respect to the law of X_n .
 2. $Y \circ \theta_n$ is a function of the future (we are looking after time n).
 3. $\mathbb{E}_\mu$ is the expectation with respect to $\mathbb{P}_\mu := \sum_{x \in S} \mu(x) \mathbb{P}_x$, where $\mathbb{P}_x$ is the unique probability on $(\Omega_0, \mathcal{A}_0)$ that makes $\{X_n\}_{n \in \mathbb{N}}$ a Markov chain.

$\triangle$

Proof. We prove the proposition for $Y = \mathbb{1}_A$. The general case follows by density of simple functions.

$\mathcal{F}_n$ is generated by $X_0 = x_0, X_1 = x_1, \dots, X_n = x_n$ with $x_0, x_1, \dots, x_n \in S$. We want to show that given $(x_0 \dots x_n) \in S$

$$\mathbb{E}_\mu[\mathbb{1}_A \circ \theta_n | X_0 = x_0, \dots, X_n = x_n] = \mathbb{P}_\mu(X_0 = x_0, \dots, X_n = x_n) P_{x_n}(A) \quad (12.1)$$

In particular if $A = \{X_0 = y_0, \dots, X_m = y_m\}$ then $\mathbb{1}_A \circ \theta_n = \mathbb{1}_{\{X_n=y_0, X_{n+1}=y_1, \dots, X_{n+m}=y_m\}}$ and (12.1) is given by proposition 12.10.3. Observe that the class of sets A of the form $A = \{X_0 = y_0, \dots, X_m = y_m\}$ or $A = \emptyset$ is a π -system that generates the σ -algebra $\mathcal{A}_0 = \sigma(X_0, X_1, \dots)$. By Dynkin's lemma (12.1) holds true for any $A \in \mathcal{A}_0$. This concludes the proof for indicator functions. The rest follows by linearity and density of simple functions. $\square$

We now look at discrete-time Markov chains with random time.

Definition 12.17. A $\mathbb{N} \cup \{\infty\}$ -valued random variable N is called **stopping time** (with respect to the filtration $\{\mathcal{F}_n\}$) if $\{N = n\} \in \mathcal{F}_n$ for every $n \in \mathbb{N}$.

Definition 12.18. Given a $\{\mathcal{F}_n\}$ -stopping time N , the σ -algebra $\mathcal{F}_N$ (of the past N) is given by

$$\mathcal{F}_N := \{A \in \mathcal{A}_0 : A \cap \{N = n\} \in \mathcal{F}_n, \forall n \in \mathbb{N}\}$$

This allows us to state the strong Markov property:

Theorem 12.19 (Strong Markov property). Let N be a $\{\mathcal{F}_n\}$ -stopping time, Y a bounded and $\mathcal{A}_0$ measurable random variable and μ a probability measure on S . Then

$$\mathbb{E}_\mu[Y \circ \theta_N | \mathcal{F}_N] = \mathbb{E}_{X_N}[Y] \quad \mathbb{P}_\mu\text{-a.s.}$$

on $\{N < \infty\}$.

Remark 12.20. 1. $Y \circ \theta_N$ is a random variable of a random variable meaning that we consider

$$(Y \circ \theta_{N(\omega)})(\omega)$$

if $N(\omega) < \infty$. If $N(\omega) = \infty$ then $Y \circ \theta_N = 0$.

2. $\mathbb{E}_{X_N}[Y]$ is $\mathbb{E}_{X_{N(\omega)}(\omega)}[Y]$ if $N(\omega) < \infty$. If $N(\omega) = \infty$ then $\mathbb{E}_{X_N}[Y] = 0$.

$\triangle$

Proof. On $\{N = n\}$, $\mathbb{E}_{X_N}[Y] = \mathbb{E}_{X_n}[Y]$ is $\mathcal{F}_n$ -measurable, implying that X_N is $\mathcal{F}_N$ -measurable. For $A \in \mathcal{F}_N$

$$\begin{aligned} \mathbb{E}_\mu[Y \circ \theta_N | A \cap \{N < \infty\}] &= \sum_{n \in \mathbb{N}} \mathbb{E}_\mu[Y \circ \theta_N | A \cap \{N = n\}] \\ &= \sum_{n \in \mathbb{N}} \mathbb{E}_\mu[Y \circ \theta_n | A \cap \{N = n\}]. \end{aligned}$$

Note that $\{N = n\} \in \mathcal{F}_n$ so $A \cap \{N = n\} \in \mathcal{F}_n$. Using the simple Markov property we get

$$\begin{aligned} \sum_{n \in \mathbb{N}} \mathbb{E}_\mu[Y \circ \theta_n | A \cap \{N = n\}] &= \sum_{n \in \mathbb{N}} \mathbb{E}_\mu[\mathbb{E}_{X_n}[Y] | A \cap \{N = n\}] \\ &= \mathbb{E}_\mu[\mathbb{E}_{X_N}[Y] | A \cap \{N < \infty\}] \end{aligned}$$

hence

$$\mathbb{E}_\mu[Y \circ \theta_N | A \cap \{N < \infty\}] = \mathbb{E}_\mu[\mathbb{E}_{X_N}[Y] | A \cap \{N < \infty\}]$$

therefore concluding the proof. $\square$

12.3 Recurrence and Transience

Let $\{X_n\}_{n \in \mathbb{N}}$ be a canonical Markov chain with state space S and transition probability $\{p_{x,y}\}_{x,y \in S}$. For $x \in S$ we define

$$\tilde{h}_x := \inf\{n \in \mathbb{N} : X_n = x\}$$

the *hitting time* of x . Observe that $\tilde{h}_x$ is a $\{\mathcal{F}_n\}$ -stopping time: Take

$$\{\tilde{h}_x = k\} = \{X_1 \neq x, \dots, X_{k-1} \neq x, X_k = x\} \in \mathcal{F}_k$$

for $k \in \mathbb{N}$. Moreover $\{\tilde{h}_x = 0\} = \emptyset \in \mathcal{F}_0$. Let $x \in S$ and define

$$h_x := \inf\{n \in \mathbb{N}_0 : X_n = x\}$$

the *entrance time* in $x \in S$. This is also a $\{\mathcal{F}_n\}$ -stopping time. It is left as an exercise to reader to verify that this is true.

Definition 12.21. For $x, y \in S$ let

$$\rho_{x,y} := \mathbb{P}_x(\tilde{h}_y < \infty),$$

the probability of the hitting time being finite.

We say that a state $x \in S$ is **recurrent** if $\rho_{x,x} = 1$ and **transient** if $\rho_{x,x} < 1$.

Let $x \in S$ and let $\tilde{h}_x^n$ be the successive time of visit of the Markov chain to the state x . Iteratively we define

$$\begin{aligned} \tilde{h}_x^0 &= 0 \\ \tilde{h}_x^1 &= \tilde{h}_x \leq \infty \\ \tilde{h}_x^{n+1} &= \tilde{h}_x^n + \tilde{h}_x \circ \theta_{\tilde{h}_x^{n-1}} \leq \infty. \end{aligned}$$

Proposition 12.22. Let $x, y \in S$. Then

$$\mathbb{P}_x(\tilde{h}_y^n < \infty) = \begin{cases} \rho_{x,y}, & n = 1 \\ \rho_{x,y}\rho_{y,y}^{n-1}, & n > 1 \end{cases}.$$

Proof. For $n = 1$ this is exactly the definition. For $n > 1$

$$\mathbb{P}_x(\tilde{h}_y^n < \infty) = \mathbb{P}_x(\tilde{h}_y^{n-1} < \infty \text{ and } \tilde{h}_y \circ \theta_{\tilde{h}_y^{n-1}} < \infty).$$

We can apply the strong Markov property and obtain

$$\mathbb{P}_x(\tilde{h}_y^n < \infty) = \mathbb{E}_x[\tilde{h}_y^{n-1} < \infty, \mathbb{P}_{X_{\tilde{h}_y^{n-1}}}[\tilde{h}_y < \infty]] = \rho_{y,y}\mathbb{P}[\tilde{h}_y^{n-1} < \infty] = \rho_{x,y}\rho_{y,y}^{n-1}.$$

□

Example 12.23 (Renewal processes). If the strong Markov property holds under $\mathbb{P}_x$ we can define

$$S_n = \tilde{h}_x^n, \quad n \geq 0.$$

This random variable S_n has the same distribution as $T_1 + \dots + T_n$, $n \in \mathbb{N}$ for i.i.d. random variables $\{T_n\}_{n \in \mathbb{N}}$ with values in $\mathbb{N} \cup \{\infty\}$ of S_n under $\mathbb{P}_x$.

Counting function: for $t \geq 0$,

$$N_t := \sum_{k \in \mathbb{N}} \mathbb{1}_{\{S_k \leq t\}} = \sup\{n \in \mathbb{N}_0 : S_n \leq t\}$$

is a renewal proces.

For $x \in S$

$$N_x := \sum_{k \in \mathbb{N}} \mathbb{1}_{\{X_k = x\}} = \sum_{k \in \mathbb{N}} \mathbb{1}_{\{\tilde{h}_x^k < \infty\}}. \quad (12.2)$$

△

Proposition 12.24. Let $y \in S$. If y is recurrent, then

$$N_y = \infty, \quad \mathbb{P}_y\text{-a.s.} \quad (12.3)$$

and if y is transient then for $x \in S$

$$\mathbb{E}_x[N_y] = \frac{\rho_{x,y}}{1 - \rho_{y,y}} < \infty \quad (12.4)$$

Proof. To prove (12.4):

$$\mathbb{P}_y(\tilde{h}_y^n < \infty) = 1$$

for $n \in \mathbb{N}$ so (12.4) follows.

To prove (12.3):

$$\mathbb{E}_x[N_y] = \sum_{n \in \mathbb{N}} \mathbb{P}_x[N_y \geq n] = \sum_{n \in \mathbb{N}} \mathbb{P}_x(\tilde{h}_y^n < \infty) = \sum_{n \in \mathbb{N}} \rho_{x,y} \underbrace{\rho_{y,y}^{n-1}}_{< 1} = \frac{\rho_{x,y}}{1 - \rho_{y,y}}$$

where we used in the last step that the sum on the right is a geometric series. □

Corollary 12.25. If S is a finite state space, then there is at least one $y \in S$ that is recurrent.

Proof. We prove this by contradiction. Assume that all $y \in S$ are transient. Then for every $x \in S$

$$\mathbb{E}_x \left[\sum_{y \in S} N_y \right] \stackrel{\text{Prop. 12.15}}{=} \sum_{y \in S} \frac{\rho_{x,y}}{1 - \rho_{y,y}} < \infty.$$

But

$$\sum_{y \in S} N_y = \sum_{y \in S} \sum_{k \in \mathbb{N}} \mathbb{1}_{\{X_k = y\}} = \sum_{k \in \mathbb{N}} 1 = \infty$$

which leads to a contradiction. $\square$

Proposition 12.26. If $x \in S$ is recurrent and let $\rho_{x,y} > 0$ for some $y \in S$. Then y is also recurrent and $\rho_{x,y} = 1$.

Proof. Let $x \neq y$. Define $k := \min\{n \in \mathbb{N} : \mathbb{P}_x(X_n = y) > 0\}$ so

$$\begin{aligned} 0 < \mathbb{P}_x(X_k = y) &= \sum_{x_1, \dots, x_{k-1} \in S \setminus \{y\}} \mathbb{P}_x(X_1 = x_1, \dots, X_{k-1} = x_{k-1}, X_k = y) \\ &= \sum_{x_1, \dots, x_{k-1} \in S \setminus \{y\}} \underbrace{p_{x,x_1} p_{x_1,x_2} \cdots p_{k-1,y}}_{>0}. \end{aligned}$$

By assumption x is recurrent which implies that

$$\begin{aligned} 0 = \mathbb{P}(\tilde{h}_x = \infty) &\geq \mathbb{P}_x(X_1 = x_1, \dots, X_k = y, \tilde{h}_x \circ \theta_k = \infty) \\ &\stackrel{\text{prop 12.10}}{=} \underbrace{p_{x,x_1} \cdots p_{k,y}}_{>0} \mathbb{P}_y(\tilde{h}_x = \infty) \end{aligned}$$

so $0 = \mathbb{P}_y(\tilde{h}_x = \infty) = 1 - \rho_{x,y}$ implying $\rho_{x,y} = 1$.

Now it is left to prove that y is recurrent. From the argumentation above follows that there is $l \geq 1$ such that $\mathbb{P}_y(X_l = x) > 0$. So for every $n \in \mathbb{N}$ and $k := \min\{n \in \mathbb{N} : \mathbb{P}_y(X_n = x) > 0\}$

$$\begin{aligned} \mathbb{P}_y(X_{l+n+k} = y) &\geq \mathbb{P}_y(X_l = x, X_{l+n} = x, X_{l+n+k} = y) \\ &\stackrel{\text{Markov-prop at time } l+n}{=} \mathbb{P}_y(X_l = x, X_{l+n} = x) \mathbb{P}_x(X_k = y) \\ &\stackrel{\text{Markov-prop at time } l}{=} \mathbb{P}_y(X_l = x) \mathbb{P}_x(X_n = x) \mathbb{P}_x(X_k = y). \end{aligned}$$

Putting this to use

$$\begin{aligned} \mathbb{E}[N_y] &\geq \sum_{n \in \mathbb{N}} \mathbb{P}_y(X_{l+n+k} = y) \geq \sum_{n \in \mathbb{N}} \mathbb{P}_y(X_l = x) \mathbb{P}_x(X_n = x) \mathbb{P}_x(X_k = y) \\ &= \underbrace{\mathbb{P}_y(X_l = x)}_{>0} \underbrace{\mathbb{P}_x(X_k = y)}_{>0} \underbrace{\sum_{n \in \mathbb{N}} \mathbb{P}_x(X_n = x)}_{=\infty} \end{aligned}$$

with $\sum_{n \in \mathbb{N}} \mathbb{P}_x(X_n = x) = \infty$ because x is recurrent. This gives us $\mathbb{E}[N_y] = \infty$ making y a recurrent point. $\square$

Remark 12.27. One can partition the state space of a Markov chain into (unions of) recurrent and transient states. $\triangle$

12.4 Stationary distributions

Definition 12.28. Let π be a probability measure (probability distribution) on the state space S . We say that π is a **stationary distribution** for the Markov chain with transitional probability $\{p_{x,y}\}$ if

$$\pi(y) = \sum_{x \in S} \pi(x) p_{x,y} \quad \text{for every } y \in S. \quad (12.5)$$

Remark 12.29. Definition 12.28 is equivalent to

$$\theta_n \circ \mathbb{P}_\pi = \mathbb{P}_\pi \quad \text{for every } n \in \mathbb{N} \quad (12.6)$$

which is exactly what it means to be stationary in time. $\triangle$

We prove that those two definitions are the same:

Proof. for $n = 1$:

- " $\Leftarrow$ ":

$$\begin{aligned} \pi(y) &= \mathbb{P}_\pi(X_0 = y) \stackrel{(12.6)}{=} \theta_1 \circ \mathbb{P}_\pi(X_0 = y) \\ &= \mathbb{P}_\pi(X_1 = y) \stackrel{\text{prop 12.10, 12.13}}{=} \sum_{x \in S} \pi(x) p_{x,y} \end{aligned}$$

which implies (12.5).

- " $\Rightarrow$ ": Let $A \in \Omega_0$ and consider $\theta_1 \circ \mathbb{P}_\pi$. Using properties of the conditional expectation we have

$$\begin{aligned} \theta_1 \circ \mathbb{P}_\pi &= \mathbb{E}_\pi[\mathbb{1}_A \circ \theta_1] = \mathbb{E}_\pi[\mathbb{E}_\pi[\mathbb{1}_A \circ \theta_1 | \mathcal{F}_1]] = \mathbb{E}_\pi[\mathbb{P}_{X_1}(A)] \\ &\stackrel{\text{prop 12.10, 12.13}}{=} \sum_{y \in S} \underbrace{\sum_{x \in S} \pi(x) p_{x,y}}_{\pi(y)} \mathbb{P}_y(A) \\ &= \sum_{y \in S} \pi(y) \mathbb{P}_y(A) = \mathbb{P}_\pi(A) \end{aligned}$$

for $n > 1$ use that $\theta_n = (\theta_1)^n$ so $\theta_1 \circ \mathbb{P}_\pi = \mathbb{P}_\pi$ implies $\theta_n \circ \mathbb{P}_\pi = \mathbb{P}_\pi$. $\square$

Remark 12.30. (12.6) motivates the name stationary: $\mathbb{P}_\pi$ is invariant under time shifts. $\triangle$

Definition 12.31. A probability π on the state space S is a **reversible distribution** of the Markov chain with transition probability $\{p_{x,y}\}$ if the detailed balance condition holds true i.e.

$$\pi(x)p_{x,y} = \pi(y)p_{y,x} \quad \text{for every } x, y \in S$$

Proposition 12.32. A reversible distribution is a stationary distribution.

Remark 12.33. Why do we call it detailed balance? If π is reversible we have

$$\pi(x)p_{x,y} = \mathbb{P}_\pi(X_0 = x, X_1 = y) = \mathbb{P}_\pi(X_0 = y, X_1 = x) = \pi(y)p_{y,x}$$

so the two probabilities in the middle are “balanced”. $\triangle$

Proof. Let us consider π to be a reversible distribution i.e. for $y \in S$

$$\sum_{x \in S} \pi(x)p_{x,y} = \sum_{x \in S} \pi(y)p_{y,x} = \pi(y) \underbrace{\sum_{x \in S} p_{y,x}}_{=1} = \pi(y)$$

which satisfies (12.5). $\square$

Example 12.34. Consider a Markov chain on a finite state space $S = \{1, \dots, N\}$ with transition probability

$$p_{x,y} = \begin{cases} 1 & x = N, y = 1 \\ 1 & x < N, y = x + 1 \\ 0 & \text{otherwise} \end{cases}.$$

We have a probability of 1 to go to the next higher point $s_n \mapsto s_{n+1} = s_n + 1$. Particularly $p_{x,y}p_{y,x} = 0$. So this Markov chain cannot be reversible i.e. cannot have a reversible distribution as it doesn't satisfy (12.5). However if we take the uniform distribution on this set S that is

$$\pi(x) = \frac{1}{N} \quad \text{for every } x \in \{1, \dots, N\}$$

then this is obviously stationary. (Look back at proposition 12.32). $\triangle$

Proposition 12.35. Consider π to be a stationary distribution. Then for $x \in S$ the following implication holds: if $\pi(x) > 0$ then x satisfies $\mathbb{E}_x[\tilde{h}_x] < \infty$ (x positive recurrent).

Proof. Next week :) $\square$

Definition 12.36. A Markov chain $\{X_n\}_{n \in \mathbb{N}}$ with state space S is **irreducible** if for every $x, y \in S$ there exists $n \geq 1$ s.t. $\mathbb{P}_x(X_n = y) > 0$.

Definition 12.37. Let $x \in S$ be a recurrent state. We say that x is **positively recurrent** if $\mathbb{E}_x[\tilde{h}_x] < \infty$. Otherwise we call x **null recurrent** (i.e. $\mathbb{E}_x[\tilde{h}_x] = \infty$).

Definition 12.38. We define

$$N_x(y) := \mathbb{E}_x \left[\sum_{n=1}^{\tilde{h}_x} \mathbb{1}_{\{X_n=y\}} \right] \quad (12.7)$$

and call this the mean number of y before coming back to x .

Proposition 12.39. Assume that $x \in S$ is a recurrent point of a Markov chain $\{X_n\}_{n \in \mathbb{N}}$. Then the following holds true

1. $N_x(x) = 1$
2. $N_x(\cdot)$ is a stationary distribution for the Markov chain $\{X_n\}_{n \in \mathbb{N}}$ and
3. If $\{X_n\}_{n \in \mathbb{N}}$ is irreducible then $N_x(y) \in (0, \infty)$.

Proof. 1. Just apply the definition and you'll see it (see def 12.38 + definition of $\tilde{h}_x$).
2.

$$\begin{aligned} N_x(y) &= \mathbb{E}_x \left[\sum_{n=1}^{\tilde{h}_x} \mathbb{1}_{\{X_n=y\}} \right] = \sum_{n=1}^{\tilde{h}_x} \mathbb{E}_x[\mathbb{1}_{\{X_n=y\}}] = \sum_{n=1}^{\infty} \mathbb{E}[\mathbb{1}_{\{X_n=y, n \leq \tilde{h}_x\}}] \\ &= \sum_{n \in \mathbb{N}} \sum_{z \in S} \mathbb{P}_x(X_n = y, X_{n-1} = z, n-1 < \tilde{h}_x) \\ &\stackrel{\text{Markov property \& exchange sums}}{=} \sum_{z \in S} \sum_{n \in \mathbb{N}} \mathbb{P}_x(X_1 = y) \mathbb{P}(X_{n-1} = z, n-1 < \tilde{h}_x) \\ &= \sum_{z \in S} \mathbb{P}_z(X_1 = y) \sum_{n \in \mathbb{N}} \mathbb{P}_x(X_{n-1} = z, n-1 < \tilde{h}_x) \\ &= \sum_{z \in S} \mathbb{P}_z(X_1 = y) \mathbb{E}_x \left[\sum_{n=1}^{\tilde{h}_x} \mathbb{1}_{\{X_{n-1}=z\}} \right] = \sum_{z \in S} \mathbb{P}_z(X_1 = y) \mathbb{E}_x \left[\sum_{k=0}^{\tilde{h}_x-1} \mathbb{1}_{\{X_k=1\}} \right] \end{aligned}$$

But x is an recurrent hence $\tilde{h}_x < \infty$ $\mathbb{P}$ -a.s., i.e.

$$\mathbb{P}_x(X_0 = X_{\tilde{h}_x} = x) = 1$$

This means

$$\sum_{k=0}^{\tilde{h}_x-1} \mathbb{1}_{\{X_k=1\}} = \sum_{k=0}^{\tilde{h}_x} \mathbb{1}_{\{X_k=1\}}, \quad \mathbb{P}_z\text{-a.s.}$$

Then

$$N_x(y) = \sum_{z \in S} \mathbb{P}_z(X_1 = y) \mathbb{E}_x \left[\sum_{k=0}^{\tilde{h}_x} \mathbb{1}_{\{X_k=1\}} \right] = \sum_{z \in S} \mathbb{P}_z(X_1 = y) N_x(z)$$

so $N_x(\cdot)$ is a stationary distribution.

3. From the definition of the stationary distribution we have if π is a stationary distribution

$$\pi(y) = \sum_{x \in S} \pi(x) \mathbb{P}_x(X_1 = y). \quad (12.8)$$

Inductively from (12.8) we get

$$\pi(y) = \sum_{x \in S} \pi(x) \mathbb{P}_x(X_k = y).$$

Moreover $\{X_n\}_{n \in \mathbb{N}}$ is irreducible by assumption meaning there exist $k, l > 0$ such that

$$\mathbb{P}_x(X_k = y) > 0 \quad \text{and} \quad \mathbb{P}_y(X_l = x) > 0.$$

From 2. we know that $N_x(\cdot)$ is a stationary distribution, hence from (12.8) follows

$$N_x(y) = \sum_{z \in S} \mathbb{P}_z(X_k = y) N_x(z) \geq \underbrace{\mathbb{P}_x(X_k = y)}_{>0} \underbrace{N_x(x)}_{=1 \text{ (see 1.)}} > 0. \quad (12.9)$$

$$N_x(x) = \sum_{z \in S} \mathbb{P}_z(X_l = x) N_x(z) \geq N_x(y) \underbrace{\mathbb{P}_y(X_l = x)}_{>0}.$$

Dividing by $\mathbb{P}_y(X_l = x)$ yields

$$N_x(y) \leq \frac{1}{\mathbb{P}_y(X_l = x)} < \infty. \quad (12.10)$$

The equations (12.9) and (12.10) imply the desired property. □

Proposition 12.40. Consider $\{X_n\}_{n \in \mathbb{N}}$, an irreducible and recurrent Markov chain and let π be its stationary distribution. Assume that $\pi(x) = 1$ for $x \in S$. Then the stationary distribution π is unique i.e. $\pi(y) = N_x(y)$ for every $y \in S$.

Proof. We know that π and N_x are stationary. (for π by definition and for N_x by the previous result). Let $\lambda = \pi - N_x$ which is also stationary and

$$\lambda(x) = \pi(x) - N_x(x) = 1 - 1 = 0.$$

We now claim that $\lambda(y) \geq 0$ for every $y \in S$.

The proof of this claim can be found in the lecture notes.

Then we pick an l that satisfies the irreducibility condition and compute

$$0 = \lambda(x) \stackrel{\text{stationary}}{=} \sum_{z \in S} \lambda(z) \mathbb{P}_z(X_l = x) \geq \underbrace{\lambda(y)}_{\geq 0} \underbrace{\mathbb{P}_y(X_l = x)}_{> 0}$$

so $\lambda \equiv 0$. □

Theorem 12.41. Let $\{X_n\}_{n \in \mathbb{N}}$ be a irreducible Markov chain. The following are equivalent:

1. There exists $x \in S$ positively recurrent
2. Any point in the state space S is positively recurrent
3. There exists stationary distribution π .

Moreover π in 3. is unique and is given by

$$\pi(y) = \frac{N_x(y)}{\mathbb{E}_x[\tilde{h}_x]} = \frac{N_x(y)}{\int_{\Omega} \tilde{h}_x d\mathbb{P}_x}.$$

Proof. • "(2) $\Rightarrow$ (1)": trivial

- "(1) $\Rightarrow$ (3)": Let x be positively recurrent, then x is recurrent which implies using prop 12.39 that N_x is a stationary distribution. Moreover

$$\sum_{y \in S} N_x(y) = \sum_{y \in S} \mathbb{E}_x \left[\sum_{n=1}^{\tilde{h}_x} \mathbb{1}_{\{X_n=y\}} \right] = \mathbb{E}_x[\tilde{h}_x].$$

This implies that $N_x(\cdot)/\mathbb{E}_x[\tilde{h}_x]$ is a stationary distribution so $\pi(y) = N_x(y)/\mathbb{E}_x[\tilde{h}_x]$.

- "(3) $\Rightarrow$ (2)": Let $x \in S$. Then there exists $y \in S$ such that $\pi(y) > 0$. By irreducibility there exists $n \in \mathbb{N}$ such that

$$\mathbb{P}_y(X_n = x) > 0.$$

In particular

$$\pi(x) = \sum_{z \in S} \pi(z) \mathbb{P}_z(X_n = x) \geq \pi(y) \underbrace{\mathbb{P}_y(X_n = x)}_{>0}.$$

Define $\lambda(y) := \frac{\pi(x)}{\pi(y)}$. Since π is stationary by assumption λ is also stationary and we have $\lambda(x) = 1$ which implies, using the same arguments as in the proof of prop 12.40, that $\lambda \geq N_x$.

$$\mathbb{E}_y[\tilde{h}_y] = \sum_{x \in S} N_y(x) \stackrel{\lambda \geq N_x}{\leq} \sum_{x \in S} \lambda(x) = \sum_{x \in S} \frac{\pi(x)}{\pi(y)} = \frac{1}{\pi(y)} \underbrace{\sum_{x \in S} \pi(x)}_{=1 \text{ (}\pi \text{ prob. meas.)}}$$

so $\pi(y) > 0$ implies $\mathbb{E}_y[\tilde{h}_y] < \infty$ which gives that y is positively recurrent.

This closes the chain of equivalences. What is left to prove is the uniqueness of the stationary distribution π : Let x be a (positive) recurrent point. Then this implies by prop ?? that

$$\sum_{y \in S} N_x(y) = \sum_{y \in S} \lambda(y)$$

hence

$$\mathbb{E}_x[\tilde{h}_x] = \frac{1}{\pi(x)}$$

implying

$$\pi(x) = \frac{1}{\mathbb{E}_x[\tilde{h}_x]}$$

hence π is the only stationary distribution. □

13 Convergence of Markov chains

We want to study the behaviour of the tail of Markov chains especially the asymptotic behaviour of these objects.

Definition 13.1 (period). Let $x \in S$ and $T(x) = \{n \in \mathbb{N} : \mathbb{P}_x(X_n = x) > 0\}$ the **times of possible returns** to x . The **period** of x is defined as $\gcd(T(x))$ (greatest common divisor of $T(x)$).

Definition 13.2. A Markov-chain is **aperiodic** if $\gcd(T(x)) = 1$ for every $x \in S$.

Theorem 13.3. Let $\{X_n\}_{n \in \mathbb{N}}$ be an irreducible aperiodic Markov-chain with stationary distribution π . Then

$$\lim_{n \rightarrow \infty} \mathbb{P}_y(X_n = x) = \pi(x) \stackrel{\text{thm 12.41}}{=} \frac{1}{\mathbb{E}_x[\tilde{h}_x]}.$$

for every $x, y \in S$.

In order to prove this result we need some technicalities. We start with the following lemma:

Lemma 13.4. For every $x, y \in S$,

$$\lim_{n \rightarrow \infty} \mathbb{E}_y \left[\frac{1}{n} \sum_{k=1}^n \mathbb{1}_{\{X_k = x\}} \right] = \frac{\rho_{y,x}}{\mathbb{E}_x[\tilde{h}_x]}$$

with $\rho_{y,x} = \mathbb{P}_x[\tilde{h}_y < \infty]$ for an irreducible Markov chain $\{X_n\}_{n \in \mathbb{N}}$.

Proof. Using the Markov property to restart the Markov chain at m we have

$$\begin{aligned} \mathbb{E}_y \left[\frac{1}{n} \sum_{k=1}^n \mathbb{1}_{\{X_k = x\}} \right] &= \mathbb{E}_y \left[\frac{1}{n} \sum_{k=1}^n \mathbb{1}_{\{X_k = x\}}, \tilde{h}_x < \infty \right] \\ &= \sum_{m \in \mathbb{N}} \mathbb{E}_y \left[\frac{1}{n} \sum_{k=1}^{\min\{n,m\}} \mathbb{1}_{\{X_k = x\}} + \frac{1}{n} \sum_{k=1}^{\max\{n-m,0\}} \mathbb{1}_{\{X_k = x\}} \circ \theta_m, \tilde{h}_x = m \right] \\ &\stackrel{\text{Markov prop. at time } m}{=} \underbrace{\frac{1}{n} \mathbb{P}_y(\tilde{h}_x < n)}_{\xrightarrow{n \rightarrow \infty} > 0} + \underbrace{\sum_{m \in \mathbb{N}} \mathbb{P}_y(\tilde{h}_x = m)}_{> 0} \underbrace{\mathbb{E}_x \left[\frac{1}{n} \sum_{k=1}^{\max\{n-m,0\}} \mathbb{1}_{\{X_k = x\}} \right]}_{\xrightarrow{n \rightarrow \infty} \frac{1}{\mathbb{E}_x[\tilde{h}_x]} \text{ (thm 12.41)}} \end{aligned}$$

so we obtain

$$\lim_{n \rightarrow \infty} \mathbb{E}_y \left[\frac{1}{n} \sum_{k=1}^n \mathbb{1}_{\{X_k=x\}} \right] = \rho_{y,x} (\mathbb{E}_x[\tilde{h}_x])^{-1}$$

hence the result. $\square$

Proposition 13.5. Consider $\{X_n\}_{n \in \mathbb{N}}$ an irreducible and aperiodic Markov chain. Then for every $x, y \in S$ there exists $\bar{n} \in \mathbb{N}_0$ such that $\mathbb{P}_x[X_m = y] > 0$ for every $m > \bar{n}$.

Proof. The easy proof is left as an exercise to the reader. $\square$

Proof of theorem 13.3. Using the Markov property we compute

$$\begin{aligned} \mathbb{P}_y(X_n = x) &= \mathbb{P}_y(X_n = x, \tilde{h}_x \leq n) = \sum_{m=1}^n \mathbb{P}_y(X_{n-m} \circ \theta_m = x, \tilde{h}_x = m) \\ &\stackrel{\text{Markov prop. at time } m}{=} \sum_{m=1}^n \underbrace{\mathbb{P}_y(\tilde{h}_x = m)}_{\xrightarrow{n \rightarrow \infty} 1} \underbrace{\mathbb{P}_x(X_{n-m} = x)}_{\mathbb{E}_x[\mathbb{1}_{\{X_{n-m}=x\}}]} \end{aligned}$$

with $\mathbb{E}_x[\mathbb{1}_{\{X_{n-m}=x\}}] \xrightarrow{n \rightarrow \infty} \mathbb{E}_x[\tilde{h}_x]^{-1}$ according to lemma 13.4. But

$$\lim_{n \rightarrow \infty} \mathbb{P}_y(X_n = x) = \frac{1}{\mathbb{E}_x[\tilde{h}_x]} = 0$$

if x is null recurrent and

$$\lim_{n \rightarrow \infty} \mathbb{P}_y(X_n = x) = \mathbb{E}_x[\tilde{h}_x]^{-1} < \infty$$

if x is positively recurrent. After using the dominated convergence thm in the case of positively recurrent points we get the result. $\square$

Remark 13.6. There are instances where we can prove this theorem also for periodic Markov chains. $\triangle$